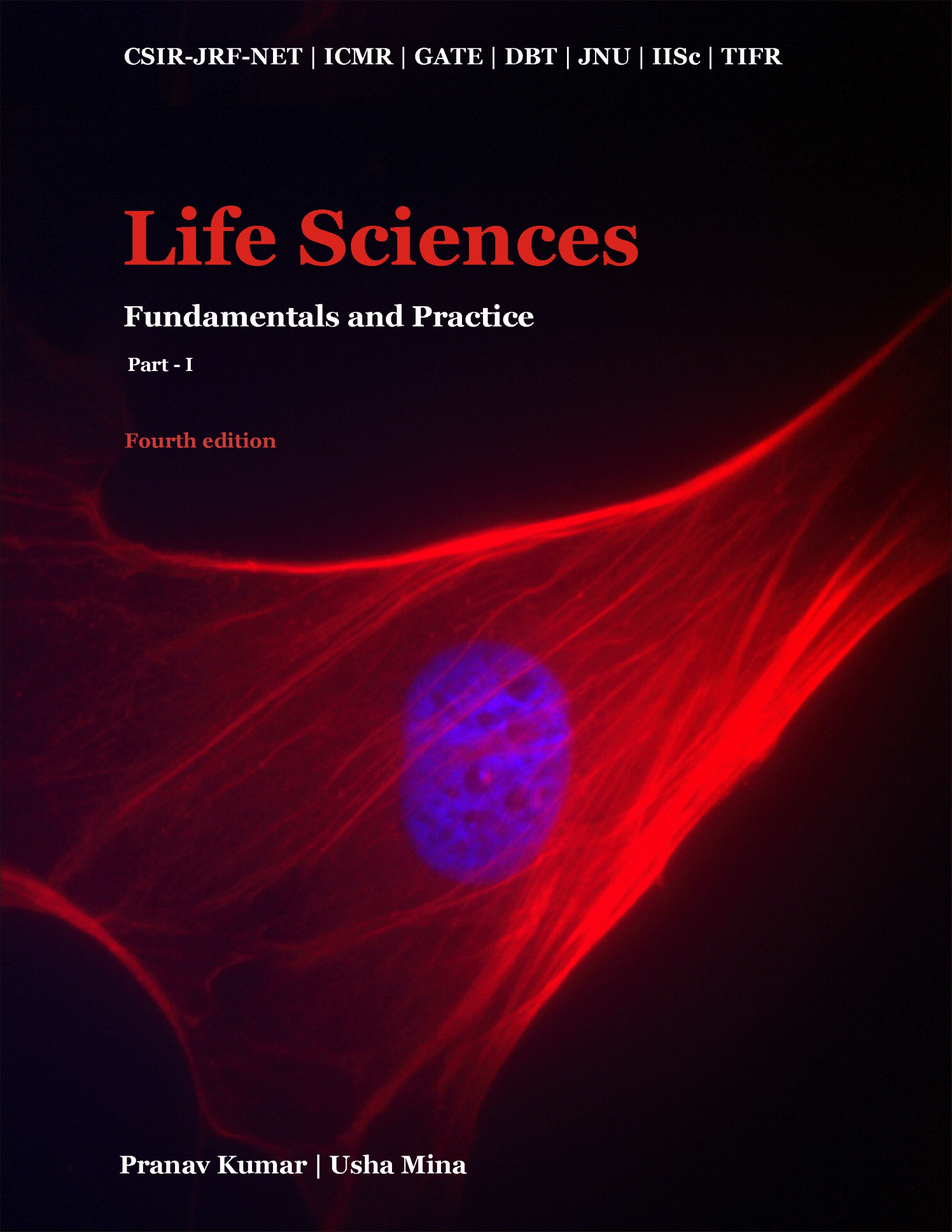
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Life Sciences

Fundamentals and Practice

Part - I

Fourth edition



Pranav Kumar | Usha Mina

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Preface

Life Sciences have always been a fundamental area of science. The exponential increase in the quantity of scientific information and the rate, at which new discoveries are made, require very elaborate, interdisciplinary and up-to-date information and their understanding. This fourth edition of Life sciences, Fundamentals and practice includes extensive revisions of the previous edition. We have attempted to provide an extraordinarily large amount of information from the enormous and ever-growing field in an easily retrievable form. It is written in clear and concise language to enhance self-motivation and strategic learning skill of the students and empowering them with a mechanism to measure and analyze their abilities and the confidence of winning. We have given equal importance to text and illustrations. The fourth edition has a number of new figures to enhance understanding. At the same time, we avoid excess detail, which can obscure the main point of the figure. We have retained the design elements that have evolved through the previous editions to make the book easier to read. Sincere efforts have been made to support textual clarifications and explanations with the help of flow charts, figures and tables to make learning easy and convincing. The chapters have been supplemented with self-tests and questions so as to check one's own level of understanding. Although the chapters of this book can be read independently of one another, they are arranged in a logical sequence. Each page is carefully laid out to place related text, figures and tables near one another, minimizing the need for page turning while reading a topic. I have given equal importance to text and illustrations as well. We hope you will find this book interesting, relevant and challenging.

Acknowledgements

Our students were the original inspiration for the first edition of this book, and we remain continually grateful to them, because we learn from them how to think about the life sciences, and how to communicate knowledge in most meaningful way. We thank, Dr. Diwakar Kumar Singh and Mr. Ajay Kumar, reviewers of this book, whose comment and suggestions were invaluable in improving the text. Any book of this kind requires meticulous and painstaking efforts by all its contributors. Several diligent and hardworking minds have come together to bring out this book in this complete form. We are much beholden to each of them and especially to Dr. Neeraj Tiwari. This book is a team effort, and producing it would be impossible without the outstanding people of Pathfinder Publication. It was a pleasure to work with many other dedicated and creative people of Pathfinder Publication during the production of this book, especially Pradeep Verma.

Pranav Kumar

Usha Mina

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Contents

Chapter 1

Biomolecules and Catalysis

1.1 Amino acids and Proteins 1
- 1.1.1 Optical properties 2
- 1.1.2 Absolute configuration 4
- 1.1.3 Standard and non-standard amino acids 5
- 1.1.4 Titration of amino acids 8
- 1.1.5 Peptide and polypeptide 11
- 1.1.6 Peptide bond 12
- 1.1.7 Protein structure 14
- 1.1.8 Denaturation of proteins 18
- 1.1.9 Solubilities of proteins 19
- 1.1.10 Simple and conjugated proteins 20

1.2 Fibrous and globular proteins 20
- 1.2.1 Collagen 21
- 1.2.2 Elastin 22
- 1.2.3 Keratins 23
- 1.2.4 Myoglobin 23
- 1.2.5 Hemoglobin 25
- 1.2.6 Models for the behavior of allosteric proteins 29

1.3 Protein folding 31
- 1.3.1 Molecular chaperones 32
- 1.3.2 Amyloid 33

1.4 Protein sequencing and assays 34

1.5 Nucleic acids 42
- 1.5.1 Nucleotides 42
- 1.5.2 Chargaff's rules 46

1.6 Structure of dsDNA 47
- 1.6.1 B-DNA 47
- 1.6.2 Z-DNA 49
- 1.6.3 Triplex DNA 49
- 1.6.4 G-quadruplex 50
- 1.6.5 Stability of the double helical structure of DNA 51
- 1.6.6 Thermal denaturation 51
- 1.6.7 Quantification of nucleic acids 53
- 1.6.8 Supercoiled forms of DNA 53
- 1.6.9 DNA: A genetic material 56

1.7 RNA 58
1.7.1 Alkali-catalyzed cleavage of RNA 60
1.7.2 RNA world hypothesis 61
1.7.3 RNA as genetic material 61
1.8 Carbohydrates 63
1.8.1 Monosaccharide 63
1.8.2 Epimers 64
1.8.3 Cyclic forms 65
1.8.4 Derivatives of monosaccharide 67
1.8.5 Disaccharides and glycosidic bond 68
1.8.6 Polysaccharides 70
1.8.7 Glycoproteins 72
1.8.8 Reducing and non-reducing sugar 73
1.9 Lipids 73
1.9.1 Fatty acids 74
1.9.2 Triacylglycerol and Wax 75
1.9.3 Phospholipids 76
1.9.4 Glycolipids 78
1.9.5 Steroid 79
1.9.6 Eicosanoid 79
1.9.7 Plasma lipoproteins 81
1.10 Vitamins 82
1.10.1 Water-soluble vitamins 82
1.10.2 Fat-soluble vitamins 86
1.11 Enzymes 89
1.11.1 Naming and classification of enzyme 90
1.11.2 What enzyme does? 91
1.11.3 How enzymes operate? 92
1.11.4 Enzyme kinetics 94
1.11.5 Enzyme inhibition 102
1.11.6 Regulatory enzymes 105
1.11.7 Isozymes 106
1.11.8 Zymogen 107
1.11.9 Ribozyme 108
1.11.10 Examples of enzymatic reactions 108

Chapter 2

Bioenergetics and Metabolism

2.1 Bioenergetics 117
2.2 Metabolism 122
2.3 Respiration 123
2.3.1 Aerobic respiration 123
2.3.2 Glycolysis 124
2.3.3 Pyruvate oxidation 129

2.3.4 Krebs cycle 131
2.3.5 Anaplerotic reaction 134
2.3.6 Oxidative phosphorylation 135
2.3.7 Inhibitors of electron transport 139
2.3.8 Electrochemical proton gradient 140
2.3.9 Chemiosmotic theory 141
2.3.10 ATP synthase 142
2.3.11 Uncoupling agents and ionophores 144
2.3.12 ATP-ADP exchange across the inner mitochondrial membrane 144
2.3.13 Shuttle systems 145
2.3.14 P/O ratio 147
2.3.15 Fermentation 148
2.3.16 Pasteur effect 150
2.3.17 Warburg effect 150
2.3.18 Respiratory quotient 151
2.4 Glyoxylate cycle 151
2.5 Pentose phosphate pathway 152
2.6 Entner-Doudoroff pathway 154
2.7 Photosynthesis 154
2.7.1 Photosynthetic pigment 155
2.7.2 Absorption and action spectra 158
2.7.3 Fate of light energy absorbed by photosynthetic pigments 160
2.7.4 Concept of photosynthetic unit 161
2.7.5 Hill reaction 162
2.7.6 Oxygenic and anoxygenic photosynthesis 162
2.7.7 Concept of pigment system 163
2.7.8 Stages of photosynthesis 165
2.7.9 Light reactions 165
2.7.10 Prokaryotic photosynthesis 171
2.7.11 Non-chlorophyll based photosynthesis 173
2.7.12 Dark reaction: Calvin cycle 174
2.7.13 Starch and sucrose synthesis 177
2.8 Photorespiration 178
2.8.1 C_4 cycle 179
2.8.2 CAM pathway 180
2.9 Carbohydrate metabolism 182
2.9.1 Gluconeogenesis 182
2.9.2 Glycogen metabolism 187
2.10 Lipid metabolism 192
2.10.1 Synthesis and storage of triacylglycerols 192
2.10.2 Biosynthesis of fatty acid 194
2.10.3 Fatty acid oxidation 198
2.10.4 Biosynthesis of cholesterol 205
2.10.5 Steroid hormones and Bile acids 206

2.11 Amino acid metabolism 208
2.11.1 Amino acid synthesis 208
2.11.2 Amino acid catabolism 211
2.11.3 Molecules derived from amino acids 217
2.12 Nucleotide metabolism 218
2.12.1 Nucleotide synthesis 218
2.12.2 Nucleotide degradation 225

Chapter 3

Cell Structure and Functions

3.1 What is a Cell? 231
3.2 Structure of eukaryotic cells 232
3.2.1 Plasma membrane 232
3.2.2 ABO blood group 239
3.2.3 Transport across plasma membrane 241
3.3 Membrane potential 248
3.4 Transport of macromolecules across plasma membrane 258
3.4.1 Endocytosis 258
3.4.2 Fate of receptor 262
3.4.3 Exocytosis 263
3.5 Ribosome 264
3.5.1 Protein targeting and translocation 265
3.6 Endoplasmic reticulum 266
3.6.1 Endomembrane system 271
3.6.2 Transport of proteins across the ER membrane 271
3.6.3 Transport of proteins from ER to cis Golgi 276
3.7 Golgi complex 277
3.7.1 Transport of proteins through cisternae 279
3.7.2 Transport of proteins from the TGN to lysosomes 280
3.8 Vesicle fusion 281
3.9 Lysosome 282
3.10 Vacuoles 284
3.11 Mitochondria 284
3.12 Plastids 287
3.13 Peroxisome 288
3.14 Cytoskeleton 289
3.14.1 Microtubules 289
3.14.2 Kinesins and Dyneins 292
3.14.3 Cilia and Flagella 292
3.14.4 Centriole 295
3.14.5 Actin filament 295
3.14.6 Myosin 297
3.14.7 Muscle contraction 298
3.14.8 Intermediate filaments 302

3.15 Cell junctions 303
3.16 Cell adhesion molecules 306
3.17 Extracellular matrix of animals 307
3.18 Plant cell wall 308
3.19 Nucleus 310
3.20 Cell signaling 313
3.20.1 Signal molecules 314
3.20.2 Receptors 315
3.20.3 GPCR and G-proteins 317
3.20.4 Ion channel-linked receptors 326
3.20.5 Enzyme-linked receptors 326
3.20.6 Nitric oxide 332
3.20.7 Two-component signaling systems 333
3.20.8 Chemotaxis in bacteria 334
3.20.9 Quorum sensing 335
3.20.10 Scatchard plot 336
3.21 Cell Cycle 338
3.21.1 Role of Rb protein in cell cycle regulation 342
3.21.2 Role of p53 protein in cell cycle regulation 343
3.21.3 Replicative senescence 344
3.22 Mechanics of cell division 344
3.22.1 Mitosis 344
3.22.2 Meiosis 351
3.22.3 Nondisjunction and aneuploidy 357
3.23 Apoptosis 358
3.24 Cancer 361
3.25 Stem cells 368

Chapter 4

Prokaryotes and Viruses

4.1 General features of Prokaryotes 373
4.2 Phylogenetic overview 374
4.3 Structure of bacterial cell 374
4.4 Bacterial genome : Bacterial chromosome and plasmid 385
4.5 Bacterial nutrition 389
4.5.1 Culture media 391
4.5.2 Bacterial growth 391
4.6 Horizontal gene transfer and genetic recombination 395
4.6.1 Transformation 396
4.6.2 Transduction 398
4.6.3 Conjugation 402
4.7 Bacterial taxonomy 407
4.8 General features of important bacterial groups 409
4.9 Archaebacteria 411

4.10 Bacterial toxins 412
4.11 Control of microbial growth 414
4.12 Virus 418
4.12.1 Bacteriophage (Bacterial virus) 419
4.12.2 Life cycle of bacteriophage 420
4.12.3 Plaque assay 423
4.12.4 Genetic analysis of phage 425
4.12.5 Animal viruses 428
4.12.6 Plant viruses 438
4.13 Prions and Viroid 439
4.13.1 Bacterial and viral disease 440

Chapter 5

Immunology

5.1 Innate immunity 443
5.2 Adaptive immunity 445
5.3 Cells of the immune system 447
5.3.1 Lymphoid progenitor 448
5.3.2 Myeloid progenitor 450
5.4 Organs involved in the adaptive immune response 451
5.4.1 Primary lymphoid organs 451
5.4.2 Secondary lymphoid organs/tissues 452
5.5 Antigens 453
5.6 Major-histocompatibility complex 457
5.6.1 MHC molecules and antigen presentation 459
5.6.2 Antigen processing and presentation 460
5.6.3 Laboratory mice 462
5.7 Immunoglobulins : Structure and function 463
5.7.1 Basic structure of antibody molecule 463
5.7.2 Different classes of immunoglobulin 466
5.7.3 Action of antibody 468
5.7.4 Antigenic determinants on immunoglobulins 468
5.8 B-cell maturation and activation 470
5.9 Kinetics of the antibody response 476
5.10 Monoclonal antibodies and Hybridoma technology 477
5.10.1 Engineered monoclonal antibodies 478
5.11 Organization and expression of Ig genes 480
5.12 Generation of antibody diversity 486
5.13 T-cells and CMI 489
5.13.1 Superantigens 499
5.14 Cytokines 500
5.15 The complement system 504
5.16 Hypersensitivity 507
5.17 Autoimmunity 510
5.18 Transplantation 510

5.19 Immunodeficiency diseases 511
5.20 Failures of host defense mechanisms 511
5.21 Vaccines 513

Chapter 6

Diversity of Life

6.1 Taxonomy 520
6.1.1 Nomenclature 520
6.1.2 Classification 521
6.1.3 Biological species concept 521
6.1.4 Phenetics 522
6.1.5 Cladistics 524
6.2 The five-kingdom system 526
6.3 Protists 528
6.3.1 Protozoan protists 528
6.3.2 Photosynthetic protists 529
6.3.3 Slime mold 530
6.3.4 Oomycetes 531
6.4 Fungi 531
6.4.1 Mycorrhiza 533
6.4.2 Lichens 533
6.5 Plantae 534
6.5.1 Plant life cycle 534
6.5.2 Algae 536
6.5.3 Life cycle of land plants 538
6.5.4 Bryophytes 539
6.5.5 Pteridophytes 541
6.5.6 Gymnosperm 542
6.5.7 Angiosperms 543
6.6 Animalia 547
6.7 Animal's classification 553
6.7.1 Phylum Porifera 554
6.7.2 Phylum Cnidaria 554
6.7.3 Phylum Platyhelminthes 554
6.7.4 Phylum Aschelminthes 555
6.7.5 Phylum Annelida 557
6.7.6 Phylum Mollusca 557
6.7.7 Phylum Arthropoda 557
6.7.8 Phylum Echinodermata 558
6.7.9 Phylum Hemichordata 558
6.7.10 Phylum Chordata 559

Answers of self test 567
Index

This is a preview. The total pages displayed will be limited.

Chapter 01

Biomolecules and Catalysis

A *biomolecule* is an organic molecule that is produced by a living organism. Biomolecules act as building blocks of life and perform important functions in living organisms. More than 25 naturally occurring chemical elements are found in biomolecules. Most of the elements have relatively low atomic numbers. Biomolecules consist primarily of carbon, hydrogen, nitrogen, oxygen, phosphorus and sulfur. The four most abundant elements in living organisms, in terms of the percentage of the total number of atoms, are hydrogen, oxygen, nitrogen, and carbon, which together make up over 99% of the mass of most cells.

Nearly all of the biomolecules in a cell are carbon compounds, which account for more than one-half of the dry weight of the cells. Covalent bonding between carbon and other elements permit formation of a large number of compounds. Most biomolecules can be regarded as derivatives of hydrocarbons. The hydrogen atoms may be replaced by a variety of functional groups to yield different families of organic compounds. Typical families of organic compounds are the alcohols, which have one or more hydroxyl groups; amines, which have amino groups; aldehydes and ketones, which have carbonyl groups; and carboxylic acids, which have carboxyl groups. Many biomolecules are polyfunctional, containing two or more different kinds of functional groups. Functional groups determine chemical properties of biomolecules.

Sugars, fatty acids, amino acids and nucleotides constitute the four major families of biomolecules in cells. Many of the biomolecules found within cells are macromolecules and mostly are polymers (composed of small, covalently linked monomeric subunits). These macromolecules are proteins, carbohydrates, lipids and nucleic acids.

Small biomolecules	*Macromolecules*
Sugars	Polysaccharide
Fatty acids	Fats/Lipids
Amino acids	Proteins
Nucleotide	Nucleic acid

Nucleic acids and proteins are informational macromolecules. Proteins are polymers of amino acids and constitute the largest fraction (besides water) of cells. The nucleic acids, DNA and RNA, are polymers of nucleotides. They store, transmit, and translate genetic information. The polysaccharides, polymers of simple sugars, have two major functions. They serve as energy-yielding fuel stores and as extracellular structural elements.

1.1 Amino acids and Proteins

Amino acids are compounds containing carbon, hydrogen, oxygen and nitrogen. They serve as monomers (building blocks) of proteins and composed of an amino group, a carboxyl group, a hydrogen atom, and a distinctive side chain, all bonded to a carbon atom, the α-carbon.

In an α-amino acid, the amino and carboxylate groups are attached to the same carbon atom, which is called the α-carbon. The various α-amino acids differ with respect to the side chain (R group) attached to their α-carbon. The general structure of an amino acid is:

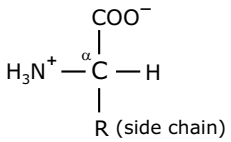


Figure 1.1 General structure of an amino acid.

This structure is common to all except one of the α-amino acids (*proline* is the exception). The R group or side chain attached to the α-carbon is different in each amino acid. In the simplest case, the R group is a hydrogen atom and amino acid is glycine.

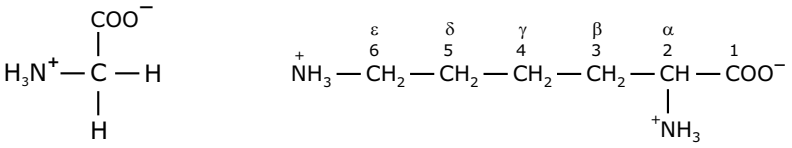


Figure 1.2 Structure of glycine and lysine.

In α-amino acids both the amino group and the carboxyl group are attached to the same carbon atom. However, many naturally occurring amino acids not found in protein, have structures that differ from the α-amino acids. In these compounds the amino group is attached to a carbon atom other than the α-carbon atom and they are called β, γ, δ, or ε amino acids depending upon the location of the C-atom to which amino group is attached.

Amino acids can act as acids and bases

When an amino acid is dissolved in water, it exists in solution as the *dipolar ion* or *zwitterion*. A zwitterion can act as either an acid (proton donor) or a base (proton acceptor). Hence, an amino acid is an **amphoteric molecule**. At high concentrations of hydrogen ions (low pH), the carboxyl group accepts a proton and becomes uncharged, so that the overall charge on the molecule is *positive*. Similarly at low concentrations of hydrogen ion (high pH), the amino group loses its proton and becomes uncharged; thus the overall charge on the molecule is *negative*.

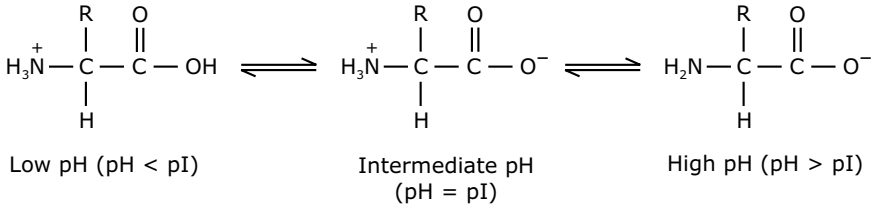


Figure 1.3 The acid-base behavior of an amino acid in solution. At low pH, the positively charged species predominates. As the pH increases, the electrically neutral zwitterion becomes predominant. At higher pH, the negatively charged species predominates.

1.1.1 Optical properties

All amino acids except glycine are optically active i.e. they rotate the plane of plane polarized light. Optically active molecules contain *chiral* carbon. A tetrahedral carbon atom with four different constituents are said to be chiral. All amino acids except glycine have chiral carbon and hence they are optically active.

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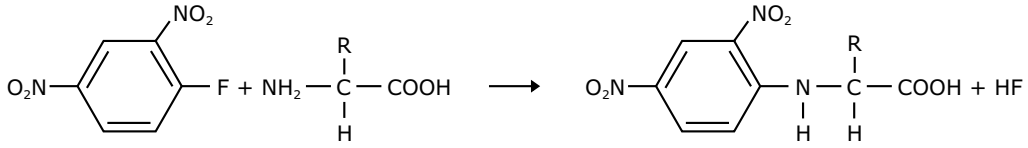
1.4 Protein sequencing and assays

Determination of amino acid compositions

Peptide bonds of proteins are hydrolyzed by either strong acid or strong base. In acid hydrolysis, the peptide can be hydrolyzed into its constituent amino acids by heating it in 6 M HCl at 110°C for 24 hours. Base hydrolysis of polypeptides is carried out in 2 to 4 M NaOH at 100°C for 4 to 8 hours. A mixture of amino acids in hydrolysates can be separated by ion exchange chromatography or by reversed phase HPLC. The identity of the amino acid is revealed by its elution volume and quantified by reaction with *ninhydrin.*

N-terminal analysis

Reagent *1-fluoro-2,4-dinitrobenzene* (FDNB) and *Dansyl chloride* are used for determination of N-terminal amino acid residue. FDNB reacts in alkaline solution (pH 9.5) with the free amino group of the N-terminal amino acid residue of a peptide to form a characteristic yellow dinitrophenyl (DNP) derivative. It can be released from the peptide by either acid or enzymic hydrolysis of the peptide bond and subsequently identified. Sanger first used this reaction to determine the primary structure of the polypeptide hormone insulin. This reagent is often referred to as *Sanger's reagent*.



FDNB Yellow-coloured derivative

Figure 1.34 FDNB reacts with free amino group to produce dinitrophenyl (DNP) derivative of amino acid.

Similarly, *Dansyl chloride* reacts with a free amino group of the N-terminal amino acid residue of a peptide in alkaline solution to form strongly fluorescent derivatives of free amino acids and N-terminal amino acid residue of peptides.

Edman degradation

Edman degradation method for determining the sequence of peptides and proteins from their N-terminus was developed by Pehr Edman. This chemical method uses phenylisothiocyanate (also termed Edman reagent) for sequential removal of amino acid residues from the *N-terminus* of a polypeptide chain.

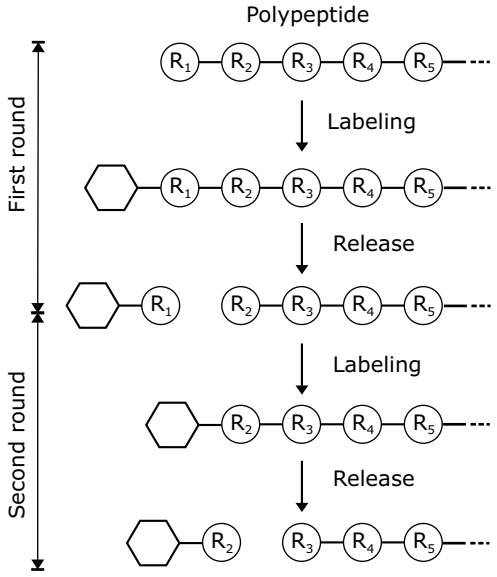
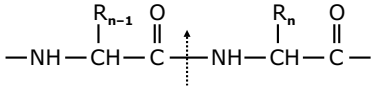


Figure 1.35
Edman degradation sequentially removes one residue at a time from the amino end of a peptide. The labeled amino-terminal residue (R_1) can be released without hydrolyzing the rest of the peptide bonds. Hence, the amino-terminal residue of the shortened peptide (R_2—R_3—R_4—R_5) can be determined in the second round.

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trypsin, chymotrypsin, elastase, thermolysin and pepsin. Various other chemicals also cleave polypeptide chains at specific locations. The most widely used is *cyanogen bromide* (CNBr), which cleaves peptide bond at C-terminal of Met residues. Similarly hydroxylamine cleaves the polypeptide chain at Asn-Gly sequences.

Table 1.8 Specificities of proteolytic enzymes.



Agents	*Site of Cleavage*
Trypsin	Carboxyl side of Lys or Arg, $R_n \neq$ Pro
Chymotrypsin	Carboxyl side of aromatic amino acid residues, $R_n \neq$ Pro
Pepsin	Amino side of aromatic amino acids like Tyr, Phe and Trp, $R_{n-1} \neq$ Pro
Elastase	Carboxyl side of Ala, Gly and Ser, $R_n \neq$ Pro

Carboxypeptidases and *aminopeptidases* are exopeptidases that remove terminal amino acid residues from C and N-termini of polypeptides, respectively. Carboxypeptidase A cleaves the C-terminal peptide bond of all amino acid residues except Pro, Lys and Arg. Carboxypeptidase B is effective only when Arg or Lys are the C-terminal residues. Carboxypeptidase C acts on any C-terminal residue. Aminopeptidases catalyze the cleavage of amino acids from the amino terminus of the protein. Aminopeptidase M catalyzes the cleavage of all free N-terminal residues.

Cleavage of disulfide bonds

If protein is made up of two or more polypeptide chains and held together by noncovalent bonds then denaturing agents, such as urea or guanidine hydrochloride, are used to dissociate the chains from one another. But polypeptide chains linked by disulfide bonds can be separated by two common methods. These methods are used to break disulfide bonds and also to prevent their reformation.

Oxidation of disulfide bonds with performic acid produces two cysteic acid residues. Because these cysteic acid side chains are ionized SO_3^- groups, electrostatic repulsion prevents S-S recombination. The second method involves the reduction by β-mercaptoethanol or dithiothreitol (Cleland's reagent) to form cysteine residues. This reaction is followed by further modification of the reactive –SH groups to prevent reformation of the disulfide bond. Acetylation by iodoacetate serves this purpose which modifies the –SH group.

Protein assays

To determine the amount of protein in an unknown sample is termed as *protein assays*. The simplest and most direct assay method for proteins in solution is to measure the absorbance at 280 nm (UV range). Amino acids containing aromatic side chains (i.e. tyrosine, tryptophan and phenylalanine) exhibit strong UV-light absorption. Consequently, proteins absorb UV-light in proportion to their aromatic amino acid content and total concentration. Several colorimetric, reagent-based protein assay techniques have also been developed. Protein is added to the reagent, producing a color change in proportion to the amount added. Protein concentration is determined by reference to a standard curve consisting of known concentrations of a purified reference protein. Some most commonly used colorimetric, reagent-based methods are:

Biuret method : Biuret method is based on the direct complex formation between the peptide bonds of the protein and Cu^{2+} ion. This method is not highly sensitive since the complex does not have a high extinction coefficient.

Folin method : The Folin assay (also called *Lowry* method) is dependent on the presence of aromatic amino acids in the protein. First, a cupric/peptide bond complex is formed and then this is enhanced by a phosphomolybodate complex with the aromatic amino acids.

Bradford method : Bradford method is based on a blue dye (Coomassie Brilliant Blue) that binds to free amino groups in the side chains of amino acids, especially Lys. This assay is as sensitive as the Folin assay.

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1.5 Nucleic acids

Nucleic acid was first discovered by Friedrich Miescher from the nuclei of the pus cells (Leukocytes) from discarded surgical bandages and called it **nuclein**. Nuclein was later shown to be a mixture of a basic protein and a phosphorus-containing organic acid, now called **nucleic acid**. There are two types of nucleic acids (*polynucleotides*): ribonucleic acid (RNA) and deoxyribonucleic acid (DNA).

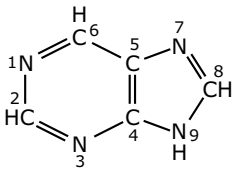
1.5.1 Nucleotides

The monomeric units of nucleic acids are called *nucleotides*. Nucleic acids therefore are also called *polynucleotides*. Nucleotides are phosphate esters of nucleosides and made up of three components:

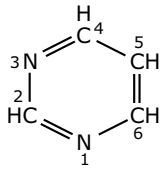
1. A base that has a nitrogen atom (nitrogenous base)
2. A five carbon sugar
3. An ion of phosphoric acid

Nitrogenous bases

Nitrogenous bases are heterocyclic, planar and relatively water insoluble aromatic molecules. There are two general types of nitrogenous bases in both DNA and RNA, pyrimidines and purines.



Purine



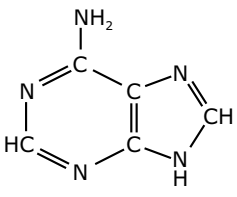
Pyrimidine

Purines

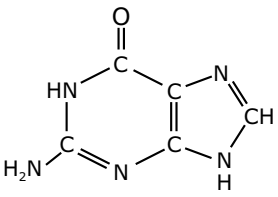
Two different nitrogenous bases with a purine ring (composed of carbon and nitrogen) are found in DNA. The two common purine bases found in DNA and RNA are *adenine* (6-aminopurine) and *guanine* (6-oxy-2-aminopurine). Adenine has an amino group ($-NH_2$) on the C6 position of the ring (carbon at position 6 of the ring). Guanine has an amino group at the C2 position and a carbonyl group at the C6 position.

Pyrimidines

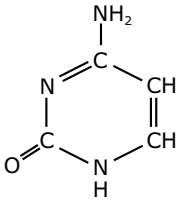
The two major pyrimidine bases found in DNA are *thymine* (5-methyl-2,4-dioxypyrimidine) and *cytosine* (2-oxy-4-aminopyrimidine) and in RNA they are uracil (2,4-dioxypyrimidine) and cytosine. Thymine contains a methyl group at the C5 position with carbonyl groups at the C4 and C2 positions. Cytosine contains a hydrogen atom at the C5 position and an amino group at C4. Uracil is similar to thymine but lacks the methyl group at the C5 position. Uracil is not usually found in DNA. It is a component of RNA.



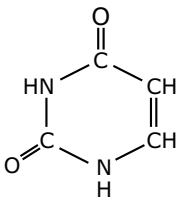
Adenine



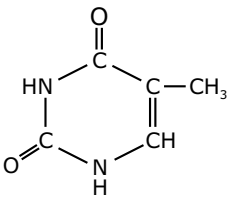
Guanine



Cytosine



Uracil



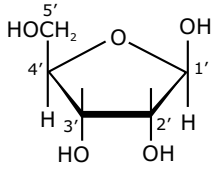
Thymine

Sugars

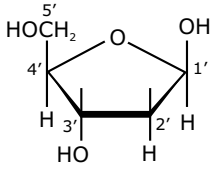
Naturally occurring nucleic acids have two types of pentose sugars: Ribose and deoxyribose sugar. All known sugars in nucleic acids have the D-stereoisomeric configuration.

Ribose sugar is found in RNA. β-D-Ribose is a five carbon sugar with a hydroxyl group (–OH) on each carbon (the carbon atoms of the ribose/deoxyribose present in nucleoside/nucleotides are designated with a prime (') mark to distinguish them from the backbone numbering in the bases).

Deoxyribose sugar is found in DNA. The hydroxyl group at 2′ position of ribose sugar is replaced by a hydrogen (–H).



β-D-Ribose

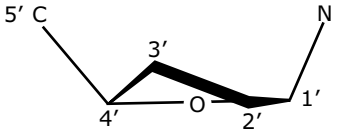


β-D-2-Deoxyribose

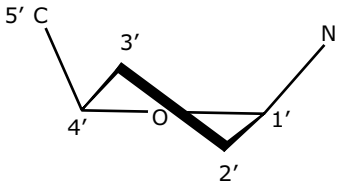
Sugar pucker

Pentose sugar is non-planar. This non-planarity is termed *puckering*. Pentose ring can be puckered in two basic conformations: *envelope* and *twisted*. In the *envelope* form, the four carbons of the pentose sugar are nearly coplanar and the fifth is away from the plane. In *twisted* form three atoms are coplanar and the other two lie away on opposite sides of this plane. Twisting the C2′ and C3′ carbons relative to the other atoms results in twisted forms of the sugar ring.

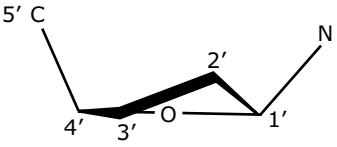
Sugar pucker can be endo or exo. C2′ or C3′ endo pucker means that C2′ or C3′ are on the same side as the base and C4′-C5′ bond. Exo-pucker describes a shift in the opposite direction. Purines show a preference for the C2′- endo pucker conformational type whereas pyrimidines favour C3′- endo. In RNA we find predominantly the C3′-*endo* conformation.



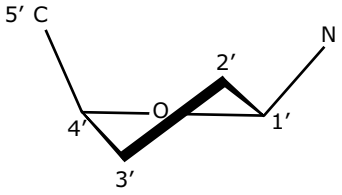
Envelope form, C3' endo



Twisted form, C3' endo and C2' exo



Envelope form, C2' endo

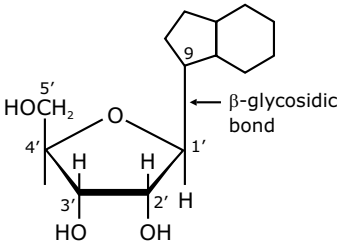


Twisted form, C2' endo and C3' exo

Figure 1.38 Sugar puckers.

Nucleoside

Sugar and nitrogenous base join to form nucleoside. The bond between the sugar and the base is called the **glycosidic bond**. This bond is said to be in the β (up) configuration with respect to the ribose sugar.



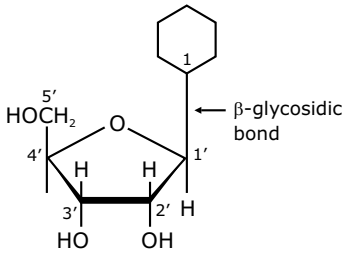


Figure 1.39 Structure of nucleoside.

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Table 1.9 Naming nucleosides and nucleotides

	Bases			
	Purines		**Pyrimidines**	
	Adenine (A)	Guanine (G)	Cytosine (C)	Uracil (U)/Thymine (T)
Nucleosides–in RNA	Adenosine	Guanosine	Cytidine	Uridine
in DNA	Deoxyadenosine	Deoxyguanosine	Deoxycytidine	Deoxythymidine
Nucleotides– in RNA	Adenylate	Guanylate	Cytidylate	Uridylate
in DNA	Deoxyadenylate	Deoxyguanylate	Deoxycytidylate	Deoxythymidylate
Nucleoside monophosphate	AMP	GMP	CMP	UMP/TMP
Nucleoside diphosphate	ADP	GDP	CDP	UDP/TDP
Nucleoside triphosphate	ATP	GTP	CTP	UTP/TTP

Polynucleotides

Polynucleotides are formed by the condensation of two or more nucleotides. The condensation most commonly occurs between the alcohol of a 5'-phosphate of one nucleotide and the 3'-hydroxyl of a second, with the elimination of H_2O, forming a *phosphodiester bond.* All nucleotides in a polynucleotide chain have the same relative orientation. The formation of phosphodiester bonds in DNA and RNA exhibits directionality. The primary structure of DNA and RNA (the linear arrangement of the nucleotides) proceeds in the 5'→3' direction. The common representation of the primary structure of DNA or RNA molecules is to write the nucleotide sequences from left to right synonymous with the 5'→3' direction as shown below.

5'-pGpApTpC-3'

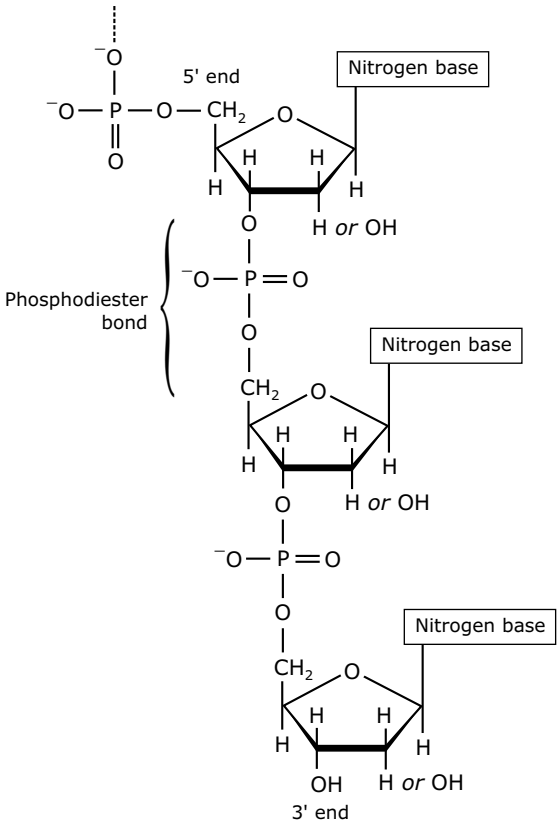


Figure 1.42 The polynucleotide has a 5' end, which is usually attached to a phosphate, and a 3' end, which is usually a free hydroxyl group. The backbones of these polynucleotide are formed by 3' to 5' phosphodiester linkages.

Pages 46 to 48 are not shown in this preview.

1.6.2 Z–DNA

Left-handed Z-DNA has been mostly found in alternating purine-pyrimidine sequences $(CG)_n$ and $(TG)_n$. Z-DNA is thinner (18 Å) than B-DNA (20 Å), the bases are shifted to the periphery of the helix, and there is only one deep, narrow groove equivalent to the minor groove in B-DNA. In contrast to B-DNA where a repeating unit is a 1 base pair, in Z-DNA the repeating unit is a 2 base pair. The backbone follows a zigzag path as opposed to a smooth path in B-DNA. The sugar and glycosidic bond conformations alternate; C2′ endo in *anti* dC and C3′ endo in *syn* dG. Electrostatic interactions play a crucial role in the Z-DNA formation. Therefore, Z-DNA is stabilized by high salt concentrations or polyvalent cations that shield interphosphate repulsion better than monovalent cations.

Z-DNA can form in regions of alternating purine-pyrimidine sequence; $(GC)_n$ sequences form Z-DNA most easily. $(GT)_n$ sequences also form Z-DNA but they require a greater stabilization energy. $(AT)_n$ sequences generally does not form Z-DNA since it easily forms cruciforms.

Table 1.10 Comparisons of different forms of DNA

Geometry attribute	*A-form*	*B-form*	*Z-form*
Helix sense	Right-handed	Right-handed	Left-handed
Repeating unit	1 bp	1 bp	2 bp
Rotation/bp (Twist angle)	33.6°	34.3°	60°/2
Mean bp/turn	10.7	10.4	12
Base pair tilt	20°	–6°	7°
Rise/bp along axis	2.3Å	3.32Å	3.8Å
Pitch/turn of helix	24.6Å	33.2Å	45.6Å
Mean propeller twist	+18°	+16°	0°
Glycosidic bond conformation	*Anti*	*Anti*	*Anti* for C, *Syn* for G
Sugar pucker	C3'-endo	C2'-endo	C:C2'-endo, G:C3'-endo
Diameter	23Å	20Å	18Å
Major groove	Narrow and deep	Wide and deep	Flat
Minor groove	Wide and shallow	Narrow and deep	Narrow and deep

1.6.3 Triplex DNA

In certain circumstances (e.g., low pH), a DNA sequence containing a long segment consisting of a polypurine strand, hydrogen bonded to a polypyrimidine strand and form a *triple helix*. The triple helix will be written as (*dT*).(dA).(dT) with the third strand in italics. Triple-stranded DNA is formed by laying a third strand into the major groove of DNA. A third strand makes a hydrogen bond to another surface of the duplex. The third strand pairs in a Hoogsteen base-pairing scheme. The central strand of the triplex must be purine rich. Thus, triple-stranded DNA requires a homopurine: homopyrimidine region of DNA. If the third strand is purine rich, it forms *reverse Hoogsteen* hydrogen bonds in an antiparallel orientation with the purine strand of the Watson-Crick helix. If the third strand is pyrimidine rich, it forms *Hoogsteen bonds* in a parallel orientation with the Watson-Crick-paired purine strand.

Triple helix can be intermolecular or intramolecular. In the *intermolecular Pu*.Pu.Py triple helix, the poly-purine third strand is organized antiparallel with respect to the purine strand of the original Watson-Crick duplex. In the intermolecular *Py*.Pu.Py triplex, the polypyrimidine third strand is organized parallel with respect to the purine strand and the phosphate backbone is positioned.

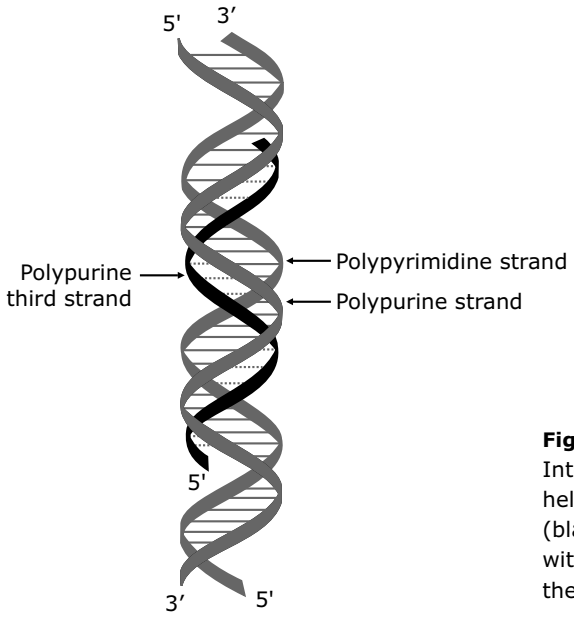


Figure 1.45
Intermolecular *Pu*.Pu.Py triple helix. The polypurine third strand (black colour) is organized antiparallel with respect to the purine strand of the original double strand DNA.

An *intramolecular triplex* (also referred to as **H-DNA**) could form within a single homopurine.homopyrimidine duplex DNA region in the supercoiled DNA. As in intermolecular triplexes, when the third strand is the pyrimidine strand, it forms Hoogsteen pairs in a parallel fashion with the central purine strand. When the third strand is the purine strand, it forms reverse Hoogsteen pairs in an antiparallel fashion with the central purine strand.

1.6.4 G−quadruplex

Nucleic acid sequences which are rich in guanine are capable of forming four-stranded structures called *G-quadruplexes* (also called G-quartat). These consist of a square arrangement of guanines (a tetrad), stabilized by Hoogsteen hydrogen bonding. The formation and stability of the G-quadruplexes is a monovalent cation-dependent. A monovalent cation is presents in the center of the tetrads. G-quadruplexes can be formed of DNA or RNA. They can be formed from one, two or four separate strands of DNA or RNA. Depending on the direction of the strands or parts of a strand that form the tetrads, structures may be described as parallel or antiparallel. All parallel quadruplexes have all guanine glycosidic angles in an *anti* conformation. Anti-parallel quadruplexes have both *syn* and *anti* conformations.

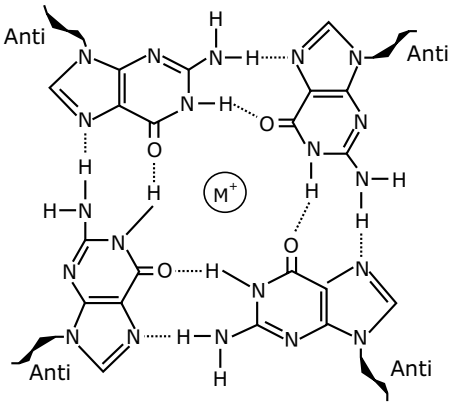


Figure 1.46
Four-stranded structures can arise from square arrangement of guanines.

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1.7 RNA

DNA contains all the information needed to maintain a cell's processes, but these precious blueprints never leave the protected nucleus. How, then, all these data are transmitted to the body of the cell itself where they are put to use? The answer: by way of RNA.

RNA molecules play essential roles in the transfer of genetic information during protein synthesis and in the control of gene expression. The diverse functions of RNA molecules in living organisms also include the enzymatic activity of ribozymes and the storage of genetic information in RNA viruses and viroids. So, RNA may be genetic or non genetic, catalytic or non-catalytic and coding (mRNA) or noncoding (like tRNA, rRNA).

Thermodynamic stability of RNA structure

Primary structure of RNA refers to the sequence of nucleotides. Secondary structure in RNA is dominated by Watson-Crick base pairing. This fundamental interaction between bases leads to the formation of double-helical structures of varying length. In RNA, double-helical tracts are generally short. RNA double helices adopt the A-form structure, which differs significantly from the canonical B-form adopted by DNA double helices. RNA's secondary structure is generally more stable than its tertiary structure. Thus, formation of the secondary structure dominates the process of RNA folding. RNA tertiary structure forms through relatively weak interactions between preformed secondary structure elements.

RNA duplexes are more stable than DNA duplexes. At physiological pH, denaturation of a double stranded helical RNA often requires higher temperatures than those required for denaturation of a DNA molecule with a comparable sequence. However, the physical basis for these differences in thermal stability is not known.

Types of RNA

Within a given cell, RNA molecules are found in multiple copies and in multiple forms. Major RNA classes are mRNA, rRNA, tRNA, snRNA, SnoRNA, miRNA, XIST, scRNA, siRNA, tmRNA and telomerase RNA. Features of few major forms of RNA present in prokaryotic and eukaryotic cells are given below.

mRNA

mRNA (messenger RNA) carries the genetic information copied from DNA in the form of a series of three-base code words, each of which specifies a particular amino acid. Most of the eukaryotic mRNAs represent only a single gene: they are *monocistronic*. mRNAs, which carry sequence coding for several polypeptides are called *polycistronic*. In these cases, a single mRNA is transcribed from a group of adjacent genes. Most of the prokaryotic mRNA are polycistronic.

All mRNAs contain two types of regions. The coding region consists of a series of codons starting with an AUG and ending with a termination codon. But the mRNA is always longer than the coding region, extra regions are present at both ends. The untranslated region at the 5' end is described as the *leader* and untranslated region at the 3' end is called the *trailer*. A polycistronic mRNA also contains *intercistronic* regions. They vary greatly in size. They may be as long as 30 nucleotides. Eukaryotic mRNA molecules often require extensive processing and transport, while prokaryotic molecules do not.

tRNA

tRNA is a small, well-characterized RNA molecules with a key role in protein biosynthesis. Transfer RNA is also known as *adaptor* RNA. The concept of an adaptor to provide the interface between nucleic acid language and protein language was introduced by Crick in 1955. tRNAs also participate in nonprotein synthetic processes such as a primer during reverse transcription in retrovirus life cycles.

tRNA is a single RNA chain of 73-93 nucleotides, present in the cytosol and organelles of all living cells. Holley and his co-workers determined the first tRNA sequence in 1965. Dictated by their primary sequence, tRNA folds into cloverleaf-like secondary structures with well-defined stems and loops that make up the acceptor arm, D arm and loop, anticodon arm and loop, and the T-arm and loop. Regardless of the length of the tRNA, the numbering of conserved nucleotides remains constant.

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maintaining the telomeres. They are always associated with specific proteins, and the complexes are referred to as *small nuclear ribonucleoproteins* (snRNP) or sometimes as snurps.

snoRNA

snoRNA (small nucleolar RNA) is a class of small RNA molecules that are involved in chemical modifications such as methylation of rRNAs and other forms of RNA in eukaryotes. snoRNAs form a component in the small nucleolar ribonucleoprotein (snoRNP), which contains snoRNA and proteins. The snoRNA guides the snoRNP complex to the modification site of the target RNA gene via sequences in the snoRNA that hybridize to the target site. The proteins then catalyze modification of the RNA gene.

gRNAs

gRNAs (guide RNA) are RNA genes that function in RNA editing. RNA editing was first reported in the mitochondria of *kinetoplastids*, in which mRNAs are edited by inserting or deleting stretches of uridylates (Us). The gRNA forms part of the *editosome* and contains sequences that hybridize to matching sequences in the mRNA, to guide the mRNA modifications.

Small silencing RNAs

Small silencing RNAs are small RNA of ~20–30 nucleotides long and make an association with members of the *Argonaute* protein family, which they guide to their regulatory targets. After the discovery of the first small silencing RNA in year 1993, several small RNA classes have been identified which differ in their biogenesis, their modes of target regulation and in the biological pathways they regulate. The most common examples of these small RNAs are siRNA (small interfering RNA), miRNA (micro RNA) and piRNA (Piwi-interacting RNA).

miRNAs are endogenous regulatory RNAs which are typically 20–25 nucleotides long, and are thought to regulate the expression of other genes. miRNAs derive from precursor transcripts called *primary miRNAs* (pri-miRNAs), which are typically transcribed by RNA polymerase II. The pri-miRNA is processed in the nucleus into a 60-70 nucleotide pre-miRNA by the activity of *Drosha*, a nuclear enzyme. The pre-miRNA molecule is then actively transported out of the nucleus into the cytoplasm by *exportin* protein. The *Dicer* enzyme, a dsRNA specific RNaseIII family endonuclease, then cuts pre-miRNA into the mature miRNA.

siRNAs are small RNA molecules of ~21 nucleotides. siRNA duplexes are produced by the Dicer. Out of two strands, the one that directs silencing of target mRNA is called *guide* RNA. Whereas the other strand which is ultimately destroyed, is the *passenger* strand. Target regulation is mediated by RISC (RNA Induced Silencing Complex). siRNA may be exo-siRNA and endo siRNA depending on the source of RNA.

tmRNA

tmRNA has a complex structure with tRNA-like and mRNA-like regions. It has currently only been found in bacteria. tmRNA recognizes ribosomes that have trouble translating or reading an mRNA and stall, leaving an unfinished protein that may be detrimental to the cell. tmRNA acts like a tRNA first, and then an mRNA that encodes a peptide tag. The ribosome translates this mRNA region of tmRNA and attaches the encoded peptide tag to the C-terminus of the unfinished protein. This attached tag targets the protein for destruction or proteolysis.

1.7.1 Alkali–catalyzed cleavage of RNA

Under alkaline conditions, RNA is hydrolyzed rapidly and generates a mixture of 2′- and 3′ nucleoside monophosphate. In the presence of a hydroxide ion, the 2′-hydroxyl group of the ribose is converted to a 2′-alkoxide ion. The 2′-alkoxide attacks the 3′-phosphodiester group, breaking the 5′-3′ phosphodiester bond and forming a cyclic 2′,3′-nucleoside monophosphate. Another hydroxide ion attacks the cyclic 2′,3′-nucleoside monophosphate, yielding a mixture of 2′-and 3′-nucleoside monophosphates. DNA is stable in basic solution because DNA lacks a 2′-hydroxyl group to carry out intramolecule catalysis.

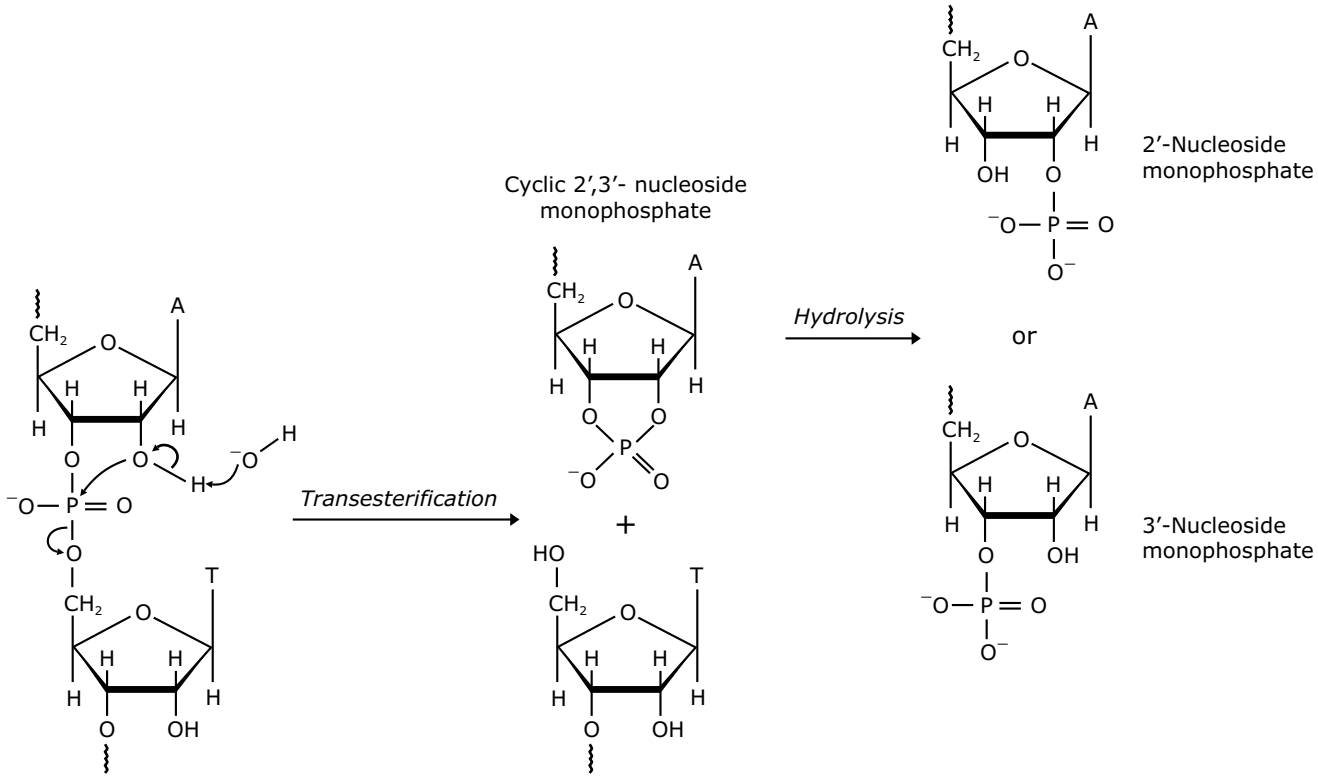


Figure 1.55 Alkali-catalyzed cleavage of RNA.

1.7.2 RNA world hypothesis

The RNA world hypothesis proposes that RNA was actually the first life-form on earth, later developing a cell membrane around it and becoming the first prokaryotic cell (the phrase *RNA World* was first used by *Walter Gilbert* in 1986). This hypothesis is supported by the RNA's ability to store, transmit, and duplicate genetic information, just like DNA does and to catalyze chemical reactions, just like protein does. RNA with catalytic activity is termed as *ribozyme*. Because RNA can perform the tasks of both genetic materials and enzymes, RNA is believed to have once been capable of independent life.

1.7.3 RNA as genetic material

Some viruses contain an RNA as genetic material. One of the first experiments that established RNA as the genetic material in RNA viruses was the *reconstitution experiment* of H.Fraenkel-Conrat and B.Singer.

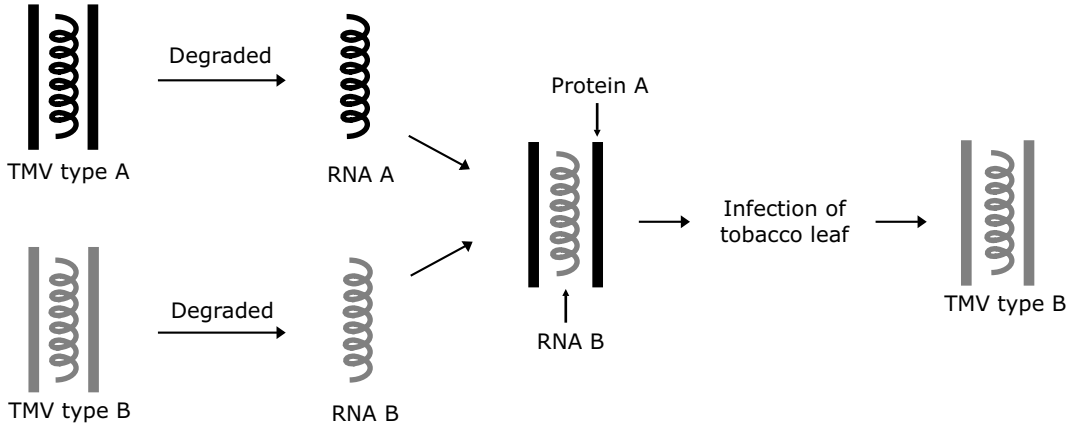


Figure 1.56 Proof that the genetic material of tobacco mosaic virus (TMV) is RNA, not protein.

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1.8 Carbohydrates

Carbohydrates are polyhydroxy aldehydes or polyhydroxy ketones, or compounds that can be hydrolyzed to them. In the majority of carbohydrates, H and O are present in the same ratio as in water, hence also called as *hydrates of carbon*. Carbohydrates are the most abundant biomolecules on Earth. Carbohydrates are classified into following classes depending upon whether these undergo hydrolysis and if so on the number of products form:

Monosaccharides are simple carbohydrates that cannot be hydrolyzed further into polyhydroxy aldehyde or ketone unit.

Oligosaccharides are polymers made up of two to ten monosaccharide units joined together by glycosidic linkages. Oligosaccharides can be classified as di-, tri-, tetra- depending upon the number of monosaccharides present. Amongst these the most abundant are the disaccharides, with two monosaccharide units.

Polysaccharides are polymers with hundreds or thousands of monosaccharide units. Polysaccharides are not sweet in taste hence they are also called *non-sugars*.

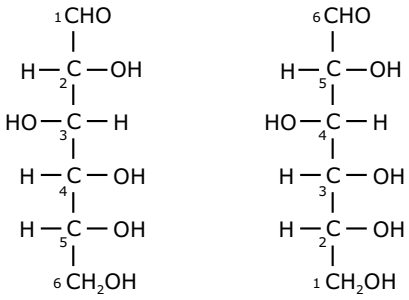
1.8.1 Monosaccharide

Monosaccharides consist of a single polyhydroxy aldehyde or ketone unit. Monosaccharides are the simple sugars, which cannot be hydrolyzed further into simpler forms and they have a general formula $C_nH_{2n}O_n$. Monosaccharides are colourless, crystalline solids that are freely soluble in water but insoluble in nonpolar solvents. The most abundant monosaccharide in nature is the D-glucose. Monosaccharides can be further sub classified on the basis of:

The number of the carbon atoms present

Monosaccharides can be named by a system that is based on the number of carbons with the suffix-*ose* added. Monosaccharides with four, five, six and seven carbon atoms are called *tetroses, pentoses, hexoses* and *heptoses,* respectively.

System for numbering the carbons : The carbons are numbered sequentially with the aldehyde or ketone group being on the carbon with the lowest possible number.



Presence of aldehydes or ketones groups

Aldoses are monosaccharides with an aldehyde group.

Ketoses are monosaccharides containing a ketone group.

The monosaccharide *glucose* is an *aldohexose;* that is, it is a six-carbon monosaccharide (-hexose) containing an aldehyde group (aldo-). Similarly *fructose* is a *ketohexose;* that is, it is a six-carbon monosaccharide (-hexose) and containing a ketone group (keto-).

Trioses are simplest monosaccharides. There are two trioses– dihydroxyacetone and glyceraldehyde. Dihydroxyacetone is called a *ketose* because it contains a *keto* group, whereas glyceraldehyde is called an *aldose* because it contains an *aldehyde* group.

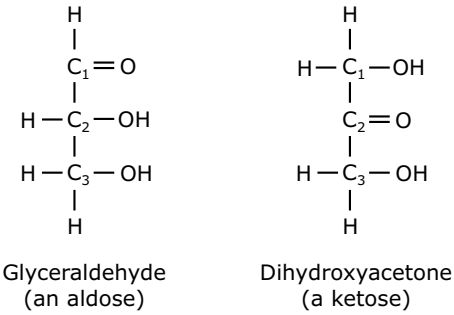


Figure 1.57 Trioses, the simplest monosaccharides.

Glyceraldehyde has a central carbon (C–2) which is chiral or asymmetrical. Chiral molecules such as glyceraldehyde can exist in two forms or configurations that are non-superimposable mirror images of each other. These two forms are called *enantiomers*. An enantiomer is identified by its absolute configuration. Glyceraldehyde has two absolute configurations. When the hydroxyl group attached to the chiral carbon is on the left in a Fischer projection, the configuration is L; when the hydroxyl group is on the right, the configuration is D.

The absolute configurations of monosaccharide containing more than one chiral centers like hexose are determined by comparing the configuration at the highest-numbered chiral carbon (the chiral carbon farthest from the aldehyde group) to the configuration at the single chiral carbon of glyceraldehyde.

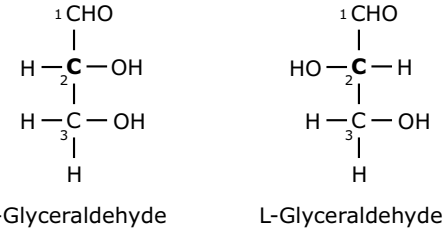


Figure 1.58 The enantiomers of glyceraldehyde. The configuration of groups around the chiral carbon 2 (shown in bold) distinguishes D-glyceraldehyde from L-glyceraldehyde. The two molecules are mirror images and cannot be superimposed on one another.

All the monosaccharides except *dihydroxyacetone* contain one or more chiral carbon atoms and thus occur in optically active isomeric forms. As the number of chiral carbon atoms increases, the number of possible stereoisomers also increases. The total number of possible isomers can be determined by using Van't Hoff's rule. A compound with 'n' chiral carbon atom has a maximum of 2^n possible stereoisomers.

1.8.2 Epimers

Many common sugars are closely related, differing only by the stereochemistry at a single carbon atom. For example, D-glucose and D-mannose differ only at carbon 2. Sugars that differ only by the stereochemistry at a single carbon (other than anomeric carbon) are called *epimers*. Similarly D-glucose and D-galactose are epimers. D-mannose and D-galactose are not epimers because their configuration differ at more than one carbon.

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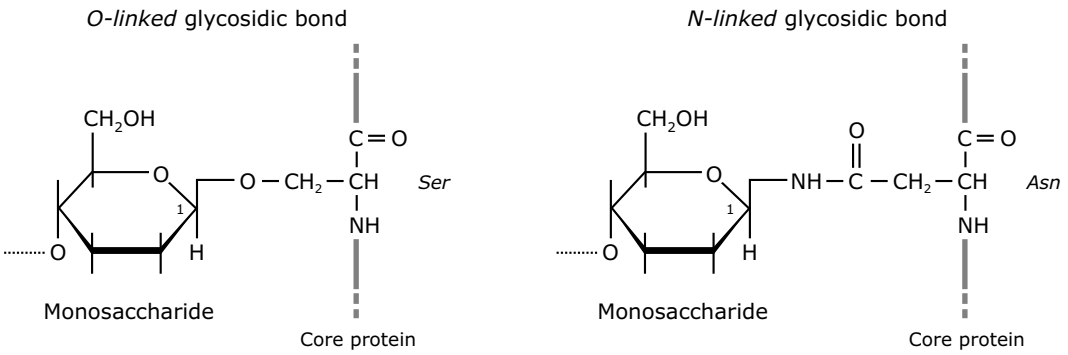


Figure 1.64 Carbohydrate groups are covalently attached to many different proteins to form *glycoproteins.* Sugars are attached either to the amide nitrogen atom in the side chain of asparagine (termed an N-*linkage*) or to the oxygen atom in the side chain of serine or threonine (termed an O-*linkage*).

1.8.8 Reducing and non–reducing sugar

Sugars capable of reducing ferric or cupric ion are called **reducing sugar**. A reducing sugar is any sugar that either has an aldehyde group or is capable of forming one in solution through isomerisation. This functional group allows the sugar to act as a reducing agent.

All monosaccharides whether aldoses and ketoses, in their hemiacetal and hemiketal form are reducing sugars. All disaccharides formed from head to tail condensation are also reducing sugar i.e. disaccharides except *sucrose, trehalose* are reducing sugars. All reducing sugars undergo mutarotation in aqueous solution.

Sugars like sucrose, trehalose not capable of reducing ferric or cupric ion are called **non-reducing sugar**. In sucrose and trehalose, anomeric carbon becomes involved in a glycosidic bond. So they donot contain free anomeric carbon atoms. Sucrose and trehalose are therefore not a reducing sugar, and have no reducing end. So it cannot be oxidized by cupric or ferric ion. In describing disaccharides or polysaccharides, the end of a chain that has a free anomeric carbon (i.e. is not involved in a glycosidic bond) is commonly called the reducing end of the chain.

1.9 Lipids

Biological lipids are a chemically diverse group of organic compounds which are insoluble or only poorly soluble in water. They are readily soluble in nonpolar solvents such as ether, chloroform, or benzene. The hydrophobic nature of lipids is due to the predominance of hydrocarbon chains ($—CH_2—CH_2—CH_2—$) in their structures. Unlike the proteins, nucleic acids, and polysaccharides, lipids are not polymers.

Functions

Biological lipids have diverse functions. The four general functions of biological lipids have been identified.

- They serve as a storage form of metabolic fuel.
- They serve as a transport form of metabolic fuel.
- They provide the structural components of membranes.
- They have protective functions in bacteria, plants, insects, and vertebrates, serving as a part of the outer coating between the body of the organism and the environment.

Apart from the general functions biological lipids serve as *pigments* (carotene), *hormones* (vitamin D derivatives, sex hormones), *signaling molecules* (eicosanoids, phosphatidylinositol derivatives), *cofactors* (vitamin K), *detergents* (bile salt) and many other specialized functions.

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The notation 18:1 denotes a 18 carbons fatty acid with one double bond, whereas 18:2 signifies that there are two double bonds.

The most commonly used systems for designating the position of double bonds in an unsaturated fatty acid is the delta (Δ) numbering system. For example, *cis*-Δ^9 means that there is a *cis* double bond between carbon atoms 9 and 10; *trans*-Δ^2 means that there is a *trans* double bond between carbon atoms 2 and 3. In this nomenclature the carboxyl carbon is designated carbon 1. For example, palmitoleic acid has 16 carbons and has a double bond between carbons 9 and 10. It is designated as 16:1:Δ^9.

There is an alternative convention for naming polyunsaturated fatty acids. In this convention, number 1 is assigned to the methyl carbon. This carbon is also designated ω. The positions of the double bonds are indicated relative to the ω carbon.

Essential fatty acids

Mammals lack the enzymes to introduce double bonds at carbon atoms beyond C-9 in the fatty acid chain. Hence, mammals cannot synthesize linoleate and linolenate. *Linoleate and linolenate are the two essential fatty acids*. The term *essential* means that they must be obtained from the diet because they are required by an organism and cannot be endogenously synthesized. Fatty acids that can be endogenously synthesized are termed as *nonessential.* They are nonessential also in the sense that they do not have to be obligatorily included in the diet.

Melting point of fatty acids

The melting point of fatty acids depend on chain length and degree of unsaturation. The longer the chain length, the higher the melting point; and the greater the number of double bonds, the lower the melting point. The presence of double bonds makes unsaturated chain more rigid. As a result, unsaturated chains cannot pack themselves in crystals efficiently and densely as saturated chain, so, they have lower melting point as compared to saturated fatty acids. Similarly, the unsaturated fatty acids with *cis* configuration have lower melting points than the unsaturated fatty acids with *trans* configuration.

Problem

Why unsaturated fatty acids have low melting points?

Solution

The presence of double bonds makes unsaturated chain more rigid. As a result, unsaturated chains cannot pack themselves in crystals efficiently and densely as saturated chain, so, they have lower melting point as compared to saturated fatty acids.

1.9.2 Triacylglycerol and Wax

Triacylglycerols (also called triglycerides) are triesters of fatty acids and glycerol. They are composed of three fatty acids and a glycerol molecule. Triacylglycerols are of two types – *simple* and *mixed* type. Those containing a single kind of fatty acids are called simple triacylglycerols and with two or more different kinds of fatty acids are called mixed triacylglycerols. The general formula of triacylglycerol is given below:

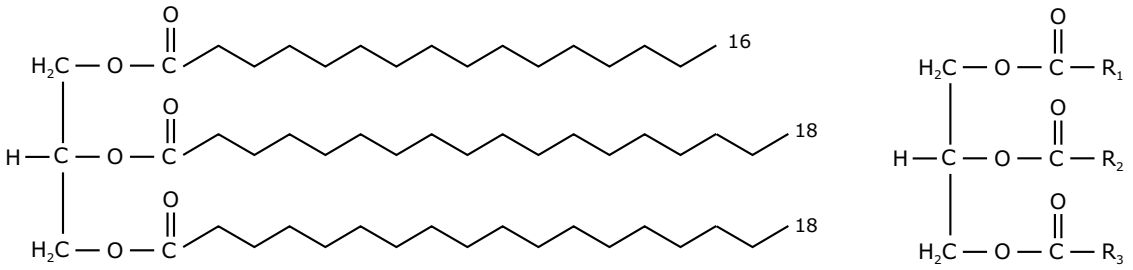


Figure 1.66 General structure of triacylglycerol.

Triacylglycerols are nonpolar, hydrophobic in nature and a major form of stored lipids. Because triacylglycerols have no charge (i.e. the carboxyl group of each fatty acid is joined to glycerol through a covalent bond), they are sometimes referred to as **neutral lipid**. Triacylglycerol molecules contain fatty acids of varying lengths, which may be unsaturated or saturated. Triacylglycerols can be distinguished as fat and oil on the basis of physical state at room temperature. *Fats,* which are solid at room temperature, contain a large proportion of saturated fatty acids. *Oils* are liquid at room temperature because of their relatively high unsaturated fatty acid content.

Hydrolysis of triacylglycerols with alkalis such as NaOH or KOH is called **saponification**. Saponification yields salts of free fatty acids (termed *soap*) and glycerol. The number of milligrams of KOH required to saponify one-gram of fat is known as *saponification number*. The saponification number measures the average molecular weight of fats. Similarly, the number of grams of iodine that can be added to 100g sample of fat or oil is called *iodine number*, which is used to determine the degree of unsaturation (i.e. extent of unsaturation).

Waxes

Natural waxes are typically esters of fatty acids and long chain alcohols. They are formed by esterification of long chain fatty acids (saturated and unsaturated) and high molecular weight monohydroxy alcohols (C14 to C36). Waxes are biosynthesized by many plants or animals. The best known animal wax is beeswax. Triacontanoylpalmitate (an ester of palmitic acid with the alcohol triacontanol) is the major component of beeswax.

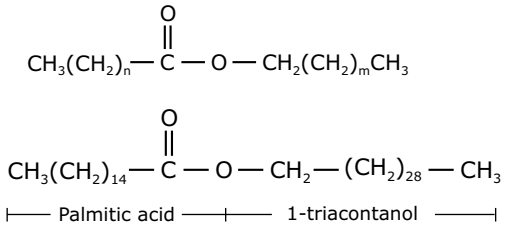


Figure 1.67 The general structure of a wax.

1.9.3 Phospholipids

A *phospholipid* is an amphipathic molecule constructed from four components: fatty acids, a platform to which the fatty acids are attached, a phosphate and an alcohol attached to the phosphate. The platform on which phospholipids are built may be *glycerol* or *sphingosine*.

Phosphoglycerides

Phospholipids derived from glycerol are called *phosphoglycerides* (or glycerophospholipids). A phosphoglyceride consists of *a* glycerol molecule, two fatty acids, a phosphate, and an alcohol (e.g. choline). Phosphoglycerides are the most numerous phospholipid molecules found in cell membranes.

In phosphoglycerides, the hydroxyl groups at C-1 and C-2 of glycerol are esterified to the carboxyl groups of the two fatty acid chains. The C-3 hydroxyl group of the glycerol backbone is esterified to phosphoric acid. When no further additions are made, the resulting compound is **phosphatidic acid**, the simplest phosphoglyceride. Phosphatidic acids are found in small amount in most natural systems. The major phosphoglycerides are derived from phosphatidic acid by the formation of an ester bond between the phosphate group and the hydroxyl group of one of several alcohols. The common alcohol moieties of phosphoglycerides are serine, ethanolamine, choline, glycerol, and the inositol. If the alcohol is choline, the phosphoglyceride molecule is called *phosphatidylcholine* (also referred to as *lecithin*) and if serine then it is called *phosphotidylserine*.

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1.10 Vitamins

Vitamins are organic compounds required by the body in trace amounts to perform specific cellular functions. They can be classified according to their solubility and their functions in metabolism. The requirement for any given vitamin depends on the organisms. Not all vitamins are required by all organisms. Vitamins are not synthesized by humans, and therefore must be supplied by the diet. Vitamins may be water soluble or fat soluble. Nine vitamins (thiamines, riboflavin, niacin, biotin, pantothenic acid, folic acid, cobalamin, pyridoxine, and ascorbic acid) are classified as water soluble, whereas four vitamins (vitamins A, D, E and K) are termed fat-soluble. Except for vitamin C, the water soluble vitamins are all precursors of coenzymes.

1.10.1 Water–soluble vitamins

Thiamine (vitamin B_1)

Thiamine pyrophosphate (TPP) is the biologically active form of the vitamin, formed by the transfer of a pyrophosphate group from ATP to thiamine. Thiamine is composed of a substituted *thiazole ring* joined to a substituted *pyrimidine* by a methylene bridge.

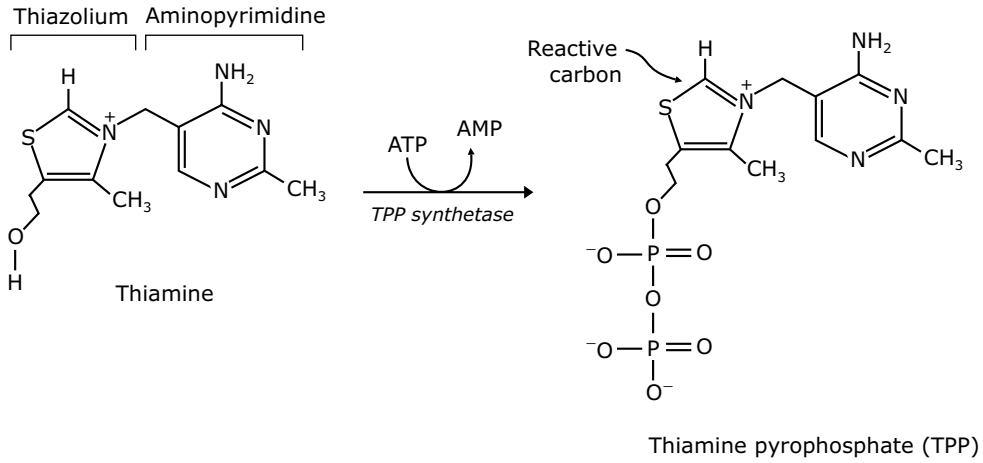


Figure 1.74 Structure of thiamine and thiamine pyrophosphate.

TPP serves as a coenzyme in the oxidative decarboxylation of α-keto acid, and in the formation or degradation of α-ketols (hydroxy ketones) by transketolase.

$$\text{Pyruvate } (\alpha\text{-keto acid}) \xrightarrow{\textit{Pyruvate decarboxylase}} \text{Acetaldehyde} + CO_2$$

$$\text{Xylulose-5-Phosphate} + \text{Ribose-5-Phosphate} \xrightarrow{\textit{Transketolase}} \text{Glyceraldehyde–3–Phosphate} + \text{Sedoheptulose–7-Phosphate}$$

Beri-Beri is a severe thiamine-deficiency syndrome found in areas where polished rice is the major component of the diet.

Riboflavin (vitamin B_2)

Riboflavin is a constituent of flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD). FMN is synthesized after the addition of phosphate in riboflavin and FAD formed by the transfer of an AMP moiety from ATP to FMN. FMN and FAD are each capable of reversibly accepting two hydrogen atoms, forming $FMNH_2$ or $FADH_2$. The oxidized form of the isoalloxazine structure absorbs light around 450 nm. The colour is lost, when the ring is reduced.

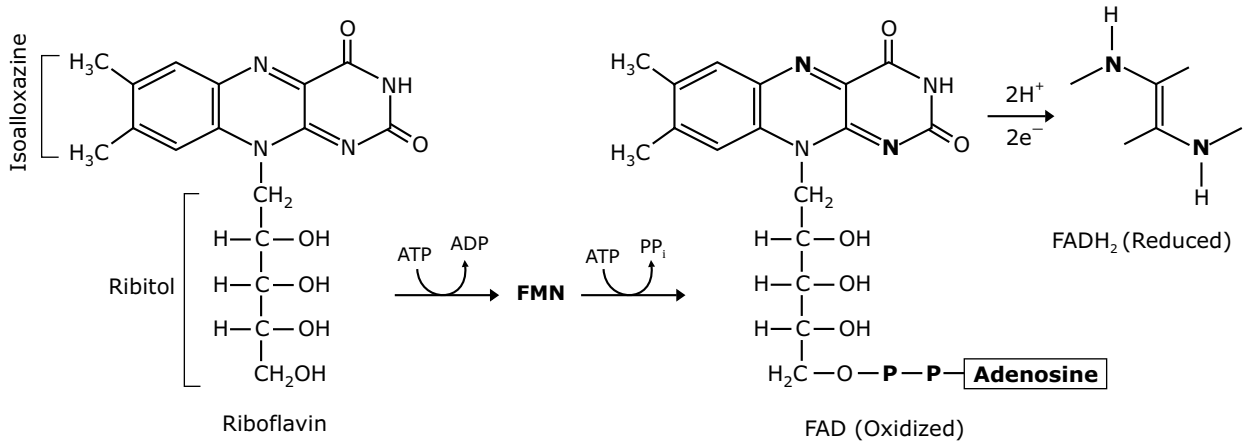


Figure 1.75 Structure and biosynthesis of flavin mononucleotide (FMN) and flavin adenine dinucleotide (FAD).

Niacin

Niacin, or nicotinic acid, is a substituted pyridine derivative. The biologically active coenzyme forms are nicotinamide adenine dinucleotide (NAD^+) and its phosphorylated derivative, nicotinamide adenine dinucleotide phosphate ($NADP^+$). Nicotinamide, is a derivative of nicotinic acid that contains an amide instead of a carboxyl group. NAD^+ and $NADP^+$ serve as coenzymes in oxidation-reduction reactions in which the coenzyme undergoes reduction of the pyridine ring by accepting a hydride ion (H^-). The reduced forms of NAD^+ and $NADP^+$ are NADH and NADPH, respectively.

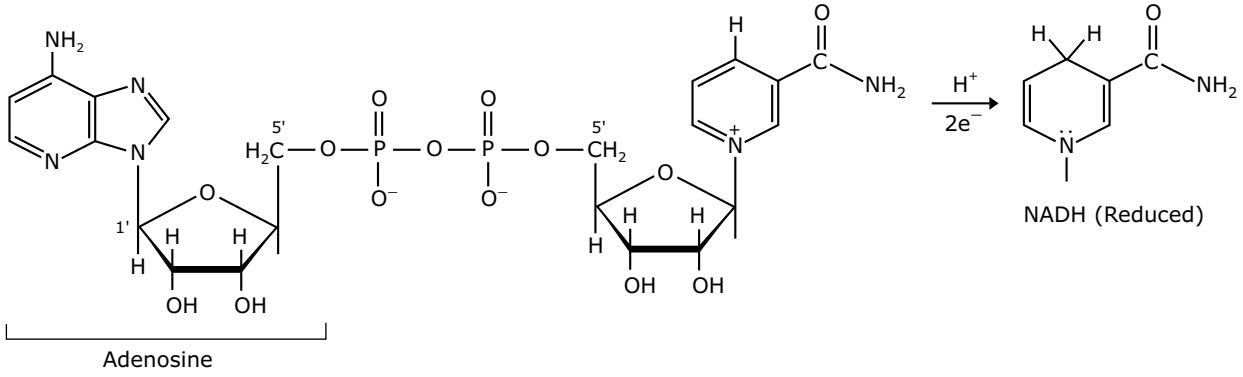


Figure 1.76 Structure of NAD^+ (Oxidized).

Deficiency of niacin causes *pellagra*, a disease involving the skin and central nervous system. The symptoms of pellagra progress through the three Ds: Dermatitis, Diarrhoea, Dementia, and, if untreated, death.

Biotin

Biotin is a coenzyme in carboxylation reactions, in which it serves as a mobile carboxyl group carrier. Biotin is covalently bound to the enzyme by an amide linkage between the carboxyl group of its valerate side chain and the ε-amino group of an enzyme *Lys* residue to form a **biocytin** (alternatively, *biotinyllysine*) residues.

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1.11 Enzymes

General features

An enzyme is a biocatalyst that increases the rate of chemical reaction without itself being changed in the overall process. It is a remarkable molecular device that determines the pattern of chemical transformations. Virtually all cellular reactions or processes are mediated by enzymes. Enzymes have several properties that make them unique.

- Most enzymes are proteins. With the exception of a small group of catalytic RNA molecules, all enzymes are proteins. Their catalytic activity depends on the integrity of their native protein conformation. If an enzyme is denatured or dissociated into its subunits, catalytic activity is usually lost.
- Enzymes are highly specific. They are highly specialized proteins and have a high degree of specificity for their substrates.
- Enzymes exhibit enormous catalytic power. It increases the rate of a reaction by lowering the activation energy. It changes only the rate at which equilibrium is achieved; it has no effect on the position of the equilibrium.

Enzymes can be divided into two general classes: **simple enzymes**, which consist entirely of amino acids and **conjugated enzymes**, contains a non-protein group called a **cofactor**, which is required for biological activity. Removal of cofactor from a conjugated enzyme leaves only protein component, called an *apoenzyme*, which generally is biologically inactive. The complete, biologically active conjugated enzyme (simple enzyme plus cofactor) is called a *holoenzyme*. A cofactor can be linked to the protein portion of the enzyme either covalently or non-covalently. Some cofactors are simple metal ions and other cofactors are complex organic groups, which are also called **coenzymes**. Cofactors which are tightly associated with the protein covalently or non-covalently are called *prosthetic group*.

Table 1.20 Vitamin B-complex and their coenzyme forms

Vitamin	*Coenzyme form*	*Reaction or process promoted*
Thiamine (B_1)	Thiamine pyrophosphate	Decarboxylation, aldehyde group transfer
Riboflavin (B_2)	FAD and FMN	Redox reaction
Pyridoxine (B_6)	Pyridoxal phosphate	Amino group transfer
Nicotinic acid (niacin)	NAD^+ and $NADP^+$	Redox reaction
Pantothenic acid (B_5)	Coenzyme A	Acyl group transfer
Biotin	Biocytin	Carboxylation
Folic acid	Tetrahydrofolic acid	One-carbon group transfer
Vitamin B_{12}	Deoxyadenosylcobalamin	Intramolecular rearrangements

The vitamins in the human diet that are coenzyme precursors are all water soluble vitamins.

Table 1.21 Example of some enzymes and their cofactors

Fe^{2+} or Fe^{3+}	Cytochrome oxidase, Catalase, Peroxidase, Xanthine oxidase
Cu^{2+}	Cytochrome oxidase, Lysyl oxidase, Superoxide dismutase
Zn^{2+}	Carbonic anhydrase, Alcohol dehydrogenase, Carboxypeptidase
Mg^{2+}	Hexokinase, Enolase, Glucose-6-phosphatase
Mn^{2+}	Arginase, Enolase, Pyruvate carboxylase
K^+	Pyruvate kinase
Ni^{2+}	Urease
Mo	Dinitrogenase, Xanthine oxidase
Se	Glutathione peroxidase

1.11.1 Naming and classification of enzyme

Many enzymes have common names. For example, trypsin, a proteolytic enzyme, is secreted by the pancreas. Common names provide little information about the reactions that enzymes catalyze. Many enzymes are named for their substrates and for the reactions that they catalyze, with the suffix-*ase* added. As for example, ATPase is an enzyme that helps in breaking down ATP, whereas ATP synthase is an enzyme that helps in synthesis of ATP. Because of the confusion that arose from these common names, an International Commission on enzymes was established to create a systematic basis for enzyme nomenclature.

The enzyme commission has developed a rule for naming enzymes. According to this rule, each enzyme is classified and named according to the type of chemical reaction it catalyzes. The *Enzyme Commission* (EC) has given each enzyme a number with four parts, like EC 2.7.1.2 (Hexokinase). The first three numbers define major *class*, *subclass*, and *sub-subclass*, respectively. The last number is a serial number in the sub-subclass, indicating the order in which each enzyme is added to the list.

Common name and EC numbers of some enzyme

Alcohol dehydrogenase	EC 1.1.1.1
Phosphofructokinase	EC 2.7.1.11
Glutamine synthetase	EC 6.3.1.2
Acetylcholinesterase	EC 3.1.1.7

Systematic classification

The first integer in the EC number designates the class of enzymes. There are six classes to which different enzymes belong. These classes are:

EC 1 Oxidoreductase

Oxidoreductase catalyzes oxidation-reduction reactions.

$A_{red} + B_{ox} \longrightarrow A_{ox} + B_{red}$

Example

Oxidases	Use oxygen as an electron acceptor but do not incorporate it into the substrate.
Dehydrogenases	Use molecules other than oxygen (e.g. NAD^+) as an electron acceptor.
Oxygenases	Directly incorporate oxygen into the substrate.
Peroxidases	Use H_2O_2 as an electron acceptor.

EC 2 Transferases

Transferases catalyze reactions that involve the transfer of groups from one molecule to another. Examples of such groups include amino, carboxyl, carbonyl, methyl, phosphoryl and acyl (RC=O). Common trivial names for the transferases often include the prefix *trans*.

$A–B + C \longrightarrow A + B–C$

Examples

Transcarboxylases	Transfers a carboxylate group to a substrate.
Transaminases	Transfer amino group from amino acids to keto acids.
Kinases	Transfer phosphate from ATP to a substrate.
Phosphorylases	Transfer inorganic phosphate to a substrate.

EC 3 Hydrolases

Hydrolases catalyze reactions in which the cleavage of bonds is accomplished by adding water.

$A–B + H_2O \longrightarrow A–H + B–OH$

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Chapter 02

Bioenergetics and Metabolism

2.1 Bioenergetics

Bioenergetics is the quantitative study of the energy transductions that occur in living cells and of the nature and functions of the chemical processes underlying these transductions.

Thermodynamic principles

The *First law of thermodynamics* states that the energy is neither created nor destroyed, although it can be transformed from one form to another i.e. the total energy of a system, including surroundings, remains constant.

Mathematically, it can be expressed as:

$$\Delta U = \Delta q - \Delta w$$

ΔU is the change in internal energy,

Δq is the heat exchanged from the surroundings,

Δw is the work done by the system.

If Δq is positive, heat has been transferred to the system, giving an increase in internal energy. When Δq is negative, heat has been transferred to the surroundings, giving a decrease in internal energy. When Δw is positive, work has been done by the system, giving a decrease in internal energy. When Δw is negative, work has been done by the surroundings, giving an increase in internal energy.

The *Second law of thermodynamics* states that the total entropy of a system must increase if a process is to occur spontaneously. Mathematically, it can be expressed as:

$$\Delta S \geq \frac{\Delta q}{T}$$ where, ΔS is the change in entropy of the system

Entropy is unavailable form of energy and it is very difficult to determine it, so a new thermodynamic term called *free energy* is defined.

Free energy

Free energy or Gibb's free energy indicates the portion of the total energy of a system that is available for useful work (also known as chemical potential). The change in free energy is denoted as ΔG.

Under constant temperature and pressure, the relationship between free energy change (ΔG) of a reacting system and the change in entropy (ΔS) is expressed by following equation:

$$\Delta G = \Delta H - T\Delta S$$

Where, ΔH is the change in enthalpy and T is absolute temperature. ΔH is the measure of change in heat content of reactants and products. The change in the free energy, ΔG, can be used to predict the direction of a reaction at constant temperature and pressure.

If ΔG is negative, the reaction proceeds spontaneously with the loss of free energy (**exergonic**),

ΔG is positive, the reaction proceeds only when free energy can be gained (**endergonic**),

ΔG is 0, the system is at **equilibrium**; both forward and reverse reactions occur at equal rates,

ΔG of the reaction A $\rightarrow$ B depends on the concentration of reactant and product. At constant temperature and pressure, the following relation can be derived:

$$\Delta G = \Delta G^0 + RT \ln \frac{[B]}{[A]}$$

Where, ΔG^0 is the standard free energy change;

R is the gas constant;

T is the absolute temperature;

[A] and [B] are the actual concentrations of reactant and product.

Standard free energy change

The actual change in free energy (ΔG) during a reaction is influenced by temperature, pressure and the initial concentrations of reactants and products, and usually differs from *standard free energy change*, ΔG^0.

The chemical reaction has a characteristic standard free energy change and it is constant for a given reaction. It can be calculated from the equilibrium constant of the reaction under standard conditions i.e., at a solute concentration of 1.0M, at temperature of 25°C and at 1.0 atm pressure. The free energy change which corresponds to this standard state is known as standard free energy change, ΔG^0.

Relationship between ΔG^0 and K_{eq}

In a reaction A $\rightarrow$ B, a point of equilibrium is reached at which no further *net* chemical change takes place–that is, when A is being converted to B, B is also being converted to A, as fast as A into B. In this state, the ratio of [B] to [A] is constant, regardless of the actual concentrations of the two compounds:

$$K_{eq} = \frac{[B]_{eq}}{[A]_{eq}}$$

where K_{eq} is the equilibrium constant, and $[A]_{eq}$ and $[B]_{eq}$ are the concentrations of A and B at equilibrium. The concentration of reactants and products at equilibrium define the *equilibrium constant*, K_{eq}. The equilibrium constant K_{eq} depends on the nature of reactants and products, the temperature and the pressure. Under standard physical conditions (25°C and 1 atm pressure, for biological systems), the K_{eq} is always the same for a given reaction, whether or not a catalyst is present.

If the reaction A $\rightleftharpoons$ B is allowed to go to equilibrium at constant temperature and pressure, then at equilibrium the overall free energy change (ΔG) is zero. Therefore,

$$\Delta G^0 = -RT \ln \frac{[B]_{eq}}{[A]_{eq}}$$

So, $\Delta G^0 = -RT \ln K_{eq}$

This equation allows some simple predictions:

K_{eq}	ΔG^0	Reaction
> 1.0	Negative	proceeds forward
1.0	Zero	is at equilibrium
< 1.0	Positive	proceeds in reverse

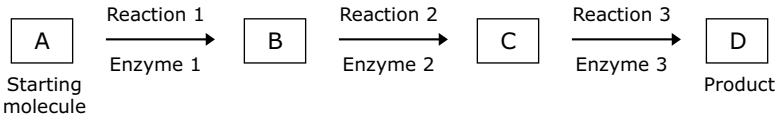
As we know, the ionic composition of an acid or base varies with pH. So, the standard free energy calculated according to the biochemistry convention is valid only at pH=7. Hence, under biochemistry convention, ΔG^0 is symbolized by $\Delta G^{0'}$ and likewise, the biochemical equilibrium constant is represented by K'_{eq}.

So, $\Delta G^{0'} = -RT \ln K'_{eq}$

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2.2 Metabolism

Metabolism (derives from the Greek word for *change*) is a series of interconnected chemical reactions occurring within a cell; the chemical compounds involved in this process are known as *metabolites*. It consists of hundreds of enzymatic reactions organized into discrete pathways. These pathways proceed in a stepwise manner, transforming substrates into end products through many specific chemical *intermediates*. Each step of metabolic pathways is catalyzed by a specific enzyme.



Metabolic pathways can be linear (such as glycolysis), cyclic (such as the citric acid cycle) or spiral (such as the biosynthesis of fatty acids). Metabolism serves two fundamentally different purposes: generation of energy to drive vital functions and the synthesis of biological molecules. To achieve these, metabolic pathways fall into two categories: anabolic and catabolic pathways. *Anabolic pathways* are involved in the synthesis of compounds and endergonic in nature. *Catabolic pathways* are involved in the oxidative breakdown of larger complex molecules and usually exergonic in nature. The basic strategy of catabolic metabolism is to form ATP and reducing power for biosyntheses. Some pathways can be either anabolic or catabolic, depending on the energy conditions in the cell. They are referred to as *amphibolic pathways*. Amphibolic pathways occur at the 'crossroads' of metabolism, acting as links between the anabolic and catabolic pathways, e.g. the citric acid cycle.

Characteristics of metabolic pathways are:

1. They are irreversible.
2. Each one has a first committed step.
3. Those in eukaryotic cells occur in specific cellular locations.
4. They are regulated. Regulation occurs in following different ways:
   I. Availability of substrate; the rate of reaction depends on substrate concentration.
   II. Allosteric regulation of enzymes by a metabolic intermediate or coenzyme.
   III. By extracellular signal such as growth factors and hormones that act from outside the cell in multicellular organisms; changes the cellular concentration of an enzyme by altering the rate of its synthesis or degradation.

A number of central metabolic pathways are common to most cells and organisms. These pathways, which serve for synthesis, degradation, interconversion of important metabolites, and energy conservation, are referred to as the *intermediary metabolism*.

Metabolic pathways involve several enzyme-catalyzed reactions. Most of the reactions in living cells fall into one of five general categories: oxidation-reductions; reactions that make or break carbon–carbon bonds; group transfers; internal rearrangements, isomerizations and eliminations; and free radical reactions.

Feedback inhibition and feedback repression

In *feedback inhibition* (or end product inhibition), the end product of a biosynthetic pathway inhibits the *activity* of the first enzyme that is unique to the pathway, thus controlling production of the end product. The first enzyme in the pathway is an allosteric enzyme. Its allosteric site will bind to the end product of the pathway which alters its active site so that it cannot mediate the enzymatic reaction.

The feedback inhibition is different from feedback repression. An inhibitory feedback system in which the end product produced in a metabolic pathway acts as a co-repressor and represses the *synthesis* of an enzyme that is required at an earlier stage of the pathway is called *feedback repression*.

2.3 Respiration

Living cells require an input of free energy. Energy is required for the maintenance of highly organized structures, synthesis of cellular components, movement, generation of electrical currents and for many other processes. Cells acquire free energy from the oxidation of organic compounds that are rich in potential energy.

Respiration is an oxidative process, in which free energy released from organic compounds is used in the formation of ATP. The compounds that are oxidized during the process of respiration are known as *respiratory substrates,* which may be carbohydrates, fats, proteins or organic acids. Carbohydrates are most commonly used as respiratory substrates.

During oxidation within a cell, all the energy contained in respiratory substrates is not released free in a single step. Free energy is released in multiple steps in a controlled manner and used to synthesise ATP, which is broken down whenever (and wherever) energy is needed. Hence, ATP acts as the energy currency of the cell.

During cellular respiration, respiratory substrates such as glucose may undergo *complete or incomplete* oxidation. The complete oxidation of substrates occurs in the presence of oxygen, which releases CO_2, water and a large amount of energy present in the substrate. A complete oxidation of respiratory substrates in the presence of oxygen is termed as *aerobic respiration*.

Although carbohydrates, fats and proteins can all be oxidized as fuel, but here processes have been described by taking glucose as a *respiratory substrate.* Oxidation of glucose is an exergonic process. An exergonic reaction proceeds with a net release of free energy. When one mole of glucose (180 g) is completely oxidized into CO_2 and water, approximately 2870 kJ or 686 kcal energy is liberated. Part of this energy is used for synthesis of ATP. For each molecule of glucose degraded to carbon dioxide and water by respiration, the cell makes up to about 30 or 32 ATP molecules, each with 7.3 kcal/mol of free energy.

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + \text{Energy (ATP + Heat)}$$

The incomplete oxidation of respiratory substrates occurs under anaerobic conditions i.e. in the absence of oxygen. As the substrate is never totally oxidized, the energy generated through this type of respiration is lesser than that during aerobic respiration.

2.3.1 Aerobic respiration

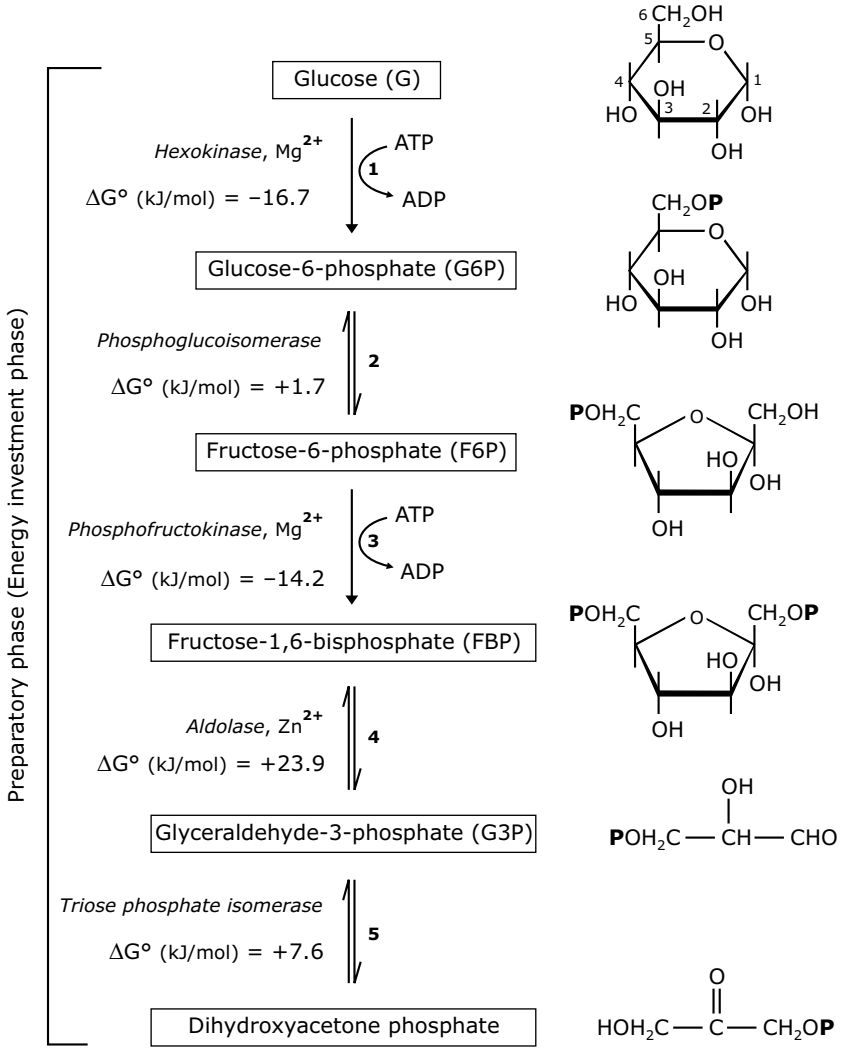
Enzyme catalyzed reactions during aerobic respiration can be grouped into three major processes: glycolysis, citric acid cycle and oxidative phosphorylation. Glycolysis takes place in the cytosol of cells in all living organisms. The citric acid cycle takes place within the mitochondrial matrix of eukaryotic cells and in the cytosol of prokaryotic cells. The oxidative phosphorylation takes place in the inner mitochondrial membrane. However, in prokaryotes, oxidative phosphorylation takes place in the plasma membrane.

Table 2.3 Intracellular location of major processes of aerobic respiration

In eukaryotes,		
Glycolysis	–	Cytosol
Citric acid cycle	–	Mitochondrial matrix
Oxidative phosphorylation	–	Inner mitochondrial membrane
In prokaryotes,		
Glycolysis	–	Cytosol
Citric acid cycle	–	Cytosol
Oxidative phosphorylation	–	Plasma membrane

2.3.2 Glycolysis

Glycolysis (from the Greek *glykys*, meaning *sweet*, and *lysis*, meaning *splitting*) also known as *Embden-Meyerhof pathway*, is an oxidative process in which one mole of glucose is partially oxidized into the two moles of pyruvate in a series of enzyme-catalyzed reactions. Glycolysis occurs in the cytosol of all cells. It is a unique pathway that occurs in both aerobic as well as anaerobic conditions and does not involve molecular oxygen.



Step 1 : (**Phosphorylation**) Glucose is phosphorylated by ATP to form a glucose 6-phosphate. The negative charge of the phosphate prevents the passage of the glucose 6-phosphate through the plasma membrane, trapping glucose inside the cell. This *irreversible* reaction is catalyzed by *hexokinase*. Hexokinase is present in all cells of all organisms. Hexokinase requires divalent metal ions such as Mg^{2+} or Mn^{2+} for activity. Hepatocytes and β-cells of the pancreas also contain a form of hexokinase called **glucokinase** (hexokinase D). Hexokinase and glucokinase are isozymes. Glucokinase is present in liver and beta-cells of the pancreas and has a high K_m and V_{max} as compared to hexokinase.

Step 2 : (**Isomerization**) A readily reversible rearrangement of the chemical structure (isomerization) moves the carbonyl oxygen from carbon 1 to carbon 2, forming a ketose from an aldose sugar. Thus, the isomerization of glucose 6-phosphate to fructose 6-phosphate is a conversion of an aldose into a ketose.

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Solution

a. Inhibition of NADH dehydrogenase by rotenone decreases the rate of electron flow through the respiratory chain, which in turn decreases the rate of ATP production. If this reduced rate is unable to meet the organism's ATP requirements, the organism dies.

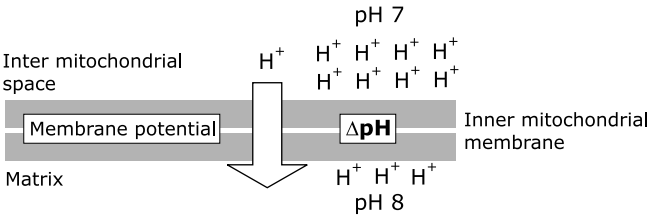
b. Antimycin A strongly inhibits the oxidation of Q in the respiratory chain, reducing the rate of electron transfer and leading to the consequences described in (a).

c. Because antimycin A blocks all electron flow to oxygen, it is a more potent poison than rotenone, which blocks electron flow from NADH, but not from $FADH_2$.

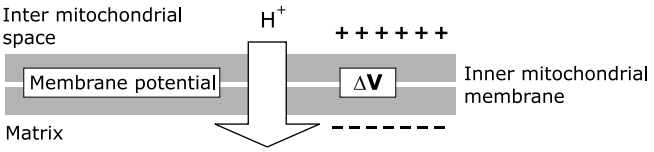
2.3.8 Electrochemical proton gradient

Transfer of electrons through the electron transport chain is accompanied by pumping of protons across inner mitochondrial membrane, from the mitochondrial matrix to inter mitochondrial space. A total of $10H^+$ ions are translocated from the matrix across the inner mitochondrial membrane per electron pair flowing from NADH to O_2. This movement of H^+ generates:

a. *pH gradient* across the inner mitochondrial membrane (with the pH higher in the matrix than in the inter mitochondrial space).



b. *Voltage gradient* (membrane potential) across the inner mitochondrial membrane (with the inside negative and outside positive).



The pH gradient (ΔpH) and voltage gradient together constitute *electrochemical proton gradient.* The electrochemical proton gradient exerts a **proton motive force** (pmf). A mitochondrion actively involved in aerobic respiration typically has a membrane potential of about 160 mV (negative inside matrix) and a pH gradient of about 1.0 pH unit (higher on the matrix side). A difference of one pH unit represents a tenfold difference in H^+ concentration and a pH gradient of one unit across a membrane is equivalent to an electric potential of 59 mV (at 20°C). The total proton-motive force across the inner mitochondrial membrane consists of a large force due to the membrane potential and a smaller force due to the H^+ concentration gradient (pH gradient). In a typical cell, the proton motive force across the inner mitochondrial membrane of a respiring mitochondrion is about 220 mV.

Determination of electric potential and pH gradient

Because mitochondria are very small, the electric potential and pH gradient across the inner mitochondrial membrane cannot be determined by direct measurement. However, the inside pH can be measured by trapping fluorescent pH-sensitive dyes inside vesicles formed from the inner mitochondrial membrane.

Similarly, the electric potential can be determined by adding radioactive K^+ ions and a trace amount of valinomycin to a suspension of respiring mitochondria. Valinomycin is an *ionophore.* Although the inner membrane is normally impermeable to K^+ ions, but valinomycin selectively binds K^+ ions in its hydrophilic interior and carries it across through the impermeable membranes. In the presence of valinomycin, K^+ ions equilibrates across the inner membrane of isolated mitochondria in accordance with the electric potential; the more negative the matrix side of the membrane,

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Experimental proof of chemiosmotic hypothesis

Experimental proof of chemiosmotic hypothesis was provided by Andre Jagendorf and Ernest Uribe in 1966. In an elegant experiment, isolated chloroplast thylakoid vesicles containing F_0F_1 particles were equilibrated in the dark with a buffered solution at pH 4.0. When the pH in the thylakoid lumen became 4.0, the vesicles were rapidly mixed with a solution at pH 8.0 containing ADP and P_i. A burst of ATP synthesis accompanied the transmembrane movement of protons driven by the electrochemical proton gradient. In similar experiments using *inside-out* preparations of submitochondrial vesicles, an artificially generated membrane electric potential also resulted in ATP synthesis.

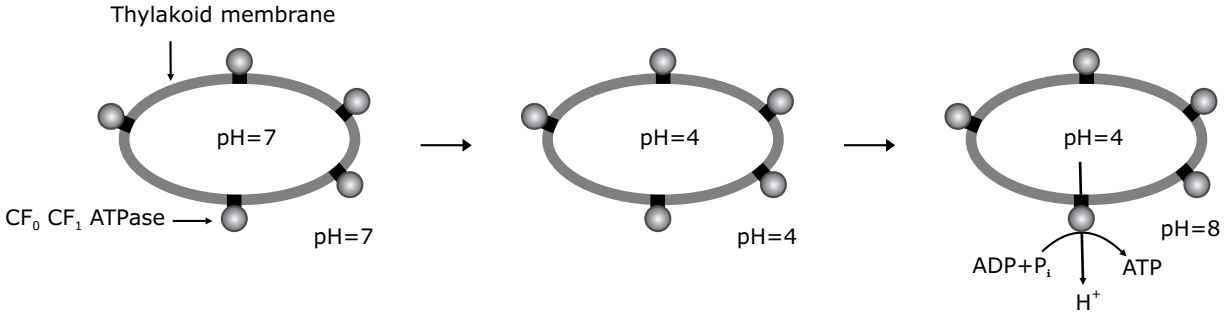


Figure 2.15 Synthesis of ATP by F_0F_1 depends on a pH gradient across the membrane.

2.3.10 ATP synthase

The use of proton motive force for ATP synthesis is catalyzed by ATP synthase. The multiprotein ATP synthase or F_0F_1 complex or complex V catalyzes ATP synthesis as protons flow back through the inner membrane down the electrochemical proton gradient. It consists of two components – F_0 component and F_1 ATPase. The F_0 component is embedded in the inner mitochondrial membrane. F_0 contains one 'a' subunit, two 'b' subunits and 9–12 'c' subunits. The c subunit consists of two α helices that span the membrane. An aspartic acid residue in the second helix lies on the center of the membrane. F_0 is a transmembrane complex that forms a regulated H^+ channel. An antibiotic **oligomycin** completely blocks ATP synthesis by blocking the flow of protons through F_0 of ATP synthase (subscript 'O' denotes its inhibition by antibiotic oligomycin). F_1 ATPase (made up of 3α, 3β, γ, δ and ε) is tightly bound to F_0 and protrudes into the matrix; it contains three β-subunits that are the sites of ATP synthesis. At the center of F_1 ATPase is the γ-subunit. The γ-subunit extends through F_1 and interacts with F_0. The $\gamma\varepsilon$ and C_{9-12} ring complex is the *rotor* (moving unit) and the a, b_2 and $\alpha_3\beta_3\delta$ complex is the *stator* (stationary unit). Rotational motion is imparted to the rotor by the passage of protons.

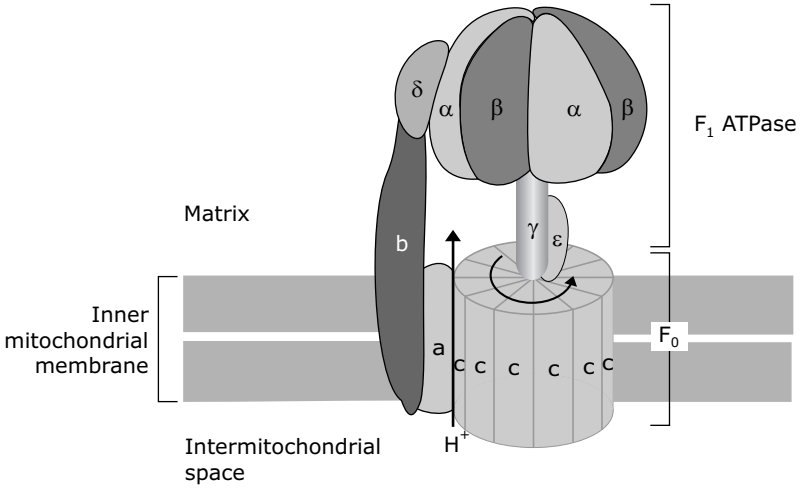


Figure 2.16 The enzyme complex consists of an F_0 component and F_1 ATPase. Proton passing through the disc of 'C' units cause it and the attached γ-subunit to rotate. The γ-subunit fits inside the F_1 ATPase of a three α and three β-subunits, which are fixed to the membrane and do not rotate.

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The three $F_1\beta$ subunits alternate between three conformational states that differ in their binding affinities for ATP, ADP and P_i :

An O state (open state) that binds ATP, ADP and P_i very weakly.

An L state (loose state) that binds ADP and P_i loosely.

A T state (tight state) that binds ADP and P_i very tightly and gives ATP.

The phosphoanhydride bond of ATP is synthesized only in the T state and ATP is released only in the O state. The free energy released on proton translocation is harnessed to interconvert three states. Change in conformations occur due to rotation of γ-subunit. A 120° rotation of γ-subunit in counterclockwise direction changes one conformation state to another. Rotation of γ-subunit relative to fixed $\alpha_3\beta_3$ occurs in discrete 120° steps and conformation of each β-subunit changes in the order: O $\rightarrow$ L $\rightarrow$ T $\rightarrow$ O.

Calculation of free energy change

The standard free energy change for the movement of protons across the membrane along the electrochemical proton gradient can be calculated from the Nernst equation:

$$\Delta G \text{ (cal/mol)} = -nF\Delta E$$

We can calculate the amount of free energy released by the passage of one mole of protons down an electrochemical gradient of 220 mV from the above equation.

$$\Delta G = -nF\Delta E; \quad \text{where, } n = 1.$$
$$= -23{,}062 \times 0.22 \text{ V} = -5074 \text{ cal/mol or } -5.1 \text{ kcal/mol}$$

The most widely accepted experimental value for the number of protons required to drive the synthesis of one ATP molecule is four. Hence, if 10 protons are pumped out per NADH, four must flow in to produce one ATP, the proton-based P/O ratio (the ratio of ATP synthesized to oxygen reduces to water molecules) is 2.5 for NADH (1.5 for $FADH_2$).

2.3.11 Uncoupling agents and ionophores

Uncoupling agents uncouple oxidation from phosphorylation. They allow the oxidation of NADH and $FADH_2$ and reduction of O_2 to continue at high levels but do not permit ATP synthesis. Thus, electron transport continues unabated, but ATP synthesis stops. Most common uncoupling agents are *2,4-dinitrophenol* (DNP), *dicoumarol* and carbonyl cyanide-p-(trifluoromethoxy) phenylhydrazone (FCCP). DNP is a weak acid that is soluble in lipid bilayer both in their protonated neutral forms and in their anionic states. DNP in an anionic state picks up protons in the inter-mitochondrial space and diffuse readily across mitochondrial membranes. After entering the matrix in the protonated form, they can release a proton, thus dissipating the proton gradient and inhibiting ATP synthesis. The energy relased by the oxidation of NADH in the presence of DNP is converted to heat. Dicoumarol and FCCP act in the same way. Similarly, **thermogenin** is a physiological uncoupler found in brown adipose tissue that functions to generate body heat, particularly for the new born and during hibernation in animals.

Ionophores are lipophilic molecules that bind specific cations and facilitate their transport through the membrane. Ionophore uncouple electron transfer from oxidative phosphorylation by dissipating the electrochemical gradient across the mitochondrial membrane. **Valinomycin**, an antibiotic, is an example of ionophore. Its addition makes inner mitochondrial membrane permeable for K^+. It causes the movement of K^+ along the concentration gradient from cytosol into the matrix. It decreases the memberane potential component of pmf (without a direct effect on the pH gradient) and thus ATP synthesis.

2.3.12 ATP–ADP exchange across the inner mitochondrial membrane

The major function of oxidative phosphorylation is to generate ATP from ADP and P_i. Most of the ATP generated by oxidative phosphorylation in mitochondria is exported to the cytoplasm. Thus mitochondria must contain mechanisms to export ATP and import ADP and P_i. However, ATP and ADP do not diffuse freely across the inner mitochondrial membrane. A specific transport protein, *ATP-ADP translocase* (also called adenine nucleotide translocase), which acts as an antiporter, mediates the transport of ATP and ADP. ADP enters the mitochondrial matrix only if ATP exits,

and vice versa. Because *ATP-ADP translocase* moves four negative charges out of every three moved in, its activity is favoured by the transmembrane electrochemical proton gradient, which gives the matrix a net negative charge. ATP-ADP translocase function is inhibited by a toxic glycoside *atractyloside*. ATP-ADP exchange is energetically expensive; proton-motive force across the inner mitochondrial membrane powers the exchange. A second membrane transport system is the *phosphate translocase*, which promotes symport of one $H_2PO_4^{2-}$ and one H^+ into the matrix. This transport process is also powered by the transmembrane proton gradient.

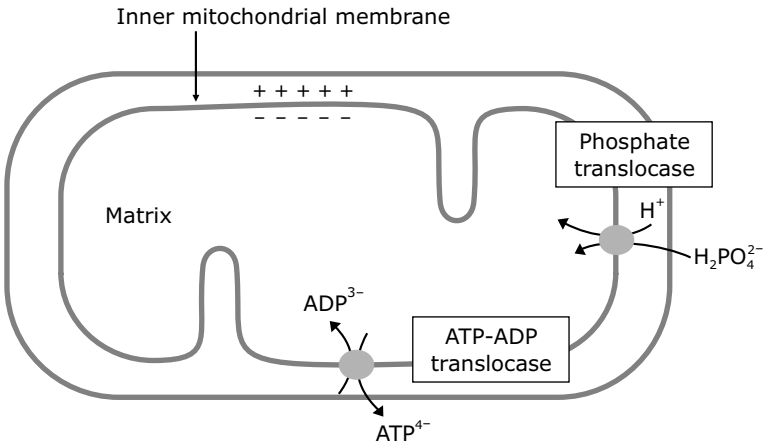


Figure 2.19 The phosphate and ATP/ADP translocase system in the inner mitochondrial membrane.

2.3.13 Shuttle systems

The glycolytic pathway is a primary source of NADH formation. NADH synthesized during the glycolytic process finally transfers the electrons to electron transport chain. But, NADH cannot cross the inner mitochondrial membrane. So, two different shuttle systems help in the transfer of electrons from NADH to the electron transport chain.

The **malate-aspartate shuttle** is the principal mechanism for the movement of NADH from the cytoplasm into the mitochondrial matrix. The electrons are *carried* into the mitochondrial matrix in the form of malate. Cytoplasmic *malate dehydrogenase* reduces oxaloacetate to malate while oxidizing NADH to NAD^+. Malate then enters the mitochondrial matrix, where the reverse reaction is carried out by mitochondria malate dehydrogenase and the regeneration of NADH occurs.

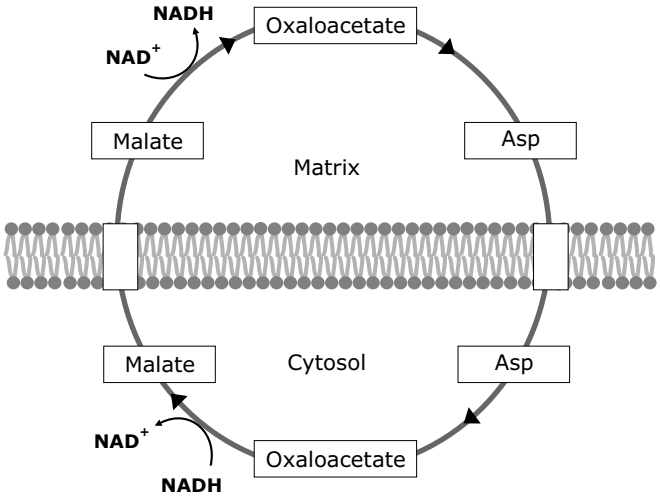


Figure 2.20 The malate-aspartate shuttle for transporting NADH from cytosol into the mitochondrial matrix. NADH in the cytosol transfers electrons to oxaloacetate, producing malate. Malate is transported across the inner membrane by the help of transporter. In the matrix, malate passes electrons to NAD^+; the resulting matrix NADH is finally oxidized by the mitochondrial respiratory chain.

Pages 146 to 153 are not shown in this preview.

integrity requires a plentiful supply of reduced glutathione (GSH), a *Cys*-containing tripeptide (γ-glutamyl-cysteinylglycine). A major function of GSH in the erythrocyte is to eliminate H_2O_2 and organic hydroperoxides. H_2O_2, a toxic product of various oxidative processes, reacts with double bonds in the fatty acid residues of the erythrocyte cell membrane to form organic hydroperoxides. These, in turn, result in premature cell lysis. Peroxides are eliminated through the action of glutathione peroxidase, yielding glutathione disulfide (GSSG). So, G6PD deficiency results in hemolytic anemia caused by the inability to detoxify oxidizing agents.

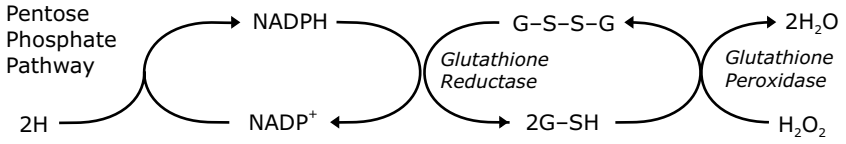


Figure 2.30 Role of the pentose phosphate pathway in the reduction of oxidized glutathione.

2.6 Entner–Doudoroff pathway

Entner-Doudoroff pathway is an alternative pathway that catabolizes glucose to pyruvate using a set of enzymes different from those used in either glycolysis or the pentose phosphate pathway. This pathway, first reported by Michael Doudoroff and Nathan Entner, occurs only in prokaryotes, mostly in gram-negative bacteria such as *Pseudomonas aeruginosa, Azotobacter, Rhizobium*.

In this pathway, glucose phosphate is oxidized to 2-keto-3-deoxy-6-phosphogluconic acid (KDPG) which is cleaved by *2-keto-3-deoxyglucose-phosphate aldolase* to pyruvate and glyceraldehyde-3-phosphate. The latter is oxidized to pyruvate by glycolytic pathway where in two ATPs are produced by substrate level phosphorylations. This process yields one ATP as well as one NADH and one NADPH for every glucose molecule.

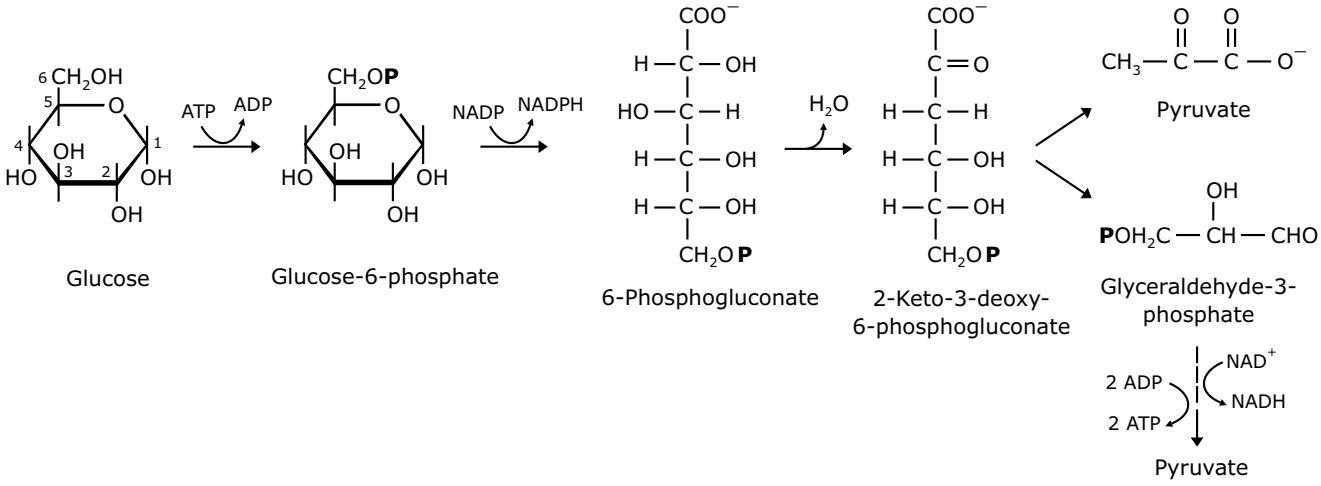


Figure 2.31 Entner-Doudoroff pathway.

2.7 Photosynthesis

Photosynthesis is a physiochemical process by which photosynthetic organisms convert light energy into chemical energy in the form of reducing power (as NADPH) and ATP, and use these chemicals to drive carbon dioxide fixation.

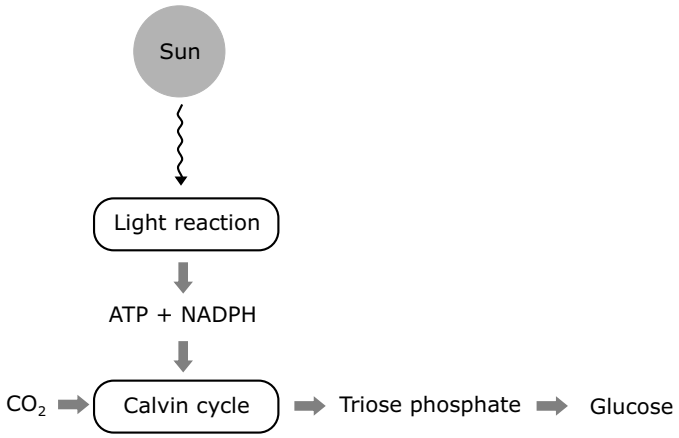


Figure 2.32 Photosynthesis is a two stage process. The first process is a light dependent one (light reactions) that requires the direct energy of light to make energy carrier molecules that are used in the second process. The calvin cycle (light independent process) occurs when the products of the light reaction are used in the formation of carbohydrate.

On the basis of generation of oxygen during photosynthesis, the photosynthetic organisms may be **oxygenic** or **anoxygenic**. Oxygenic photosynthetic organisms include both eukaryotes as well as prokaryotes whereas anoxygenic photosynthetic organisms include only prokaryotes.

Oxygenic photosynthetic organisms

Eukaryotes – Plants and Photosynthetic protists

Prokaryotes – Cyanobacteria

Anoxygenic photosynthetic organisms

Prokaryotes – Green and purple photosynthetic bacteria

In oxygenic photosynthetic organisms, photosynthetic oxygen generation occurs via the light-dependent oxidation of water to molecular oxygen. This can be written as the following simplified chemical reaction:

$$nCO_2 + nH_2O \xrightarrow[Chl]{Light} (CH_2O)_n + nH_2O + nO_2$$

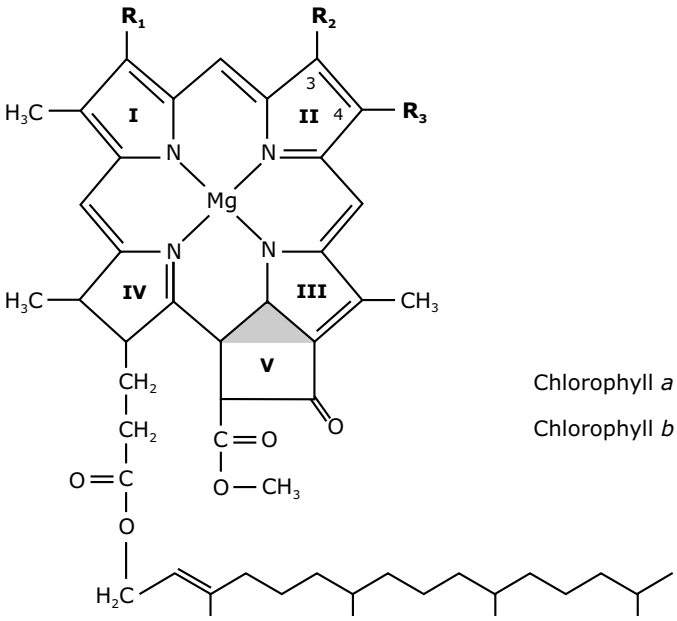
2.7.1 Photosynthetic pigment

The solar energy required for photosynthesis is captured by photosynthetic pigment molecules. Different types of pigments, described as photosynthetic pigment, participate in this process. The major photosynthetic pigment is the chlorophyll.

Chlorophylls

Chlorophyll, a light-absorbing green pigment, contains a polycyclic, planar tetrapyrrole ring structure. Chlorophyll is a *lipid soluble* pigment. It has the following important features:

1. The central metal ion in chlorophyll is Mg^{2+}.
2. Chlorophyll has a cyclopentanone ring (ring V) fused to pyrrole ring III.
3. The propionyl group on a ring IV of chlorophyll is esterified to a long-chain *tetraisoprenoid alcohol*. In chlorophyll a and b it is phytol.



	R_1	R_2	R_3
Chlorophyll *a*	— CH = CH_2	— CH_3	— CH_2CH_3
Chlorophyll *b*	— CH = CH_2	— CHO	— CH_2CH_3

Figure 2.33 Chlorophyll structures. Chlorophyll is composed of two parts; the first is a porphyrin ring with magnesium at its center, the second is a hydrophobic phytol tail. The tail is a 20 carbon chain that is highly hydrophobic.

Table 2.8 Differences between Chlorophyll *a* and Chlorophyll *b*

Chlorophyll a	*Chlorophyll b*
1. It is $C_{55}H_{72}O_5N_4Mg$.	1. It is $C_{55}H_{70}O_6N_4Mg$.
2. In the pure state, chlorophyll *a* is blue-green.	2. In the pure state, chlorophyll *b* is olive-green.
3. It is an essential photosynthetic pigment.	3. It is accessory photosynthetic pigment.
4. Pyrrole ring II contains methyl (—CH_3) group.	4. Pyrrole ring II contains aldehyde (—CHO) group.
5. It absorbs more red wavelengths than violet-blue wavelength of light.	5. It absorbs more violet-blue wavelength than red wavelength of light.

Oxygenic photosynthetic organisms contain different types of chlorophyll molecules like Chl a, Chl b, Chl c and Chl d. These chlorophyll molecules differ by having different substituent groups on the tetrapyrrole ring. Anoxygenic photosynthetic organisms contain bacteriochlorophyll molecules. They are related to chlorophyll molecules. Different groups of anoxygenic photosynthetic organisms contain different types of bacteriochlorophyll: BChl a, BChl b, BChl c, BChl d and BChl e. Bacteriochlorophyll molecules absorb light at longer wavelengths as compared to chlorophyll molecules.

Accessory pigments

Besides the major light-absorbing chlorophyll molecules, there are two groups of accessory pigments which absorb light in the wavelength region, where chlorophylls do not absorb strongly. The two types of accessory pigments are *carotenoids* and *phycobilins*.

Carotenoids are long-chain, conjugated hydrocarbons containing a string of isoprene residues and distinguished from one another by their end groups. They are generally C_{40} terpenoid compounds formed by the condensation of eight isoprene units. Carotenoids are *lipid soluble* pigments and can be subdivided into two classes, *xanthophylls* (which contain oxygen) and *carotenes* (which are purely hydrocarbons, and contain no oxygen). The characteristic

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Glycogen storage diseases

Glycogen storage diseases are caused by a genetic deficiency of one or another of the enzymes of glycogen metabolism. Many diseases have been characterized that result from an inherited deficiency of the enzyme. These defects are listed in the table.

Table 2.17 Glycogen storage diseases

Name	*Enzyme deficiency*
Von Gierke's disease	Liver glucose-6-phosphatase
Pompe's disease	Lysosomal $\alpha 1 \rightarrow 4$ and $\alpha 1 \rightarrow 6$ glucosidase (acid maltase)
Hers' disease	Liver phosphorylase
Tarui's disease	Muscle and erythrocyte phosphofructokinase 1
McArdle's disease	Muscle glycogen phosphorylase
Andersen's disease	Amylo (1,4 → 1,6) transglycosylase (Branching enzyme)

2.10 Lipid metabolism

2.10.1 Synthesis and storage of triacylglycerols

All animals and plants have the ability to synthesize triacylglycerol (TAG). In animals, many cell types and organs have the ability to synthesise triacylglycerols, but the liver and intestines are most active. Within all cell types, even those of the brain, triacylglycerols are stored as cytoplasmic *lipid droplets* (also termed *fat globules*, *oil bodies*, *lipid particles*, *adiposomes*, etc.) enclosed by a monolayer of phospholipids and hydrophobic proteins, such as the perilipins in adipose tissue or oleosins in seeds. Two main biosynthetic pathways are known, the *sn-glycerol-3-phosphate pathway*, which predominates in liver and adipose tissue, and a *monoacylglycerol pathway* in the intestines. The most important route to triacylglycerol biosynthesis is the *sn*-glycerol-3-phosphate or Kennedy pathway.

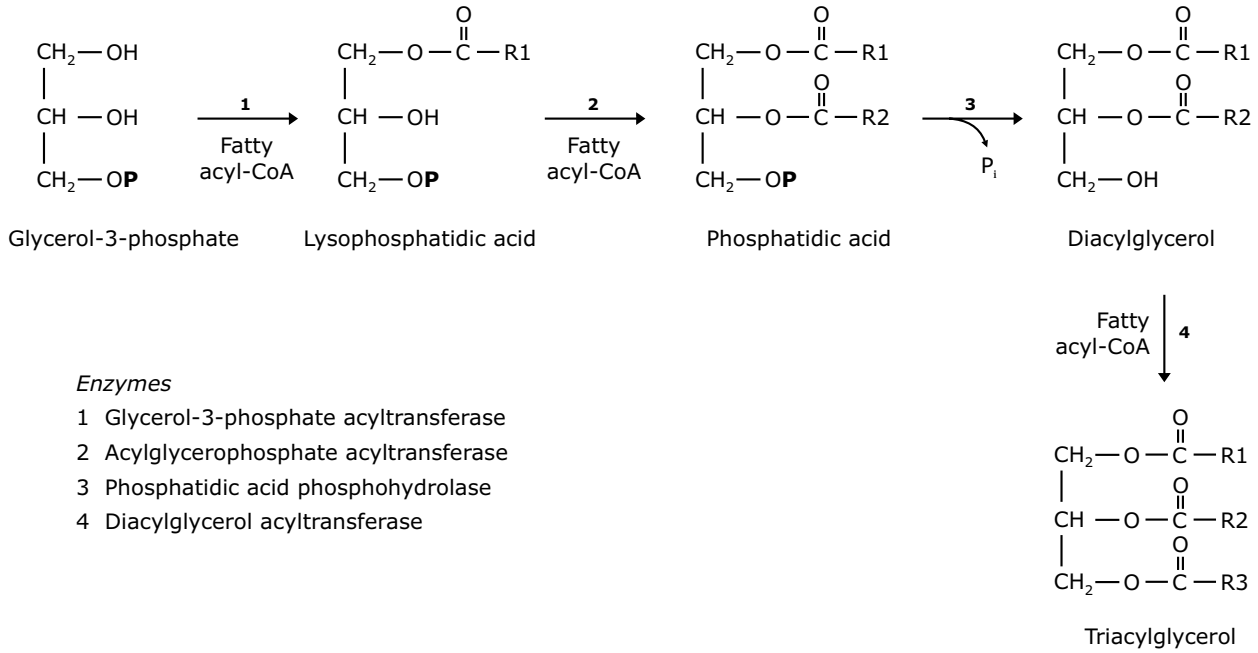
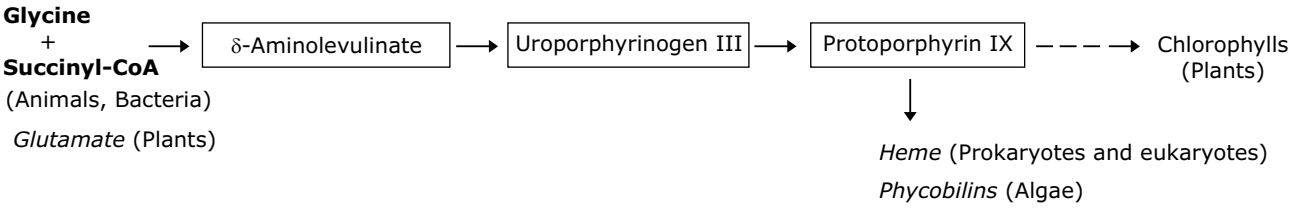


Figure 2.68 Triacylglycerol biosynthetic pathway.

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In animals, all of the nitrogen of heme is derived from glycine and all of the carbons is derived from succinate and glycine. Hence, this synthesis is often called the *succinate-glycine pathway*. Porphyrin biosynthesis involves three distinct processes:

a. Synthesis of a substituted pyrrole compound, *porphobilinogen* from ALA.
b. Condensation of four porphobilinogen molecules to yield a partly reduced precursor called a *porphyrinogen*.
c. Modification of the side chains, dehydrogenation of the ring system, and the introduction of iron, to give the porphyrin product, heme.

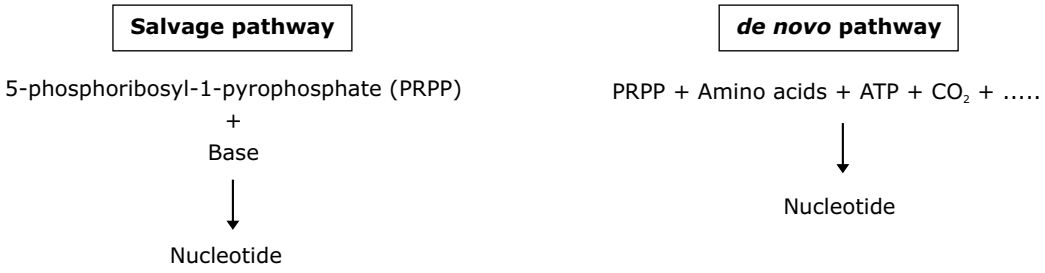
2.12 Nucleotide metabolism

2.12.1 Nucleotide synthesis

The pathways for the biosynthesis of nucleotides fall into two classes: *de novo pathways* and *salvage pathways*.

In *de novo* (means *anew*) pathways, the nucleotide bases are assembled from simpler compounds. The framework for a *pyrimidine* base is assembled first and then attached to ribose. In contrast, the framework for a *purine* base is synthesized piece by piece directly onto a ribose-based structure.

In *salvage pathways*, preformed bases are recovered and reconnected to a ribose unit.



Both *de novo* and *salvage pathways* lead to the synthesis of *ribo*nucleotides. All deoxyribonucleotides are synthesized from the corresponding ribonucleotides. The deoxyribose sugar is generated by the reduction of ribose within a fully formed nucleotide. Furthermore, the methyl group that distinguishes the thymine of DNA from the uracil of RNA is added at the last step in the pathway.

de novo synthesis

Pyrimidine ribonucleotide

In *de novo* synthesis of pyrimidines, the ring is synthesized first and then it is attached to ribose to form a *pyrimidine nucleotide*.

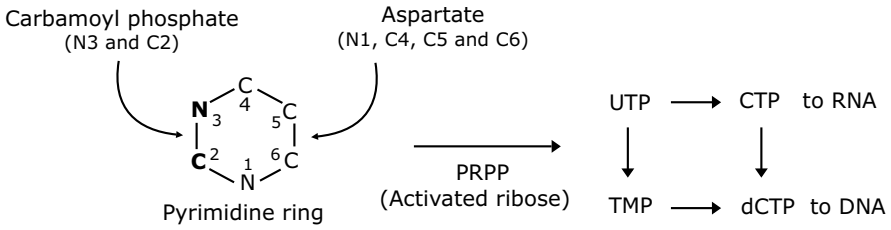
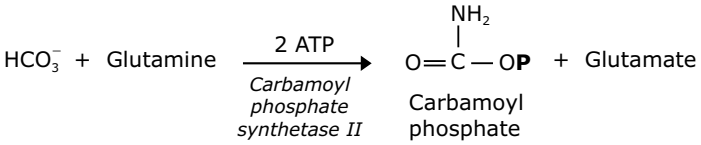


Figure 2.89 *de novo* pathway for pyrimidine nucleotide synthesis. The C-2 and N-3 atoms in the pyrimidine ring come from carbamoyl phosphate, whereas the other atoms of the ring come from aspartate.

Pyrimidine rings are synthesized from carbamoyl phosphate and aspartate. The precursor of carbamoyl phosphate is bicarbonate and ammonia. The synthesis of *carbamoyl phosphate* from bicarbonate and ammonia occurs in a multistep process, requiring the cleavage of two molecules of ATP. This reaction is catalyzed by cytosolic *carbamoyl phosphate synthetase II*.



Carbamoyl phosphate reacts with aspartate to form carbamoylaspartate in a reaction catalyzed by *aspartate transcarbamoylase*. Carbamoylaspartate then cyclizes to form dihydroorotate which is then oxidized to form orotate.

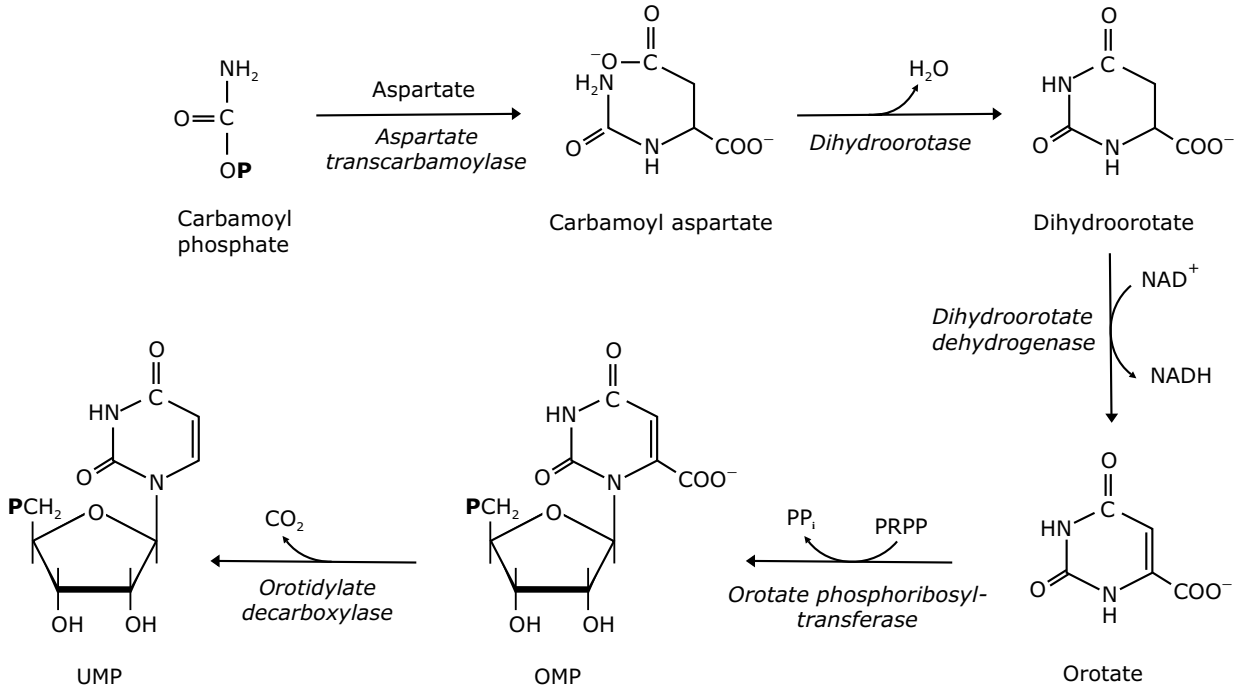


Figure 2.90 Pyrimidine synthesis.

Orotate couples to ribose, in the form of *5-phosphoribosyl-1-pyrophosphate* (PRPP), a form of ribose activated to accept nucleotide bases (PRPP is synthesized from ribose-5-phosphate, formed by the pentose phosphate pathway,

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Chapter 03

Cell Structure and Functions

3.1 What is a Cell?

The basic structural and functional unit of cellular organisms is the *cell*. It is an aqueous compartment bound by cell membrane, which is capable of independent existence and performing the essential functions of life. All organisms, more complex than viruses, consist of cells. Viruses are *noncellular* organisms because they lack cell or cell-like structure. In the year 1665, Robert Hooke first discovered cells in a piece of cork and also coined the word *cell*. The word cell is derived from the Latin word *cellula,* which means small compartment. Hooke published his findings in his famous work, *Micrographia*. Actually, Hooke only observed cell walls because cork cells are dead and without cytoplasmic contents. Anton van Leeuwenhoek was the first person who observed living cells under a microscope and named them animalcules, meaning *little animals.*

On the basis of the internal architecture, all cells can be subdivided into two major classes, *prokaryotic cells* and *eukaryotic cells*. Cells that have unit membrane bound nuclei are called eukaryotic, whereas cells that lack a membrane bound nucleus are prokaryotic. Eukaryotic cells have a much more complex intracellular organization with internal membranes as compared to prokaryotic cells. Besides the nucleus, the eukaryotic cells have other membrane bound **organelles** (*little organs*) like the endoplasmic reticulum, Golgi complex, lysosomes, mitochondria, microbodies and vacuoles. The region of the cell lying between the plasma membrane and the nucleus is the **cytoplasm**, comprising the cytosol (or cytoplasmic matrix) and the organelles. The prokaryotic cells lack such unit membrane bound organelles.

Cell theory

In 1839, Schleiden, a German botanist, and Schwann, a British zoologist, led to the development of the *cell theory* or *cell doctrine*. According to this theory *all living things are made up of cells* and *cell is the basic structural and functional unit of life.* In 1855, Rudolf Virchow proposed an important extension of cell theory that *all living cells arise from pre-existing cells* (*omnis cellula e cellula*). The cell theory holds true for all cellular organisms. Non-cellular organisms such as virus do not obey cell theory. Over the time, the theory has continued to evolve. The modern cell theory includes the following components:

- All known living things are made up of one or more cells.
- The cell is the structural and functional unit of life.
- All cells arise from pre-existing cells by division.
- Energy flow occurs within cells.
- Cells contain hereditary information (DNA) which is passed from cell to cell.
- All cells have basically the same chemical composition.

Evolution of the cell

The earliest cells probably arose about 3.5 billion years ago in the rich mixture of organic compounds, the *primordial soup*, of prebiotic times; they were almost certainly chemoheterotrophs. Primitive heterotrophs gradually acquired

the capability to derive energy from certain compounds in their environment and to use that energy to synthesize more and more of their own precursor molecules, thereby becoming less dependent on outside sources of these molecules-less extremely heterotrophic. A very significant evolutionary event was the development of photosynthetic ability to fix CO_2 into more complex organic compounds. The original electron (hydrogen) donor for these photosynthetic organisms was probably H_2S, yielding elemental sulfur as the byproduct, but at some point, cells developed the enzymatic capacity to use H_2O as the electron donor in photosynthetic reactions, producing O_2. The cyanobacteria are the modern descendants of these early photosynthetic O_2 producers.

One important landmark along this evolutionary road occurred when there was a transition from small cells with relatively simple internal structures - the so-called prokaryotic cells, which include various types of bacteria - to a flourishing of larger and radically more complex *eukaryotic* cells such as are found in higher animals and plants. The fossil record shows that earliest eukaryotic cells evolved about 1.5 billion years ago. Details of the evolutionary path from prokaryotes to eukaryotes cannot be deduced from the fossil record alone, but morphological and biochemical comparison of modern organisms has suggested a reasonable sequence of events consistent with the fossil evidence.

Three major changes must have occurred as prokaryotes gave rise to eukaryotes. *First*, as cells acquired more DNA, mechanisms evolved to fold it compactly into discrete complexes with specific proteins and to divide it equally between daughter cells at cell division. These DNA-protein complexes called chromosomes become especially compact at the time of cell division. *Second,* as cells became larger and intracellular membrane organelles developed. Eukaryotic cells have a *nucleus* which contains most of the cell's DNA, enclosed by a double layer of membrane. The DNA is, thereby, kept in a compartment separate from the rest of the contents of the cell, the cytoplasm, where most of the cell's metabolic reactions occur.

Finally, primitive eukaryotic cells, which were incapable of photosynthesis or of aerobic metabolism, pooled their assets with those of aerobic bacteria or photosynthetic bacteria to form symbiotic associations that became permanent. Some aerobic bacteria evolved into the mitochondria of modern eukaryotes, and some photosynthetic cyanobacteria became the chloroplasts of modern plant cells.

3.2 Structure of eukaryotic cells

3.2.1 Plasma membrane

Plasma membrane is a dynamic, fluid structure and forms the external boundary of cells. It acts as a *selectively permeable* membrane and regulates the molecular traffic across the boundary. The plasma membrane exhibits selective permeability; that is, it allows some solutes to cross it more easily than others. Different models were proposed to explain the structure and composition of plasma membranes. In 1972, Jonathan Singer and Garth Nicolson proposed **fluid-mosaic model**, which is now the most accepted model. In this model, membranes are viewed as quasi-fluid structures in which proteins are inserted into lipid bilayers. It describes both the *mosaic* arrangement of proteins embedded throughout the lipid bilayer as well as the *fluid* movement of lipids and proteins alike.

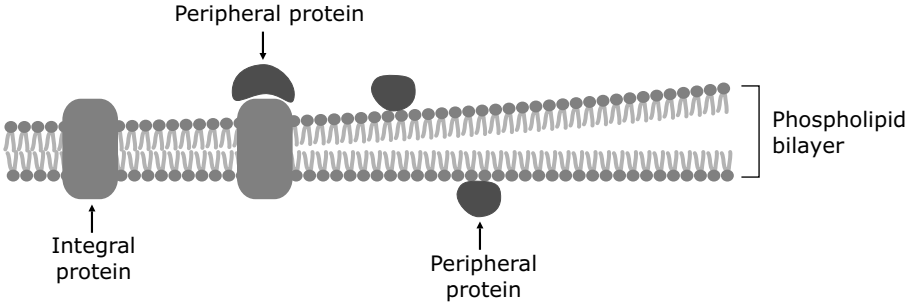


Figure 3.1 Fluid mosaic model for membrane structure. The fatty acyl chains in the lipid bilayer form a fluid, hydrophobic region. Integral proteins float in this lipid bilayer. Both proteins and lipids are free to move laterally in the plane of the bilayer, but movement of either from one face of the bilayer to the other is restricted.

Chemical constituents of plasma membrane

All plasma membranes, regardless of source, contain proteins as well as lipids. The ratio of protein to lipid varies enormously depends on cell types. For example, the plasma membrane of human RBC contains 43% lipid and 49% protein by weight whereas plasma membrane of mouse liver cells contain 54% lipid and 46% protein by weight. Carbohydrates are also present in plasma membranes and comprise only 5 to 10% of membrane mass. Carbohydrates bound either to proteins as constituents of glycoproteins or to lipids as constituents of glycolipids. Carbohydrates are especially abundant in the plasma membranes of eukaryotic cells.

Lipid bilayer

The basic structure of the plasma membrane is the lipid bilayer. This bilayer is composed of two *leaflets* of amphipathic lipid molecules, whose polar head groups are in contact with the intra- or extracellular aqueous phase, whereas their non-polar tails face each other, constituting the hydrophobic interior of the membrane. The primary physical forces for organizing lipid bilayer are *hydrophobic interactions*. Three classes of lipid molecules present in lipid bilayer - phospholipids, glycolipids and sterol.

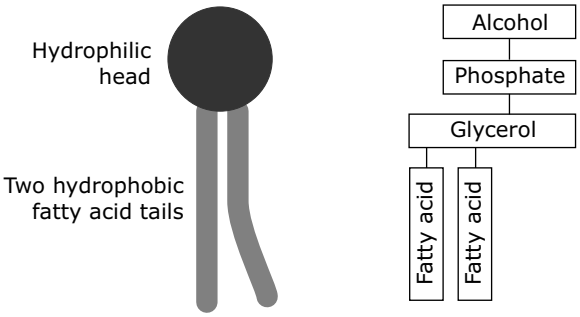


Figure 3.2 Phospholipid structure. The hydrophilic unit, also called the polar head group, is represented by a circle, whereas the hydrocarbon tails are depicted by straight lines.

Phospholipids

Phospholipids are made up of four components: an alcohol (glycerol or *sphingosine),* fatty acids, phosphate, and an alcohol attached to the phosphate. The fatty acid components are hydrophobic, whereas the remainder of the molecule has hydrophilic. There are two types of phospholipids: *glycerophospholipids* and *sphingophospholipids*.

Phospholipids derived from glycerol are called *glycerophospholipids.* Glycerophospholipids (or phosphoglycerides) contain glycerol, fatty acids, phosphate and an alcohol (e.g. choline). Phosphoglyceride molecules are classified according to the types of alcohol linked to the phosphate group. For example, if the alcohol is choline, the molecule is called phosphatidylcholine (also referred to as *lecithin*) and if serine, then it is called phosphotidylserine. Phosphoglycerides are the most numerous phospholipid molecules found in plasma membranes.

Sphingophospholipids contain an amino alcohol called **sphingosine** instead of glycerol, a fatty acid, phosphate and an alcohol attached to the phosphate. In sphingophospholipid, the amino group of the sphingosine backbone is linked to a fatty acid by an amide bond. *Sphingomyelin is the most abundant sphingophospholipid.*

The plasma membrane of animal cells contains four major phospholipids, such as *phosphatidylcholine* (the most abundant glycerophospholipids in the plasma membrane), *phosphatidylserine*, *phosphatidylethanolamine* and *sphingomyelin*. At neutral pH, the polar head group may have no net charge (phosphatidylcholine and phosphatidyl-ethanolamine) or it may have net negative charges (phosphatidylinositol and phosphatidylserine). Rarer phospholipids have a net positive charge.

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3.3 Membrane potential

Electrical character of ion transport may be **electroneutral** i.e. electrically silent either by symport of the oppositely charged ions or antiport of similarly charged ions or **electrogenic** i.e. result in charge separation across the membrane. Electrogenic transport affects and can be affected by the membrane potential. For example, the Na^+–K^+ pump imports $2K^+$ and simultaneously exports $3Na^+$; that is, it moves 1 positive charge out of the cell. Its electrogenic operation directly contributes to the negative inside membrane potential, which is evidenced by the fact that stopping the pump using an alkaloid inhibitor, *ouabain*, causes an immediate and slight depolarization of the cell membrane.

All cells have an electrical potential difference, or *membrane potential*, across their plasma membrane. Electrical potential across cell membranes is a function of the electrolyte concentrations in the intracellular and extracellular solutions and of the selective permeabilities of the ions. Active transport of ions by ATP-driven ion pumps, generate and maintain ionic gradients. In addition to ion pumps, which transport ions against concentration gradients, plasma membrane contains channel protein that allows ions to move through it at different rates down their concentration gradient. Ion concentration gradients and selective movements of ions create a difference in electric potential or voltage across the plasma membrane. This is called *membrane potential*.

How membrane potentials arise?

To help explain how an electric potential across the plasma membrane can arise, we first consider a set of simplified experimental systems in which a membrane, which is only permeable for K^+ separates a 1 M KCl solution on the left from a 1 M KCl solution on the right. Because the concentrations of K^+ across the membrane are equal, there is no net flow of ions across the membrane and thus no electric potential is generated. If the concentration of K^+ ions across the membrane is different as shown in the figure, then K^+ ions tend to move down their concentration gradient from the left side to the right, leaving an excess of negative Cl^- ions compared with K^+ ions on the left side and generate an excess of positive K^+ ions compared with Cl^- ions on the right side. The resulting separation of charge across the membrane constitutes an electric potential, or voltage, with the left side of the membrane having excess negative charge with respect to the right. However, continued left-to-right movement of the K^+ ions eventually is inhibited by the mutual repulsion between the excess positive charges accumulated on the right side of the membrane and by the attraction of K^+ ions to the excess negative charges built up on the left side. The system soon reaches an equilibrium point at which the two opposing factors that determine the movement of K^+ ions—the membrane electric potential and the ion concentration gradient—balance each other out. At equilibrium, no net movement of K^+ ions occurs across the membrane.

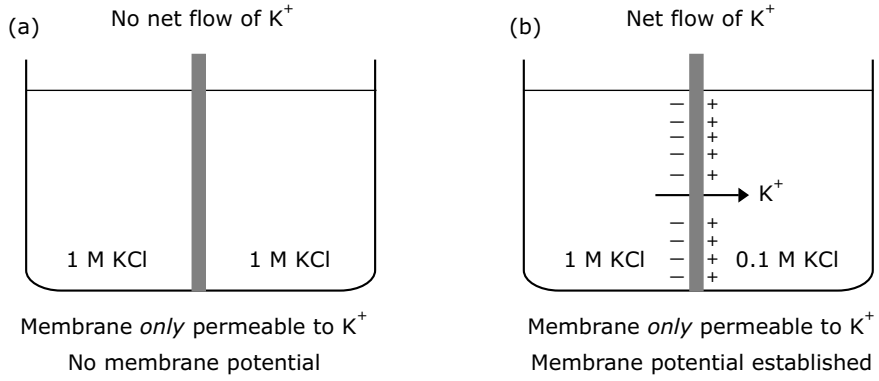


Figure 3.15 Two compartments are separated by a membrane permeable only to K^+ ions. (a) Because the concentrations in the two compartments are equal, there is no net flow of ions across the membrane and no electrical potential. (b) A difference in concentration causes K^+ ions to move from the left compartment to the right one. At equilibrium, an electrical potential is established across the membrane due to an accumulation of negative charges on the left side and positive charges on the right.

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Let us now consider the changes in potential during an action potential, and the permeability and ion movements responsible for generating this change in potential. Movement of ions occurs through ion channels. Ion channels may be either leaky channels or gated channels. Leaky channels, which are open all the time, permit unregulated leakage of specific ion across the membrane. Gated channels, in contrast, have gates that can be open or closed, permitting ion passage through the channels when open and preventing ion passage through the channels when closed. Movement of K^+ through K^+ leaky channels makes the greatest contribution to the establishment of the resting potential, because the membrane at rest is considerably more permeable to K^+. Action potentials are the direct consequence of the voltage-gated cation channels. During an action potential, rapid fluxes of Na^+ occur through voltage gated Na^+ channels down their electrochemical gradients. This voltage-gated Na^+ channel can exist in three conformations: 1. closed but capable of opening; 2. open or activated and 3. closed and not capable of opening or inactivated. The channel undergoes through these various conformations as a result of voltage changes that take place during an action potential.

At resting potential (about –70 mV), Na^+ voltage-gated channels are closed. Therefore, Na^+ cannot pass through these voltage-gated channels at resting potential. A stimulus that causes sufficient depolarization promptly causes voltage-gated Na^+ channels to open, allowing a small amount of Na^+ to enter the cell down its electrochemical gradient. The influx of positive charge depolarizes the membrane further, thereby opening more Na^+ channels, which admit more Na^+ ions, causing still further depolarization. This changes the membrane potential value of about –70 mV to about +30 mV within a fraction of a second.

The Na^+ channels have an automatic inactivating mechanism, which causes the channels to reclose rapidly even though the membrane is still depolarized. The Na^+ channels remain in this inactivated state, unable to reopen, until a few milliseconds after the membrane potential returns to its initial negative value.

In addition to the inactivation of Na^+ channels, another channels operate to help bring the activated plasma membrane more rapidly back toward its original negative potential. It is voltage-gated K^+ channels. The opening and consequent efflux of K^+ quickly drives the membrane back toward the K^+ equilibrium potential. This process is called *repolarization*. Voltage-gated K^+ channels respond to changes in membrane potential in much the same way as the Na^+ channels do, but with slightly slower kinetics; for this reason they are sometimes called *delayed K^+ channels*. Unlike voltage-gated Na^+ channels, most voltage-gated K^+ channels do not exhibit an inactivated state. Instead, they alternate between closed and open states.

Hence, an action potential has two main phases: a depolarizing phase and a repolarizing phase. During the *depolarizing phase,* the negative membrane potential becomes less negative, reaches zero, and then becomes positive. During the *repolarizing phase,* the membrane potential is restored to the resting state of –70 mV. Following the repolarizing phase there may be an *after-hyperpolarizing phase*, during which the membrane potential temporarily becomes more negative (about –90 mV) than the resting level.

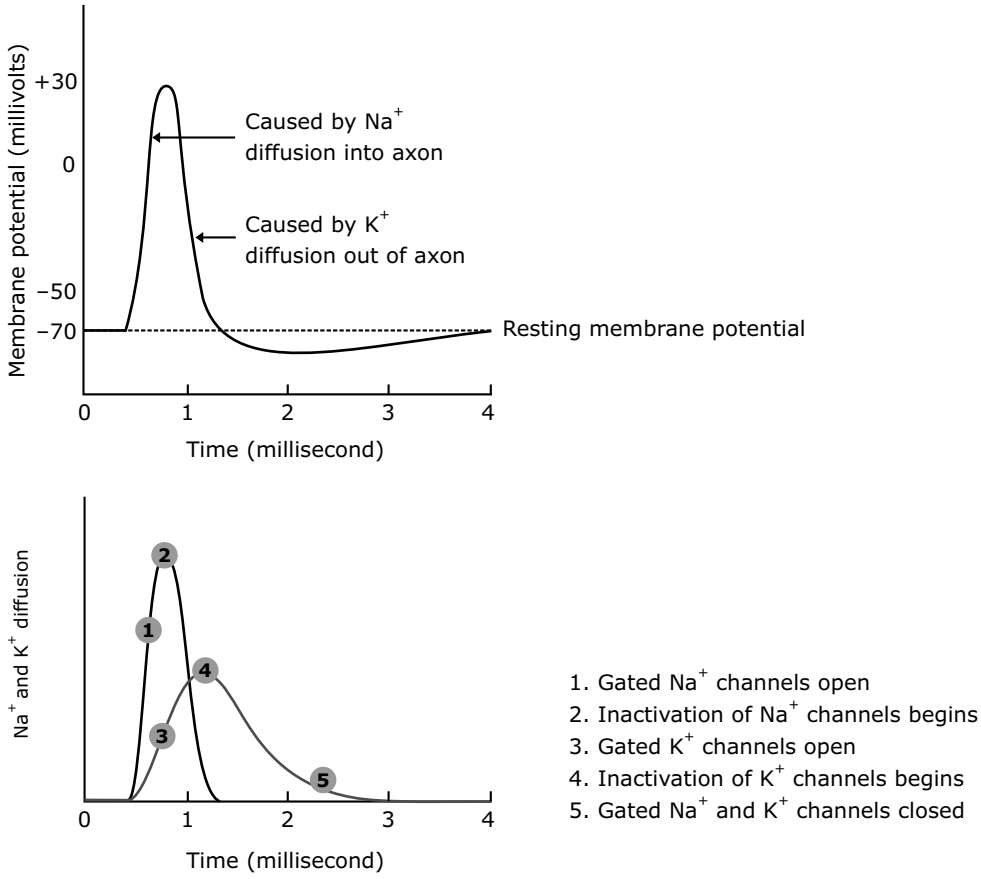


Figure 3.16 Membrane potential changes and ion movements during an action potential. The top graph depicts an action potential. The bottom graph depicts the net diffusion of Na^+ and K^+ during the action potential. The x-axis for time is the same in both graphs, so that the depolarization, repolarization, and after-hyperpolarization in the top graph can be correlated with events in the Na^+ and K^+ channels and their effects on ion movements in the bottom graph. The inward movement of Na^+ drives the membrane potential toward the Na^+ equilibrium potential during the depolarization (rising) phase of the action potential, whereas the outward movement of K^+ drives the membrane potential toward the potassium equilibrium potential during the repolarization (falling) phase of the action potential.

The period of time after an action potential begins during which an excitable cell cannot generate another action potential in response to a *normal* threshold stimulus is called the **refractory period**. It can be absolute or relative. During the *absolute refractory period*, even a very strong stimulus cannot initiate a second action potential. This period coincides with the period of Na^+ channel activation and inactivation. Inactivated Na^+ channels cannot reopen; they first must return to the resting state. The *relative refractory period* is the time period during which a second action potential can be initiated, but only by a larger-than normal stimulus. It coincides with the period when the voltage gated K^+ channels are still open after inactivated Na^+ channels have returned to their resting state. The refractory period limit the number of action potentials that can be produced by an excitable membrane in a given period of time.

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plasma membrane at the opposite side. An example of transcytosis is the movement of maternal antibodies across the intestinal epithelial cells of the newborn rat. A newborn rat obtains antibodies from its mother's milk by transporting them across the epithelium of its gut. The lumen of the gut is acidic, and, at this low pH, the antibodies in the milk bind to specific receptors on the apical (absorptive) surface of the gut epithelial cells. The receptor-antibody complexes are internalized via clathrin coated vesicles and are delivered to early endosomes. The complexes remain intact and are retrieved in transport vesicles that bud from the early endosome and subsequently fuse with the basolateral domain of the plasma membrane. On exposure to the neutral pH of the extracellular fluid that bathes the basolateral surface of the cells, the antibodies dissociate from their receptors and eventually enter the newborn's bloodstream.

3.4.3 Exocytosis

Transport vesicles destined for the plasma membrane undergo fusion with the plasma membrane and release the contents outside the cell in the process called exocytosis. It may be a *constitutive secretory pathway* (carried out by all cells) or *regulated secretory pathway* (carried out by specialized cells). Examples of proteins released by such constitutive (or continuous) secretion include collagen by fibroblasts, serum proteins by hepatocytes, and antibodies by activated B-lymphocytes.

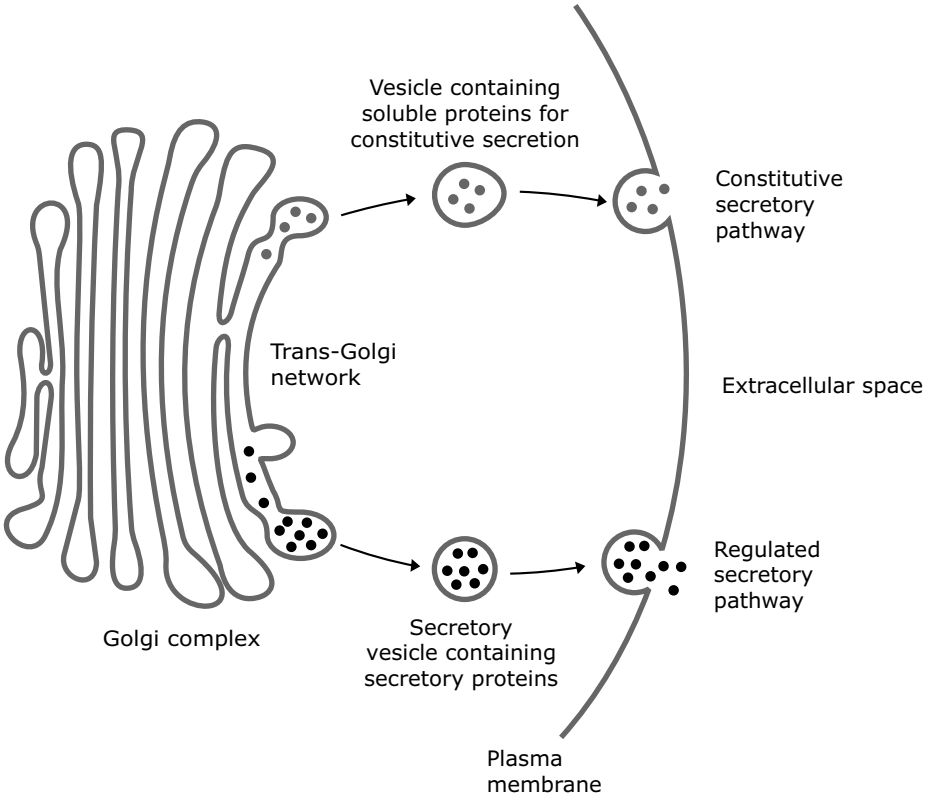


Figure 3.26 Constitutive and regulated secretory pathways. The two pathways diverge in the *trans* Golgi network. The constitutive secretory pathway operates in all cells. Many soluble proteins are continually secreted from the cell by this pathway. This pathway also supplies the plasma membrane with newly synthesized lipids and proteins. Specialized secretory cells also have a regulated secretory pathway, by which selected proteins in the *trans* Golgi network are diverted into *secretory vesicles*, where the proteins are concentrated and stored until an extracellular signal stimulates their secretion. The regulated secretion of small molecules, such as histamine and neurotransmitters, occurs by a similar pathway.

The regulated secretory pathway is found mainly in cells specialized for secreting products rapidly on demand such as hormones, neurotransmitters, or digestive enzymes. In this secretory pathway, secretory vesicles form from the *trans* Golgi network, and they release their contents to the cell exterior by exocytosis in response to specific signals. The secreted product can be either a small molecule (such as histamine) or a protein (such as a hormone or digestive enzyme). Proteins destined for secretion (called *secretory proteins*) are packaged into appropriate secretory vesicles in the *trans* Golgi network. The signal that directs secretory proteins into such vesicles is not known.

3.5 Ribosome

The ribosomes are large *ribonucleoproteins* consisting of RNAs and proteins, ubiquitous in all cells, that translate genetic information stored in the messenger RNA into polypeptides. The ribosome is approximately globular structure, its average diameter ranging from 2.5 nm (*Escherichia coli*) to 2.8 nm (mammalian cells). The functional ribosomes consist of two subunits of unequal size, known as the large and small subunits. Ribosomes consist of *rRNA* and *r-proteins*. The r-proteins are termed as L or S depending on whether the protein is from the large or small subunit.

Table 3.5 Ribosome structure and chemical composition

Property	*Prokaryote*	*Eukaryote*
Overall size	70S	80S
Small subunit	30S	40S
Number of proteins	~21	~30
RNA size (number of bases)	16S (1500)	18S (2300)
Large subunit	50S	60S
Number of proteins	~34	~50
RNA size (number of bases)	23S (2900)	28S (4200)
	5S (120)	5.8S (160)
		5S (120)

'S' stands for the *sedimentation coefficient*. It is the ratio of a velocity to the centrifugal acceleration. The sedimentation coefficient has units of second. A sedimentation coefficient of 1×10^{-13} second is defined as one Svedberg, S.

rDNA organization

In prokaryotes such as *Escherichia coli*, there are three ribosomal RNAs (16S, 23S and 5S), which are organized as a single transcription unit. In all eukaryotes studied so far, the organization of the ribosomal RNA genes is recognizably similar to that of prokaryotes, but with major differences; the size of the small subunit RNA has increased from 16S to 18S, and that of the large subunit from 23S and 28S; a new small 5.8S rRNA has become interspersed between the 18S and the 28S rRNA, and the 5S rRNA has become separated from the other rRNAs in a different transcription unit. The former transcription unit is generally referred to as the rRNA gene or the ribosomal DNA (rDNA). 5S genes are transcribed by a different RNA polymerase from rRNA genes (RNA polymerase III rather than RNA polymerase I). There are generally more copies of the 5S genes than of the rRNA genes. The human genome contains about 100 copies of rRNA genes per haploid set. Many other species, including most plants, have several thousand copies. The rRNA gene is transcribed to give a precursor the 45S pre-rRNA, which is processed in a series of post-transcriptional modifications to the mature rRNA species.

Table 3.6 Different types of ribosomes and their rRNAs

Ribosome source	*Sedimentation coefficient*	*rRNA* (large subunit/small subunit)
Bacterial	70S	5S, 23S/16S
Chloroplast	70S	5S, 23S/16S
Mitochondria (human)	55S	16S/12S
Archaebacteria	70S	5S,23S/16S
Eukaryotes (cytosol)	80S	5S, 5.8S, 28S/18S

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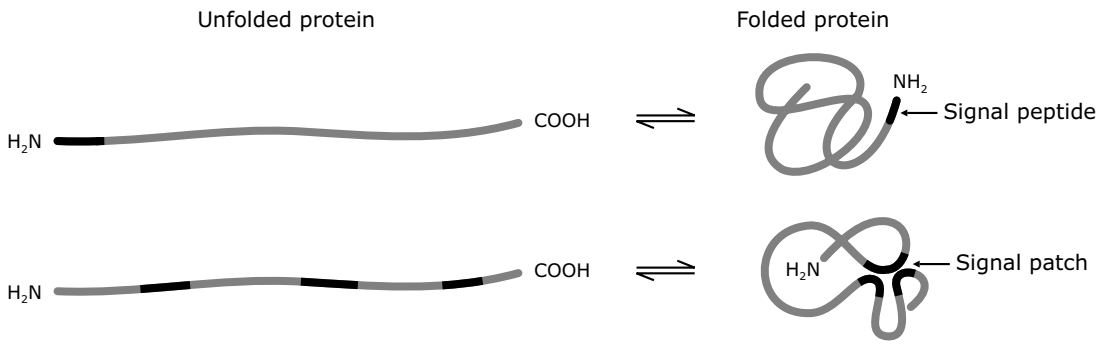


Figure 3.27 Signal peptide and signal patch.

Protein translocation describes the movement of a protein across a membrane. Within the cell, translocation of proteins from cytosol to specific organelle or organelle to cytosol and from one organelle to another occur in three different ways:

1. *Gated transport* : The protein translocation between the cytosol and nucleus occurs through the nuclear pore complexes. This process is called gated transport because the nuclear pore complexes function as selective gates that can actively transport specific macromolecules.
2. *Transmembrane transport* : In transmembrane transport, membrane-bound *protein translocators* directly transport specific proteins across a membrane from the cytosol into a organelle. The transport of selected proteins from the cytosol into the ER lumen or into mitochondria is an example of *transmembrane transport*.
3. *Vesicular transport* : In vesicular transport, proteins move from one organelle to another through transport vesicles. The transfer of proteins from the endoplasmic reticulum to the Golgi apparatus, for example, occurs in this way.

Protein translocation may occur co-translationally or post-translationally. Proteins synthesized by membrane bound ribosomes are translocated co-translationally. All proteins synthesized by membrane free ribosomes are translocated post-translationally.

3.6 Endoplasmic reticulum

Endoplasmic reticulum (ER) is the largest single membrane bound intracellular compartment. It is an extensive network of closed and flattened membrane-bound structure. The enclosed compartment is called the ER *lumen*. ER membranes are physiologically active, interact with the cytoskeleton and contain differentiated domains specialized for distinct functions.

ER membranes are differentiated into rough and smooth regions (RER and SER, respectively), depending on whether ribosomes are associated with their cytoplasmic surfaces. Regions of ER that lack bound ribosomes are called SER (sometime also called *transitional* ER). The membranes and luminal spaces of the ER are normally continuous throughout the cell and that RER and SER form an interconnected membrane system. When cells are disrupted by homogenization, the ER breaks into fragments and reseals into small vesicles called **microsomes**. Microsomes derived from RER are studded with ribosomes on the outer surface and are called *rough microsomes*. Microsomes lacking attached ribosomes are called *smooth microsome*.

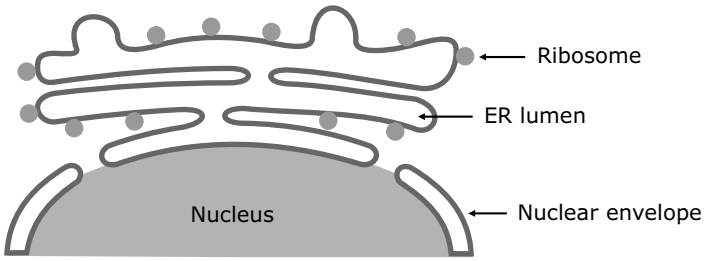


Figure 3.28 An endoplasmic reticulum is a membranous network of cisternae (sac-like structures). The cisternal space (or lumen) remains continuous with the perinuclear space.

Function of ER

Proteins synthesized by ribosomes associated with the membrane of RER enter into the lumen and membrane of RER by the process of co-translational translocation. In the lumen of the RER, five principal modifications of proteins occur before they reach their final destinations: addition and processing of carbohydrates (*N*-linked glycosylation), formation of disulfide bonds, proper folding, specific proteolytic cleavages and assembly into multimeric proteins. The SER acts as the site of lipid biosynthesis, detoxification and calcium regulation.

N-linked glycosylation of proteins

N-linked glycosylation is the attachment of a sugar molecule to a nitrogen atom in an amino acid residue in a protein. In the RER, this process involves the addition of a large preformed oligosaccharide precursor to a protein. This precursor oligosaccharide is linked by a pyrophosphoryl residue to *dolichol,* a long-chain (75–95 carbon atoms) polyisoprenoid lipid that is firmly embedded in the RER membrane and acts as a carrier for the oligosaccharide. The structure of *N*-linked oligosaccharide is the same in plants, animals and single-celled eukaryotes - a branched oligosaccharide, containing three glucose (Glc), nine mannose (Man) and two *N*-acetylglucosamine (GlcNAc) molecules which is written as $Glc_3\ Man_9\ GlcNAc_2$.

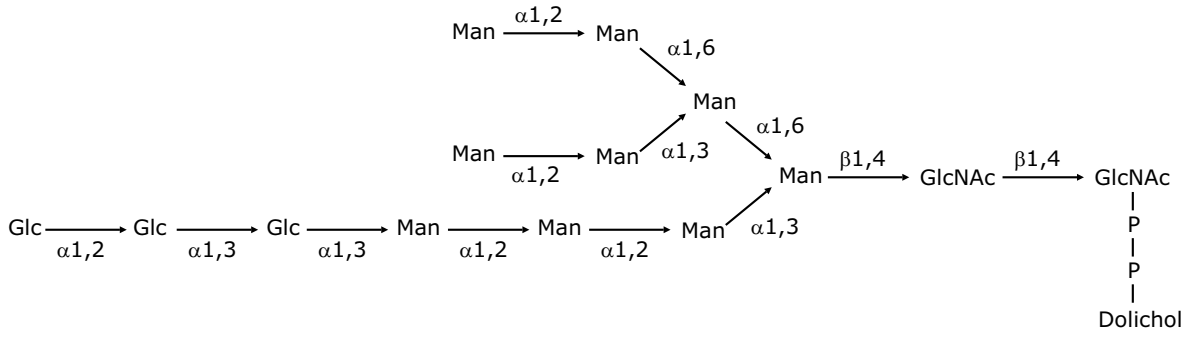


Figure 3.29 Structure of dolichol—P—P—oligosaccharide.

Biosynthesis of oligosaccharide begins on the cytosolic face of the ER membrane with the transfer of N-acetyl glucosamine to dolichol phosphate. Two *N*-acetylglucosamine (GlcNAc) and five mannose residues are added one at a time to a dolichol phosphate on the cytosolic face of the ER membrane. The first sugar, N-acetyl glucosamine, is linked to dolichol by a pyrophosphate bridge. This high-energy bond activates the oligosaccharide for its transfer from the dolichol to an asparagine side chain of a nascent polypeptide on the luminal side of the rough ER. **Tunicamycin**, an antibiotic, blocks the first step in this pathway and thus inhibits the synthesis of oligosaccharide. After the seven-residue dolichol pyrophosphoryl intermediate is flipped to the luminal face. The remaining four mannose and all three glucose residues are added one at a time in the luminal side. The sugar molecules participate

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Table 3.8 Coated vesicles found within eukaryotic cells

Coated vesicle	*Coat proteins*	*Transport*
Clathrin	Clathrin, AP1	Golgi complex to endosome
Clathrin	Clathrin, AP2	Plasma membrane to endosome
COPI	COPI	Golgi complex to the ER or intra Golgi complex
COPII	COPII	ER to Golgi complex

The coat proteins surrounding transport vesicles that move from the late endosome to lysosomes and to the plasma membrane have not yet been identified.

ER-resident proteins often are retrieved from the *Cis*-Golgi

As we have mentioned in the previous section that proteins entering into the lumen of the ER are of two types- *resident* proteins and *export* proteins. How, then, are resident proteins retained in the ER lumen to carry out their work?

The answer lies in a specific C-terminal sequence present in resident ER proteins. Most ER-resident proteins have a *Lys-Asp-Glu-Leu* (KDEL in the one-letter code) sequence at their C-terminus. Several experiments demonstrated that the *KDEL sequence* which acts as sorting signal, is both necessary and sufficient for retention in the ER. If this *ER retention signal* is removed from BiP, for example, the protein is secreted from the cell; and if the signal is transferred to a protein that is normally secreted, the protein is now retained in the ER. The KDEL sorting signal is recognized and bound by the *KDEL receptor* found on the ER and the *cis*-Golgi. The KDEL receptor acts mainly to retrieve proteins with the KDEL sorting signal that have escaped to the *cis*-Golgi network and returns them to the ER. The finding that most KDEL receptors are localized to the membranes of small transport vesicles shuttling between the ER and the *cis*-Golgi also supports this concept. The KDEL receptor acts mainly to retrieve soluble proteins containing the KDEL sorting signal. The retention of transmembrane proteins in the ER is carried out by short C-terminal sequences that contain two lysine residues (KKXX sequences).

The affinity of the KDEL receptor for proteins with KDEL sorting signal changes in different compartments. How can the affinity of the KDEL receptor change depending on the compartment in which it resides? The answer may be related to the differences in pH. In the low-pH environment of *cis*-Golgi and transport vesicles, the KDEL receptor has greater binding affinity with the KDEL sorting signal whereas in the neutral-pH environment of the ER, the ER proteins dissociate from the KDEL receptor due to lesser affinity.

Clearly, the transport of newly synthesized proteins from the RER to the Golgi cisternae is a highly selective and regulated process. The selective entry of proteins into membrane-bound transport vesicles is an important feature of protein targeting as we will encounter them several times in our study of the subsequent stages in the maturation of secretory and membrane proteins.

3.7 Golgi complex

The Golgi complex was first discovered in 1897 by Italian physician Camillo Golgi. The Golgi complex, also termed as *Golgi body* or *Golgi apparatus,* is a single membrane bound organelle and part of endomembrane system. It consists of five to eight flattened membrane-bound sacs called the **cisternae**. Each stack of cisternae is termed as *Golgi stack* (or dictyosome). The cisternae in *Golgi stack* vary in number, shape and organization in different cell types. The typical diagrammatic representation of three major cisternae (cis, medial and trans) as shown in the figure 3.41 is actually a simplification. In some unicellular flagellates, however, as many as 60 cisternae may combine to make up the Golgi stack. The number of Golgi complexes in a cell varies according to its function. A mammalian cell typically contains 40 to 100 stacks. In mammalian cells, multiple Golgi stacks are linked together at their edges.

Each Golgi stack has two distinct faces: a ***cis* face** (or entry face or forming face) and a ***trans* face** (or maturing face). Both *cis* and *trans* faces are closely associated with special compartments: the ***cis* Golgi network** (CGN)

and the ***trans* Golgi network** (TGN), respectively. TGN was earlier known as GERL (Golgi apparatus - ER - lysosome). Proteins and lipids enter the *cis* Golgi network in vesicular tubular clusters arriving from the ER and exit from the *trans* Golgi network. Both networks are thought to be important for protein sorting. As we have seen, proteins entering the CGN can either move onward in the Golgi apparatus or be returned to the ER. Similarly, proteins exiting from the TGN can either move onward and be sorted according to whether they are destined for lysosomes, secretory vesicles, or the cell surface, or be returned to an earlier compartment.

The Golgi apparatus is especially prominent in cells that are specialized for secretion, such as the goblet cells of the intestinal epithelium, which secrete large amounts of polysaccharide-rich mucus into the gut. In such cells, unusually large secretory vesicles are found on the *trans* side of the Golgi apparatus. Secretory vesicles form from the trans Golgi network, and they release their contents to the cell exterior by exocytosis. The membrane proteins and the lipids in these vesicles provide new components for the cell's plasma membrane, while the soluble protein and other compounds inside the vesicles are secreted to the extracellular space.

The Golgi apparatus is often considered the distribution and shipping department for the cell's chemical products. It modifies proteins and lipids that have been built in the endoplasmic reticulum and prepares them for export outside of the cell or for transport to other locations in the cell. Proteins and lipids from the smooth and rough endoplasmic reticulum bud off in tiny bubble-like vesicles that move through the cytoplasm until they reach the Golgi apparatus. The vesicles fuse with the Golgi membranes and release their internally stored molecules into the organelle. Once inside, the compounds are further processed by the Golgi apparatus. When completed, the product is extruded from the Golgi apparatus in a vesicle and directed to its final destination inside or outside the cell.

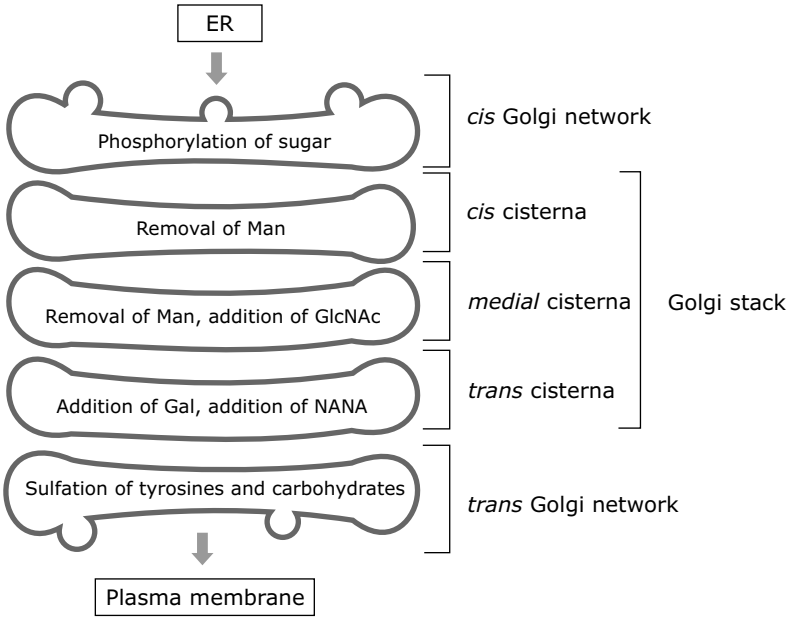


Figure 3.41 The functional compartmentalization of the Golgi apparatus.

The modifications to molecules that take place in the Golgi apparatus occur in an orderly fashion. Substances from ER enter into the *cis* face of a Golgi stack for processing and exit from *trans* face. Consequently, the cis face is found near the endoplasmic reticulum and the trans face is positioned near the plasma membrane of the cell. The chemical make-up of each face is different and the enzymes contained in the cisternae between the faces are distinctive.

Glycosylation of proteins

N-linked oligosaccharide chains on proteins are altered as the proteins pass through the Golgi cisternae *en route* from the ER. Further modifications of *N*-linked oligosaccharide in the Golgi apparatus gives two broad classes of *N*-linked oligosaccharides, the *complex oligosaccharides* and the *high-mannose oligosaccharides.* High-mannose

Pages 279 to 312 are not shown in this preview.

termed as *heterochromatin.* Because of its condensed state, this heterochromatin is generally believed to be transcriptionally silent.

Centromere

The constricted region of linear chromosomes is known as the centromere. Although this constriction is called the centromere, it is usually not located exactly in the center of the chromosome and, in some cases, is located almost at the chromosome's end. The regions on either side of the centromere are referred to as the chromosome's arms. The centromeres serve both as the sites of association of sister chromatids and as the attachment sites for microtubules of the mitotic spindle.

Telomere

Telomeres are specialized structures, which cap the ends of eukaryotic chromosomes. It consists of a long array of short, tandemly repeated sequences. Sequencing of telomeres from different organisms has shown that most are repetitive sequences with a high G content in the strand with its 3' end.

Origin of replication

The origin of replication (also called the *replication origin*) is a particular sequence in a chromosome at which replication is initiated. One chromosome contains multiple origin of replication.

Chromosome number

All eukaryotic cells have multiple linear chromosomes. Every cell maintains a characteristic number of chromosomes. Depending on the eukaryotic organism, the number of chromosomes varies from 2 to several hundreds. The number of chromosomes in a species has no specific significance nor does it indicate any relationship between two species which may have the same chromosome number. The majority of eukaryotic cells are diploid; that is, they contain two copies of each chromosome.

Table 3.16 Chromosome number in different eukaryotic organisms.

Species	*Haploid number of chromosome*
Saccharomyces cerevisiae (budding yeast)	16
Schizosaccharomyces pombe (fission yeast)	03
Caenorhabditis elegans	06
Arabidopsis thaliana	05
Drosophila melanogaster	04
Tetrahymena thermophilus	Micronucleus 5, Macronucleus 225
Mus musculus (mouse)	19 + X and Y
Homo sapiens	22 + X and Y

3.20 Cell signaling

All cells receive and respond to signals from their surroundings. This is accomplished by a variety of signal molecules that are secreted or expressed on the surface of one cell and bind to receptors expressed by other cells, thereby integrating and coordinating the functions of the many individual cells that make up organisms. Each cell is programmed to respond to specific extracellular signal molecules. Extracellular signaling usually involves the following steps:

1. Synthesis and release of the signaling molecule by the signaling cell;
2. Transport of the signal to the target cell;
3. Binding of the signal by a specific receptor leading to its activation;
4. Initiation of signal-transduction pathways.

In animals, extracellular signaling by signal molecules can be classified into four categories—endocrine, paracrine, autocrine and juxtacrine signaling.

In *endocrine* signaling*,* the signaling molecules act on target cells distantly located from their site of synthesis. It is a long-range signaling in which signal molecule is transported by the blood stream.

In *paracrine* signaling*,* the signaling molecules released by a cell affect target cells only in close proximity. An example of this is the action of neurotransmitters in carrying signals between nerve cells at a synapse.

In *autocrine* signaling*,* the signaling molecules produce an effect on same cell that produces it. One important example of such is the response of cells of the vertebrate immune system to foreign antigens. Certain types of T-lymphocytes respond to antigenic stimulation by synthesizing a growth factor that drives their own proliferation, thereby increasing the number of responsive T-lymphocytes and amplifying the immune response.

In *juxtacrine* signaling, signal molecules do not diffuse from the cell producing it and cell bearing signal molecules interact with receptor proteins of adjacent responding cells. Unlike other modes of cell signaling, *juxtacrine* signaling requires physical contact between the cells involved. Notch signalling and classical cadherin signalling are examples of juxtacrine signaling.

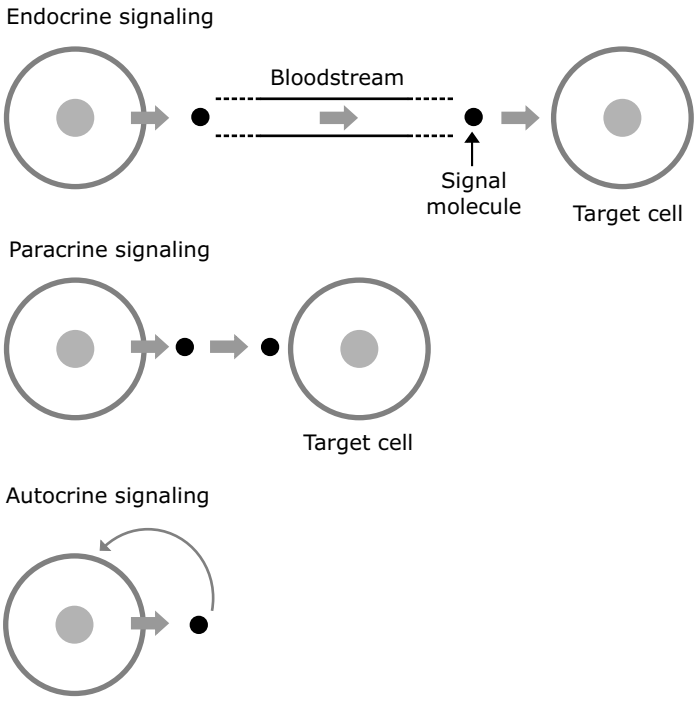


Figure 3.70 Long-range signaling between cells is called endocrine when the signal molecule is transported by the bloodstream (typical for hormones), paracrine when the signal diffuses between neighboring cells across the extracellular matrix (typical for neurotransmitters and many so-called tissue hormones or local mediators), and autocrine when the signal re-acts on the transmitter cell.

3.20.1 Signal molecules

Signal molecules are chemically heterogenous compounds. These molecules are divided into two categories – membrane bound and secretory signal molecules. Membrane bound signal molecules remain bound to the surface of the cells and mediate *contact dependent* signaling. In most cases, signal molecules are secreted by signaling cells. Secreted extracellular signal molecules are further divided into three general categories based on the distance over which signals are transmitted: *endocrine, paracrine* and *autocrine* signal molecules.

Pages 315 to 360 are not shown in this preview.

pro-apoptotic. Mammalian Bcl2 family of proteins regulate the intrinsic pathway of apoptosis mainly by controlling the release of cytochrome *c* and other intermembrane mitochondrial proteins into the cytosol.
The anti-apoptotic Bcl2 proteins include Bcl2 itself and Bcl-X_L. Bcl2 was the first protein shown to cause an inhibition of apoptosis. It is the mammalian homologue of the CED-9 in *C. elegans*. The pro-apoptotic Bcl2 proteins consist of two subfamilies - the BH123 proteins and the BH3-*only* proteins. The main BH123 proteins are Bax and Bak, which are structurally similar to Bcl2. Important members of the BH3-*only* proteins are Bid, Bim, Bik, Bad and Bmf.
When an apoptotic stimulus triggers the intrinsic pathway, the pro-apoptotic BH123 proteins become activated and induces the release of cytochrome c and other intermembrane proteins by an unknown mechanism. In the absence of an apoptotic stimulus, anti-apoptotic Bcl2 proteins bind to and inhibit the BH123 proteins on the mitochondrial outer membrane and in the cytosol. In the presence of an apoptotic stimulus, BH3-*only* proteins are activated and bind to the anti-apoptotic Bcl2 proteins so that they can no longer inhibit the BH123 proteins. Some activated BH3-*only* proteins may stimulate mitochondrial protein release more directly by binding to and activcting the BH123 proteins.

3.24 Cancer

A normal cell undergoes regulated division, differentiation and apoptosis (programmed cell death). When normal cells have lost the usual control over their division, differentiation and apoptosis they become tumor cells. So, a tumor is the result of an abnormal proliferation of cells without differentiation and apoptosis. Tumor or neoplasm (any abnormal proliferation of cells) may be of two types: *Benign tumor* and *Malignant tumor*.

Benign and malignant tumor

In benign tumor, neoplastic cells remain clustered together in a single mass and cannot spread to other sites. It contains cells that closely resemble normal cells and that may function like normal cells.

Neoplastic cells that don't remain localized and encapsulated and becomes progressively invasive and malignant are described as *malignant tumors*. They invade surrounding normal tissues (called *invasiveness*) and spread throughout the body through circulatory or lymphatic systems (called *metastasis*). The term *cancer* refers specifically to malignant tumors.

Table 3.20 Comparison of benign and malignant tumours

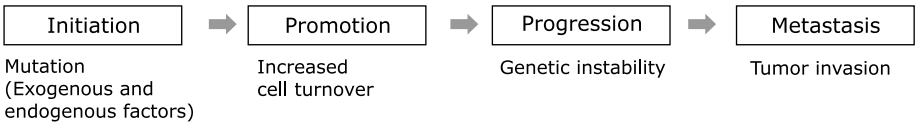
Characteristics	*Benign*	*Malignant*
Differentiation	Well differentiated	Lack differentiation
Rate of growth	Slow	Rapid
Invasiveness	Absent	Present
Metastasis	Absent	Present

Both benign and malignant tumors are classified according to the type of cell from which they arise. Most cancers fall into three main groups– *Carcinomas* (tumors that arise from endodermal or ectodermal tissues), *Sarcomas* (malignancies of mesodermal connective tissues) and *Leukemia/Lymphomas* (from blood forming tissues and from cells of the immune system).

Most cancers originate from single abnormal cell i.e. *monoclonal origin.* Cancers are probably initiated by changes in the cell's DNA sequence (*genetic changes*) or change in pattern of gene expression without a change in DNA sequences (*epigenetic changes*).

Most cancers are initiated by genetic changes and majority of them are caused by changes in somatic cells and therefore are not transmitted to the next generation. About 1% of all cancers is due to genetic changes in germinal cells and is therefore inherited. About 80% of these inherited cancers are dominant in nature.

The transition of a normal cell into a tumor cell is referred to as *transformation.* The transition from a normal to a transformed state is a multisteps process involving genetic/epigenetic changes and selection of cells with the progressively increasing capacity for proliferation, invasion and metastasis. Conceptually, this process can be divided into three distinct stages: *initiation, promotion* and *progression.* The first step in the process is *tumor initiation*. It is a process in which normal cells are changed so that they are able to form tumors. *Promotion* is generally associated with increased proliferation of initiated cells, which increases the population of initiated cells. *Progression* refers to the process of acquiring additional genetic changes that lead to malignancy and metastasis. Additional genetic changes in cancer critical genes are the force that drives tumorigenesis. Each successive genetic change is thought to provide the developing tumor cell with important growth advantages that allow cell clones to outgrow their more normal neighboring cells.



Properties of cancer cells

Cancer cells typically display several abnormal properties as compared to normal cells that provide a description of malignancy at the cellular level.

Density dependent inhibition

Normal cells in culture display *density dependent inhibition* of cell proliferation and proliferate until they reach a finite cell density, which is determined partly by availability of growth factors in culture medium. But the proliferation of cancer cells is not sensitive to density dependent inhibition.

Contact inhibition

Normal cells migrate across the surface of a culture dish until they make contact with a neighboring cell. Further cell migration is then inhibited and normal cells adhere to each other forming an orderly array of cells on the culture dish surface. This inhibition of growth after contact is called *contact inhibition*. Tumor cells in contrast continue moving after contact with their neighbouring cells migrating over adjacent cells and growing in disordered multilayered patterns.

Immortalization

Normal cells have a limited capacity to grow and divide both *in vivo* and *in vitro*. Even if provided with optimal growth conditions, *in vitro* normal cells will cease dividing after about 50 generations and then senesce and die. In contrast, malignant cells are immortal and can grow indefinitely.

Invasiveness and Metastasis

One of the most important characteristics of transformed cells is their *invasiveness.* Invasion refers to the direct migration and penetration by cancer cells into neighboring tissues. Metastasis refers to the ability of cancer cells to penetrate into lymphatic and blood vessels, circulate through the bloodstream, and then invade normal tissues elsewhere in the body.

There are two properties of cancer cells that play important roles in invasion and metastasis. *First*, malignant cells generally secrete proteases that digest extracellular matrix components, allowing the cancer cells to invade adjacent normal tissues. *Second*, cancer cells also secrete growth factors that promote the formation of new blood vessels (**angiogenesis**). Angiogenesis is needed to support the growth of a tumor by supplying oxygen and nutrients to the proliferating tumor cells.

Loss of anchorage dependence

Most normal cells must be attached to a rigid substratum (i.e. they must be anchored) in order to grow. Transformed cells can grow even when they are not attached to the substratum, as for example when they are suspended in a semisolid medium containing agar or methyl cellulose.

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Chapter 04

Prokaryotes and Viruses

4.1 General features of Prokaryotes

Prokaryotes (*pro* means before and *karyon* means kernel or nucleus) consist of *eubacteria* and the *archaea* (also termed as archaebacteria or archaeobacteria). The term eubacteria refer specifically to *bacteria*. *The informal name bacteria are occasionally used loosely in the literature to refer to all the prokaryotes, and care should be taken to interpret its meaning in any particular context.* Prokaryotes can be distinguished from eukaryotes in terms of their cell structure and molecular make-up. Prokaryotic cells have a simpler internal structure than eukaryotic cells. Although many structures are common to both cell types, some are unique to prokaryotes. Most prokaryotes lack extensive, complex, internal membrane systems. The major distinguishing characteristics of prokaryotes and eukaryotes are as follows:

Features of prokaryotic organisms

True membrane bound nucleus	–	Absent
DNA complexed with histone	–	Absent
Number of chromosomes	–	One (mostly)
Mitosis and meiosis	–	Absent
Genetic recombination	–	Partial (unidirectional transfer of DNA)
Sterol in plasma membrane	–	Absent (Except *Mycoplasma*)
Ribosome	–	70S
Unit membrane bound organelles	–	Absent
Cell wall	–	Present in *most* of prokaryotic cells. In eubacteria, it is made up of peptidoglycan.

Features of eukaryotic organisms

True membrane bound nucleus	–	Present
DNA complexed with histone	–	Present
Number of chromosomes	–	More than one
Mitosis and meiosis	–	Present
Genetic recombination	–	By crossing over during meiosis
Sterol in the plasma membrane	–	Present
Ribosome	–	80S (in cytosol) and 70S (in organelles)
Unit membrane bound organelles	–	Present
Cell wall	–	Made up of cellulose in plant and chitin in fungi. Absent in animal cells.

Prokaryotic cells show similarities with eukaryotic organelles like mitochondria and chloroplast. The endosymbiotic theory (Margulis, 1993) proposes that the mitochondria and chloroplasts of eukaryotic cells originated as symbiotic prokaryotic cells. The presence of circular, covalently closed DNA and 70S ribosomes in mitochondria and chloroplast support this theory.

Table 4.1 Similarities between prokaryotic cells and eukaryotic organelles

	Prokaryotic cells	*Eukaryotic organelles*
Nature of DNA	ds circular	ds circular
Histone protein	Absent	Absent
Ribosome type	70S	70S
Growth	Binary fission	Binary fission

4.2 Phylogenetic overview

Historically, prokaryotes were classified on the basis of their phenotypic characteristics. Prokaryotic taxonomy therefore involved measuring a large number of characteristics, including morphology and biochemical characteristics (e.g. ability to grow on different substrates, cell wall structure, antibiotic sensitivities, and many others). This contrasts with the classification of eukaryotic organisms, for which phylogenetic (evolution-based) classification was possible through the availability of fossil evidence.

A major revolution occurred with the realization that evolutionary relationships could be deduced on the basis of differences in gene sequence. The most important gene for prokaryote phylogeny is the 16S ribosomal RNA (rRNA) gene, which is present in all cells. The gene is approximately 1500 bp in length and possesses *signature sequences.* These sequences are conserved and found in the organisms of one taxonomic group but not in other groups.

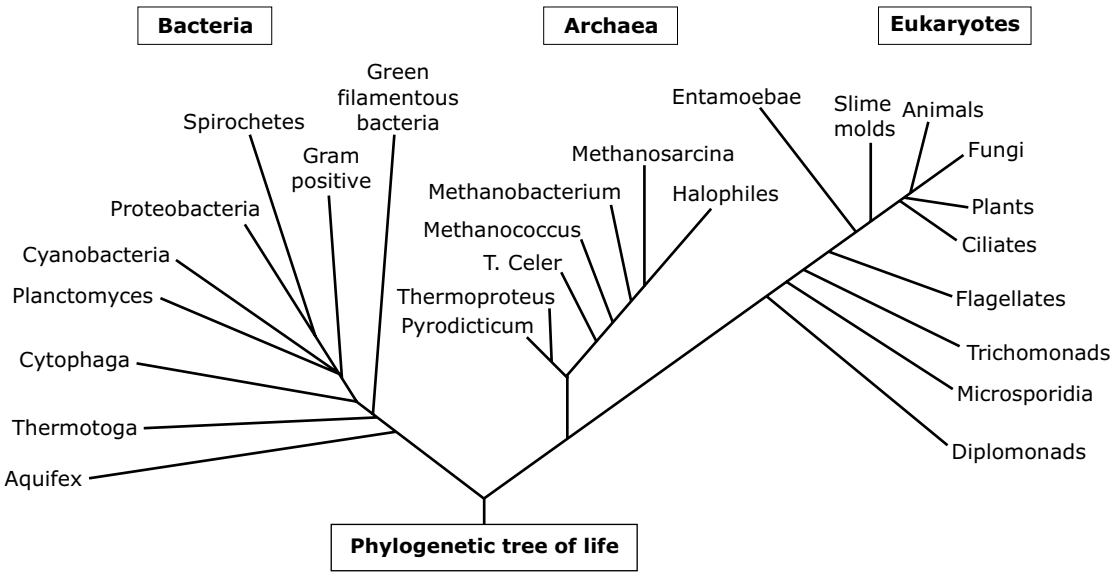


Figure 4.1 A phylogenetic tree of living things, based on RNA data (proposed by Carl Woese), showing the separation of bacteria, archaea, and eukaryotes from a common ancestor.

Based on ribosomal RNA signature sequences, Carl Woese proposed a radical reorganization of the five kingdoms into three domains. In his classification system, Woese placed all four eukaryotic kingdoms (protista, fungi, plantae, animalia) into a single domain called *Eukarya*, also known as the eukaryotes. He then split the former kingdom of Monera into the *Eubacteria* and the *Archaea* domains. Unlike Whittaker's five kingdom system, Woese's three domain system organizes biodiversity by evolutionary relationships.

4.3 Structure of bacterial cell

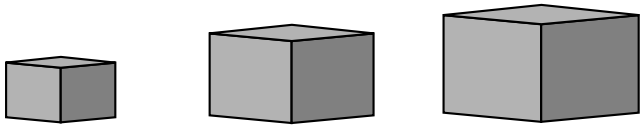
Bacteria (eubacteria) are microscopic, relatively simple, prokaryotic organisms whose cells lack a nucleus. Prokaryotes can be distinguished from eukaryotes in terms of their cell structure and molecular make-up. Prokaryotic cells are

morphologically simpler than eukaryotic cells. Although many structures are common to both cell types, some are unique to prokaryotes. Most prokaryotes lack extensive, complex, internal membrane systems.

Size, shape and arrangement of bacterial cells

Bacteria range in size from 0.2—2 micrometres in diameter, and 0.5—5.0 micrometres in length. However, a few species — for example, *Thiomargarita namibiensis* and *Epulopiscium fishelsoni* — are very large and visible to the unaided eye. The rod shaped *E. fishelsoni* is about 600 micrometres in length and 75 micrometres in diameter. The smallest known bacteria are members of the genus *Mycoplasma* (about 0.3 micrometres in diameter).

Most bacteria are very small in size. Because of their small size, bacteria have a large surface area-to-volume ratio. The large surface area-to-volume ratio of bacteria also means that no internal part of the cell is very far from the surface and the nutrients can easily and quickly reach all parts of the cell. It also allows rapid uptake and intracellular distribution of nutrients and excretion of wastes. At low surface area-to-volume ratio, the diffusion of nutrients and waste products across the cell membrane limit the rate at which microbial metabolism occurs, making the cell less evolutionarily fit.



	1 mm cube	2 mm cube	4 mm cube
Surface area	6 mm^2	24 mm^2	96 mm^2
Volume	1 mm^3	8 mm^3	64 mm^3
Surface area to volume ratio	6/1	3/1	1.5/1

Cell shape is generally characteristic of a given bacterial species. The four basic shapes of bacteria are *bacillus* (rod like), *coccus* (spherical or ovoid), *vibrio* (comma shaped) and *spirilla* (spiral or helical shaped). The term 'Bacillus' has two meanings. Bacillus refers to a bacterial shape as well as genus. When capitalized and italicized, it refers to a specific genus. For example, *Bacillus anthracis,* the causative agent of anthrax.

When cocci divide, cells can remain attached to one another. Cocci that remain in pairs after dividing are called *diplococci*; those that divide and remain attached in chain like patterns are called *streptococci*. Some cocci divide in two planes and remain in groups of four, forming squares are known as *tetrads*. Those that divide in three planes and remain attached in cubelike groups of eight are called *sarcinae*. Those that divide in multiple planes and form grapelike clusters are called *staphylococci*. Most bacilli appear as single rods. *Diplobacilli* appear in pairs after division, and *streptobacilli* occur in chains. Bacteria that look like curved rods are called *vibrios*. Spirilla have a helical shape and fairly rigid bodies. Another group of spirals are helical and flexible; they are called *spirochetes.*

Most bacteria maintain a single shape (*monomorphic*). Some bacteria like *Corynebacterium,* have many shapes, and lack a single, characteristic shape. These bacteria are called *pleomorphic*. *The modern definition of pleomorphism in the context of bacteriology is based on variation of shape of the cell, rather than a change of shape.*

Staining

Staining is a biochemical technique of colouring specimens. Dyes are used to stain specimens. Generally, dyes are organic compounds consisting of two functional chemical groups – one group gives dyes its characteristic colour; while the another group contains an ionizable chemical structure, which helps to solubilize the dye and facilitates its binding to different structures. Previously, dyes were classified as acidic or basic, depending upon whether the dye was negatively or positively charged at neutral pH. More accurately, dyes can be referred to as anionic or cationic. Cationic or *basic* dyes (such as crystal violet, methylene blue, safranin, malachite green) will react with groups that have a negative charge. Anionic or *acidic* dyes (such as eosin, acid fuchsin) will react with groups that have a positive charge.

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Table 4.5 Difference between gram-negative and gram-positive bacteria

Property	*Gram-negative bacteria*	*Gram-positive bacteria*
Cell wall	2 to 7 nm thick	20 to 80 nm thick
Teichoic acids	Absent	present
Periplasmic space	Prominent	Negligible
Outer membrane	Present	Absent
Motility	Motile or non-motile.	Mostly non-motile
Appendages	Usually bears appendages like pili, fimbriae	Usually lack appendages
Flagellar structure	4 rings in basal body	2 rings in basal body
Toxin produced	Endotoxins and exotoxins	exotoxins
Endospores	Cannot form endospores	Some genera form endospores

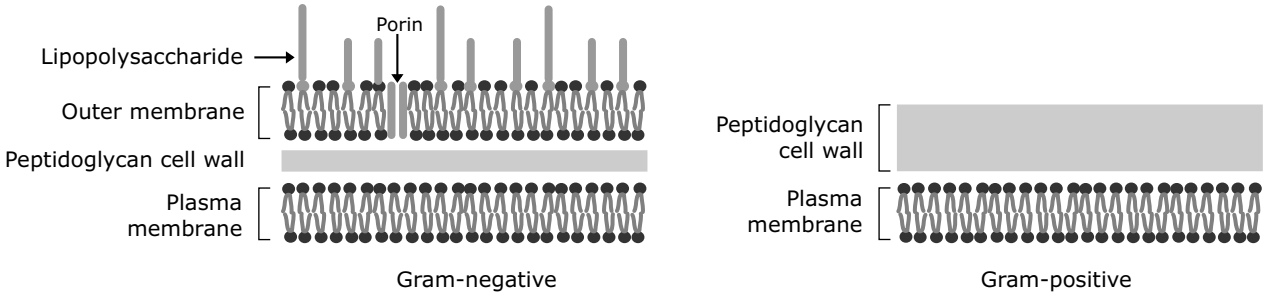


Figure 4.13 A comparison of bacterial cell wall.

4.4 Bacterial genome : Bacterial chromosome and plasmid

Nucleoid

The nucleoid is the region in a bacterial cell that contains the bacterial chromosome. It lacks a limiting membrane. The bacterial chromosome is a circular dsDNA molecule. It does not contain basic histone proteins, but is associated with low-molecular-weight polyamines. The bacterial chromosome is organized into independently supercoiled loops called *domains*. The domain organization enables the chromosomal DNA to undergo structural changes during different cellular processes (replication, transcription, and segregation) that take place simultaneously in a bacterial cell. Nucleoid-associated proteins such as HU (Heat-Unstable nucleoid protein), H-NS (Histone-like Nucleoid Structuring), IHF (Integration Host Factor) and SMC (Structural Maintenance of Chromosomes) facilitate compaction of chromosomal DNA by bending, bridging, and wrapping.

In addition, SMC protein is also involved in chromosome segregation. The primary role of the H-NS protein is to maintain the structure of the bacterial nucleoid. H-NS binds to DNA and then the H-NS proteins bind to each other, forming aggregates of four or more H-NS units, thus helping the condensation of the DNA into the nucleoid. HU protein and IHF are examples of proteins that are involved in bending DNA (as opposed to H-NS, which binds to DNA already bent as a result of its sequence). They help in integration, inversion and recombination events by bending DNA.

Most bacteria, like *E. coli*, contain a single circular DNA molecule as a chromosome. The recent studies have indicated that some bacteria contain as many as four linear or circular chromosomes. For example, *Vibrio cholerae*, the bacterium that causes cholera, contains two circular chromosomes. One of these chromosomes contains the genes involved in metabolism and virulence, while the other contains the remaining essential genes. An even more extreme example is provided by *Borrelia burgdorferi*, the bacterium that causes Lyme disease. It contains up to 11 copies of a single linear chromosome. Unlike *E. coli*, *Borrelia* cannot supercoil its linear chromosomes into a tight ball within the nucleoid; rather, these strands are diffused throughout the cell. The DNA molecules of Archaea share

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are very rare and reported in few plants, fungi and even animals. Bacterial plasmids are usually much smaller than the bacterial chromosome, varying from less than 5 kbp to more than several hundred kbp, though plasmids as large as 2 Mbp also occur in some bacteria. They usually encode traits that are not essential for bacterial viability, and replicate independently of the chromosome. Most plasmids are circular, negatively supercoiled, dsDNA molecules, but linear plasmids have also been reported in genera *Borrelia* and *Streptomyces*.

Types of plasmid

Conjugative, mobilizable and self-transmissible plasmid

Conjugative plasmids are large plasmids (>40 kbp) that carry genes for the formation of specific donor recipient pairs. These plasmids have genes for the synthesis of pili that help in the process of conjugation.

A plasmid that is not self-transmissible due to the absence of *tra* genes but can be transferred by a self-transmissible plasmid is called *mobilizable plasmid*. The process by which it is transferred is called *mobilization*. It relies on the product of *tra* genes specified by a self-transmissible plasmid to achieve the transfer. A mobilizable plasmid must contain an oriT site.

A *self-transmissible* plasmid encodes all the proteins that is required for transfer between donor and recipient. It is both conjugative and mobilizable. A plasmid that lacks genes necessary for effective contact and DNA transfer is termed *non-transmissible* plasmid. Most naturally occurring plasmids are either self-transmissible or mobilizable.

F-plasmid

The *F-plasmid* is an example of self-transmissible plasmid. Bacteria that possess a copy of F-plasmid are called F-positive or F-plus (denoted F^+). Cells that lack F plasmids are called F-negative or F-minus (F^-). The F-plasmid is also an *episome.* A plasmid that can exist autonomously in a cell or can integrate itself into the bacterial chromosome is termed as an episome. A bacterial cell that contains an F-plasmid integrated with the bacterial chromosome is referred to as an **Hfr** (High frequency recombination) cell. Integration involves homologous recombination between two covalently closed circular DNA molecules forming one circular molecule containing both of the original DNA structures. It is thought that the insertion sequences (IS sequence) present in the F-plasmid and those in the host chromosome serve as regions of homology for the insertional event.

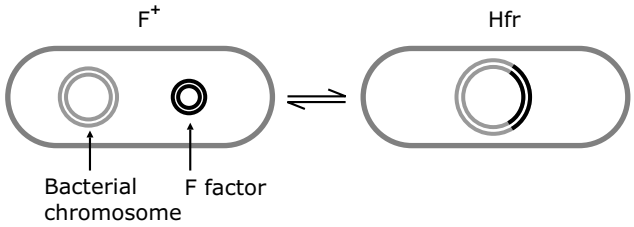


Figure 4.14 The F-plasmid is an episome and can integrate into the bacterial chromosome at several different locations by recombination between homologous insertion sequences present on both the plasmid and host chromosomes. Such a donor bacterial cell is called an Hfr strain (for high frequency of recombination) because it exhibits a very high efficiency of chromosomal gene transfer in comparison with F-cells.

R-plasmid (*resistance plasmid*)

R-plasmids make the host cell resistant to one or more antibiotics. They typically have genes that code for enzymes capable of destroying or modifying antibiotics. Some R-plasmids have only a single antibiotic resistance gene, whereas others have many. Most R-plasmids are mobilizable plasmids.

Col (*colicinogenic*) *plasmids*

Col plasmids synthesize proteins termed as *bacteriocin* that prevent the growth of susceptible bacterial strains that do not contain a Col plasmid. Bacteriocins are proteins that inhibit the growth of other closely related bacterial

Pages 388 to 401 are not shown in this preview.

a DNA fragment that is too large. The co-transduction can also be used to map genes present in the bacterial chromosome. Mapping of genes by the process of co-transduction is based on the fact that the closer the genes to each other, greater the probability of their co-transduction. That is, the frequency of co-transduction is inversely proportional to the distance between two genes. For example, consider three genes a, b and c for which the frequencies of co-transduction determined by two factor crosses are: a-b, 90%; a-c, 33%; and b-c, 42%. The values indicate that the b is closer to a than to c. A mathematical formula (proposed by Wu) is used to relate co-transduction frequencies to map distance:

$$C = (1-d/L)^3$$

Where, C is the co-transduction frequency, d is the distance in minutes between two genes and L is the size of the transducing fragment in minutes.

4.6.3 Conjugation

In 1946, Joshua Lederberg and Edward Tatum showed that bacteria undergo conjugation, a process in which the genetic information from one bacteria is transferred to, and recombined with, that of other bacteria. In conjugation, direct contact between the donor and recipient bacteria leads to the establishment of a cytoplasmic bridge between them and transfer of a part or all of the donor genome to the recipient takes place. Donor ability is determined by the presence of self transmissible /conjugative plasmids called fertility plasmids or sex plasmids. Self transmissible plasmids exist in both gram positive and gram negative bacteria. A recipient cell that has received DNA as a result of conjugation is called *transconjugant.* Transfer of plasmids is generally intraspecific; however, many plasmids have transfer system that enable them to transfer DNA between unrelated species. These are known as *promiscuous plasmids.*

$F^+ - F^-$ conjugation

The F-plasmid (also called F-factor) of *E. coli* is the prototype for fertility plasmids in Gram-negative bacteria. Strains of *E. coli* with an extrachromosomal F plasmid are called F^+ and function as *donors*, whereas strains that lack the F-plasmid are F^- and behave as *recipients*. The conjugative functions of the F-plasmid are specified by a cluster of at least 25 transfer (*tra*) genes which determine expression of sex pili, synthesis and transfer of DNA during mating, interference with the ability of F^+ bacteria to serve as recipients, and other functions. The *tra* genes can be divided into two groups: those whose products are involved in the mating pair formation (*Mpf*) and those whose products are involved in processing the plasmid DNA for transfer (*Dtr*). The Mpf component includes a sex pilus that extrudes from the cell and holds mating cells together. The Mpf system also includes the channel in the membrane through which DNA and protein pass.

Each F^+ bacterium has 1 to 3 sex pili that binds to a specific outer membrane protein on recipient bacteria to initiate mating. An intercellular cytoplasmic bridge is formed, and one strand of the F-plasmid DNA is transferred from donor to recipient, beginning at a unique origin and progressing in the 5′ to 3′ direction. The site on the plasmid DNA at which transfer initiates is called the origin of transfer (*ori*T). *Relaxase* protein makes a single strand cut at the *ori* T site in the plasmid. The transferred strand is converted to circular double-stranded F-plasmid DNA in the recipient bacterium. Both of the exconjugant bacteria are F^+, and the F-plasmid can therefore spread by infection among genetically compatible populations of bacteria.

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F′ – F⁻ conjugation

Integrated F-plasmids in Hfr strains can sometimes be excised from the bacterial chromosome. If excision precisely reverses the integration process, F^+ cells are produced. On rare occasions, however, excision occurs by recombinations involving insertion sequences or other genes on the bacterial chromosome that are located at some distance from the original integration site. In such cases segments of the bacterial chromosome can become incorporated into hybrid F plasmids that are called **F′ plasmids**.

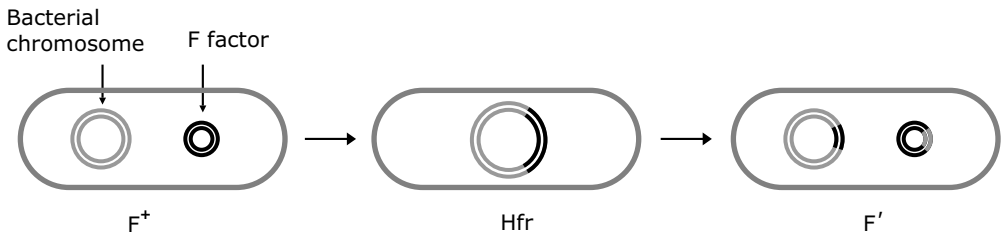


Figure 4.25 The F-factor is an episome and can integrate into the bacterial chromosome. However, this integration is reversible. It can leave the bacterial chromosome. When the F-factor leaves an Hfr chromosome, it occasionally picks up some bacterial genes to become an F′-plasmid.

F′– F⁻ conjugation is virtually identical with F^+ – F⁻ conjugation. Bacterial genes on the F′ plasmid are transferred with it and need not be incorporated into the recipient chromosome to be expressed. The recipient becomes F′ and is a partially diploid merozygote since it has two sets of the genes carried by the plasmid. Such transfer of bacterial genes is often called *sexduction.*

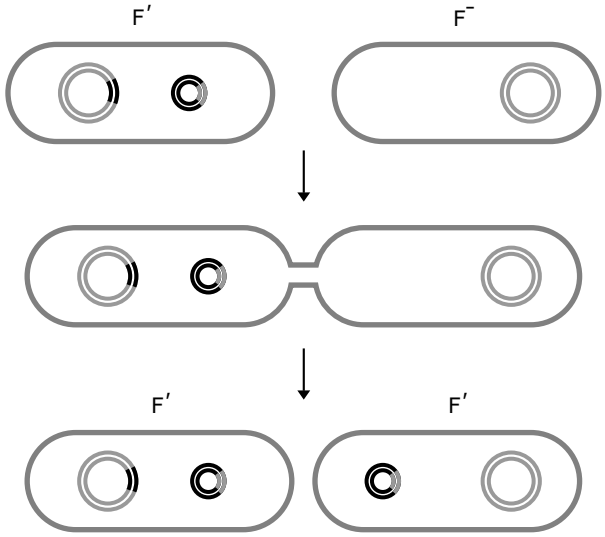


Figure 4.26 F′ conjugation. Due to an error in excision, the chromosomal gene of an Hfr cell is picked up by the F-factor. This gene is then transferred to a recipient during conjugation.

Interrupted mating experiment

Conjugation between different Hfr and F⁻ strains can be used to map the relative positions of genes in the bacterial chromosome. The mapping procedure is based on the fact that genes closer to the origin of transfer will be transferred at a higher frequency than genes farther away from the origin.

In an *interrupted mating experiment*, E. Wollman and F. Jacob mixed Hfr and F⁻ strains with suitable marker gene and assessed the recombination of specific genes at different time. To accomplish this, a culture containing a mixture of Hfr and F⁻ strain was incubated, samples were removed at various intervals and each was placed in an

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Chapter 05

Immunology

Immunology is the science that is concerned with immune response to foreign challenges. **Immunity** (derived from Latin term *immunis*, meaning *exempt*), is the ability of an organism to resist infections by pathogens or state of protection against foreign organisms or substances. The array of cells, tissues and organs which carry out this activity constitute the **immune system**. Immunity is typically divided into two categories—*innate* and *adaptive* immunity.

5.1 Innate immunity

Innate (native/natural) immunity is present since birth and consists of many factors that are relatively nonspecific—that is, it operates against almost any foreign molecules and pathogens. It provides the *first line of defense* against pathogens. It is not specific to any one pathogen but rather acts against all foreign molecules and pathogens. It also does not rely on previous exposure to a pathogen and response is functional since birth and has no memory.

Elements of innate immunity

Physical barriers

Physical barriers are the *first line of defense* against microorganisms. It includes skin and mucous membrane. Most organisms and foreign substances cannot penetrate intact skin but can enter the body if the skin is damaged. Secondly, the acidic pH of sweat and sebaceous secretions and the presence of various fatty acids and hydrolytic enzymes like lysozyme inhibit the growth of most microorganisms. Similarly, respiratory and gastrointestinal tracts are lined by mucous membranes. Mucus membranes entrap foreign microorganisms. The respiratory tract is also covered by cilia, which are hair like projections of the epithelial-cell membranes. The synchronous movement of the cilia propels mucus-entrapped microorganisms out of these tracts. Similarly, the conjunctiva is a specialized, mucus-secreting epithelial membrane that lines the interior surface of each eyelid. It is kept moist by the continuous flushing action of tears (lacrimal fluid) from the lacrimal glands. Tears contain *lysozyme, lactoferrin, IgA* and thus provide chemical as well as physical protection.

Microorganisms do occasionally breach the epithelial barricades. It is then up to the innate and adaptive immune systems to recognize and destroy them, without harming the host. In case of innate immune response several antimicrobial chemicals and phagocytic cells provide protection against pathogens.

Chemical mediator

A variety of chemicals mediate protection against microbes during the period before adaptive immunity develops. The molecules of the innate immune system include complement proteins, cytokines, pattern recognition molecules, acute-phase proteins, cationic peptides, enzyme like lysozyme and many others.

Complement proteins

The complement proteins are soluble proteins/glycoproteins that are mainly synthesized by liver and circulate in the blood and extracellular fluid. They were originally identified by their ability to amplify and complement the

action of antibodies; hence, the name complement. It also bridges innate and adaptive immunity and removes immune complexes. The complement system is composed of over 30 serum proteins. Activation of complement proteins in response to certain microorganisms results in a controlled enzymatic cascade, which targets the membrane of pathogenic organisms and leads to their destruction.

Cytokines

The term *cytokine* is a generic term for any low molecular weight soluble protein or glycoprotein released by one cell population which acts as an intercellular mediator. It includes *monokines, lymphokines, interleukins, interferons* and others. Cytokines are required for immunoregulation of both innate as well as adaptive immune responses.

Interferons are cytokines made by cells in response to virus infection, which essentially induce a generalized antiviral state in surrounding cells.

Chemokines are small, positively charged secreted proteins that have a central role in guiding the migrations of various types of white blood cells. They bind to the surface of endothelial cells, and to negatively charged proteoglycans of the extracellular matrix in organs. By binding to G-protein-linked receptors on the surface of specific blood cells, chemokines attract these cells from the bloodstream into an organ, guide them to specific locations within the organ, and then help stop migration.

Pattern recognition molecule

Many molecules involved in innate immunity have the ability to recognize a given class of molecules i.e. recognize pattern. Patterns are conserved structures and invariant among microorganisms of a given class. Pattern recognition molecules that recognize *P*athogen-*A*ssociated *M*olecular *P*attern (PAMP) may be soluble, circulating proteins or cell surface receptors. Many PAMPs are recognized by pattern recognition molecules present on the surface of phagocytic cells. Mannose-binding lectin (MBL) and C-reactive protein (CRP) are soluble pattern recognition molecules that bind to microbial surfaces and promote their opsonization. Toll-Like Receptors (TLRs) are a class of pattern recognition molecules that function exclusively as signaling receptors. It was originally identified as a protein involved in the establishment of dorso-ventral polarity in developing fly embryos. It is also involved, however, in the adult fly's resistance to fungal infections. There are at least 10 distinct TLRs in humans, which recognize lipopolysaccharide, peptidoglycan, zymosan and CpG DNA. For example, TLR-4 signals the presence of bacterial lipopolysaccharide (LPS) and heat-shock proteins. TLR-2 signals the presence of bacterial lipoproteins and peptidoglycans. The TLR family proteins consist of extracellular leucine rich repeat (LRR) motifs and a cytoplasmic tail containing a Toll/IL-1 receptor homology (TIR) domain. The LRR motifs are responsible for ligand recognition and the TIR domain is essential for triggering intracellular signaling pathways.

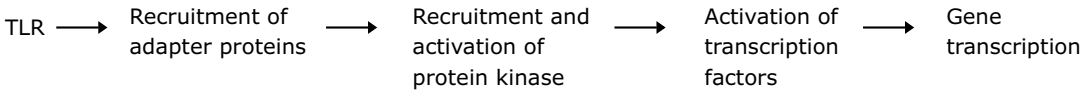


Figure 5.1 TLR and basic signaling mechanisms.

Acute phase proteins are a heterogeneous group of plasma proteins mainly produced in the liver as the result of a microbial stimulus. They include C-reactive protein (CRP), serum amyloid protein A (SAA) and mannose binding protein (MBP). Cytokines (IL-1, IL-6, IL-8, etc.) released by macrophages upon activation by bacteria stimulate the liver to rapidly produce acute-phase proteins. These proteins maximize activation of the complement system and opsonization of invading microbes.

Cellular defenses

Many specialized cell types like *neutrophils, macrophages, monocytes, natural killer cells* participate in innate host defense mechanisms. Once a pathogen evades the physical and chemical barriers, these specialized cells play an important role in protection. Phagocytosis is a fundamental protective mechanism carried out by these cell types,

neutrophils, macrophages, monocytes and dendritic cells. In vertebrates, macrophages reside in tissues throughout the body. Macrophages are long lived cells, which patrol the tissues of the body. The second major type of phagocytic cells in vertebrates, the neutrophils, are short lived cells which are abundant in blood but are not present in normal healthy tissues. *Phagocytosis* is the ingestion of invading foreign particles, such as bacteria by individual cell. Phagocytosis may be enhanced by a variety of factors collectively referred to as **opsonins** (Greek word meaning 'prepared food for'), which consist of antibodies and various serum components of complement. The process by which particulate antigens are rendered more susceptible to phagocytosis is called **opsonization**. After ingestion, the foreign particle is entrapped in a phagocytic vacuole (phagosome), which fuses with lysosomes forming the *phagolysosome*. The antimicrobial and cytotoxic substances present within the lysosome destroy the phagocytosed microorganisms in the following ways:

Oxygen dependent killing mechanisms

During phagocytosis, a metabolic process known as the *respiratory burst* occurs in activated phagocytes. It results in a transient increase in oxygen consumption by cell. Activated phagocytes generate a number of toxic products such as reactive oxygen intermediates (such as hydroxyl radicals, hypochlorite anion, superoxide anions, hydrogen peroxide) and reactive nitrogen intermediates (like NO, NO_2, $HNO_2^{\bullet}$) which have potent antimicrobial activity.

Oxygen independent killing mechanisms

Activated macrophages also synthesize *lysozyme*, *defensins* (cysteine rich cationic peptides containing 29–35 amino acid residues) and various hydrolytic enzymes/cytotoxic peptides whose degradative activities do not require oxygen.

Inflammatory barriers

Inflammation is an important nonspecific defense reaction to cell injury. The hallmark signs of inflammation are pain, redness (erythema), swelling (edema) and heat. Each of these is the result of specific changes in the local blood vessels. Erythema is caused by *increased vascular diameter*, which leads to *increased blood flow*, thereby causing heat and redness in the area. The blood vessels become permeable to fluid and proteins, leading to local swelling and an accumulation of blood proteins that aid in defense. At the same time, the endothelial cells lining the local blood vessels are stimulated to express cell adhesion proteins that facilitate the attachment and *extravasion* (movement of blood cells through the vessel wall into the surrounding tissue) of white blood cells, including neutrophils, lymphocytes, and monocytes.

The inflammatory response is mediated by a variety of signaling molecules. Activated macrophages produce chemoattractants (known as **chemokines**). Some of these attract neutrophils, which are the *first* cells recruited in large numbers to the site of the new infection. Others later attract monocytes and dendritic cells. The dendritic cells pick up antigens from the invading pathogens and carry them to nearby lymph nodes, where they present the antigens to lymphocytes to marshal the forces of the adaptive immune system. Two principal mediators of the inflammatory response are **histamine** (released by a variety of cells in response to tissue injury) and **kinins** (present in blood plasma in an inactive form). Both cause vasodilation and increased permeability of capillaries. Kinins are also very potent nerve stimulators and are the molecules most responsible for pain associated with inflammation.

5.2 Adaptive immunity

Adaptive immunity, also known as *specific* or *acquired* immunity, is capable of recognizing and selectively eliminating specific foreign antigens. It does not come into play until there is an antigenic challenge to the organism. Adaptive immunity displays four characteristic features:

1. *Antigenic specificity* : It is the ability to discriminate among different epitopes/antigens.
2. *Immunologic memory* : It is the ability to recall previous contact with a foreign molecule and respond to it in a learned manner-that is, with a more rapid and larger response.

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5.5 Antigens

Adaptive immune responses arise as a result of exposure to foreign compounds. The compound that evokes the response is referred to as **antigen**, a term initially coined due to the ability of these compounds to cause *anti*body responses to be *gene*rated. An antigen is any agent capable of binding specifically to T-cell receptor (TCR) or an antibody molecule (membrane bound or soluble). The ability of a compound to bind with an antibody or a TCR is referred to as **antigenicity**. There is a functional distinction between the term antigen and immunogen. An **immunogen** is any agent capable of inducing an immune response and is therefore **immunogenic**. The distinction between the terms is necessary because there are many compounds that are incapable of inducing an immune response, yet they are capable of binding with components of the immune system that have been induced specifically against them. Thus all immunogens are antigens, but not all antigens are immunogens.

Requirements for immunogenicity

A substance must possess the following characteristics to be immunogenic:

1. *Foreignness*

   The most important feature of an immunogen is that an effective immunogen must be *foreign* with respect to the host. The adaptive immune system recognizes and eliminates only foreign (nonself) antigens. Self antigens are not recognized and thus individuals are *tolerant* to their own self molecules, even though these same molecules have the capacity to act as immunogens in other individuals of the same species.

2. *Size*

   The second requirement for being immunogenic is that the compound must have a certain minimal molecular weight. There is a relationship between the size of immunogen and its immunogenicity. In general, small compounds with a molecular weight <1000 Da (e.g. penicillin, aspirin) are not immunogenic; those of molecular weight between 1000 and 6000 Da (e.g. insulin, adrenocorticotropic hormone) may or may not be immunogenic; and those of molecular weight >6000 Da (e.g. albumin, tetanus toxin) are generally immunogenic. The most active immunogens tend to have a molecular mass of 100,000 Da or more. In short relatively small substances have decreased immunogenicity, whereas large substances have increased immunogenicity.

3. *Chemical complexity*

   The third characteristic necessary for a compound to be immunogenic is a certain degree of chemical complexity. For example, homopolymers of amino acids or sugars are seldom good immunogens regardless of their size. Similarly, a homopolymer of poly-γ-D-glutamic acid (the capsular material of *Bacillus anthracis*) with a molecular weight of 50,000 Da is not immunogenic. The absence of immunogenicity is because these compounds, although of high molecular weight, are not sufficiently chemically complex.

   Virtually all proteins are immunogenic. Furthermore, the greater the degree of complexity of the protein, the more vigorous will be the immune response to that protein. Carbohydrates are immunogenic only if they have a complex polysaccharide structure or part of complex molecules such as glycoproteins. Nucleic acids and lipids are poor immunogens by themselves, but they become immunogenic when they are conjugated to protein carriers.

4. *Dosage and route of administration*

   The insufficient dose of immunogen may not stimulate an immune response either because the amount administered fails to activate enough lymphocytes or because such a dose renders the responding cells unresponsive. Besides the need to administer a threshold amount of immunogen to induce an immune response, the number of doses administered also affects the outcome of the immune response generated.

   The route of administration also affects the outcome of the immunization because this determines which organs and cell populations will be involved in the response. Immunogens can be administered through a number of common routes: *Intravenous* (into a vein); *intradermal* (into the skin); *subcutaneous* (beneath the skin);

intramuscular (into the muscle). Antigens administered via the most common route namely, *subcutaneously*, generally elicit the strongest immune responses. This is due to their uptake, processing, and presentation to effector cells by Langerhans cells present in the skin, which are among the most potent APCs. Antigens administered subcutaneously moves first to local lymph nodes. *Intravenously* administered antigens are carried first to the spleen, where they can either induce immune unresponsiveness or tolerance, or if presented by APCs, generate an immune response. Orally administered antigens (*gastrointestinal route*) elicit local antibody responses within the intestinal lamina propria.

Haptens

Substances called **haptens** (from the Greek *hapten*, which means *to grasp*) fail to induce immune responses in their native form because of their low molecular weight and their chemical simplicity. Haptens are antigenic but not immunogenic. These compounds become immunogenic when they are conjugated to high molecular weight, physiochemically complex **carriers**. Thus, a hapten is a compound that, by itself, is incapable of inducing an immune response. However, when hapten is conjugated to a carrier, an immune response can be induced against it.

Antigen-antibody interactions

Ag-Ab interaction is highly specific and occurs in a similar way as a bimolecular association of an enzyme-substrate. The binding between antigens and immune components involves weak non-covalent interactions. The binding forces are relatively weak and *reversible* and consist mainly of van der Waals forces, electrostatic forces, and hydrophobic forces, all of which require a very close proximity between the interacting moieties. The smallest unit of antigen that is capable of binding with antibodies is called an **antigenic determinant** (or epitope). The corresponding area on the Ab molecule combining with the epitope is called **paratope**. The number of epitopes on the surface of an antigen is its *valence*. The valence determines the number of antibody molecules that can combine with the antigen at one time. If one epitope is present, the antigen is **monovalent**. Most antigens, however, have more than one copy of the same epitope and are termed **polyvalent**.

Affinity and avidity

The intrinsic association constant that characterizes the noncovalent interaction between single antigen binding site of an antibody (paratope) with an epitope is termed **affinity**. Low-affinity antibodies bind antigen weakly and tend to dissociate readily, whereas high-affinity antibodies bind antigen more tightly and remain bound longer.

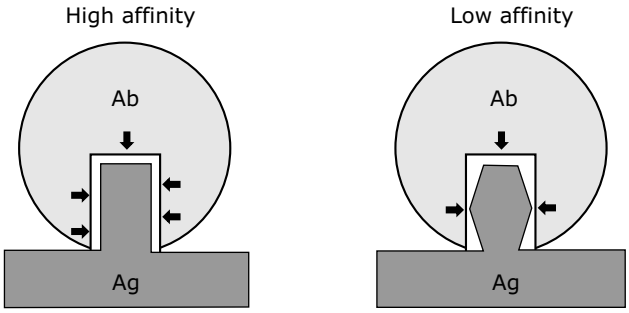


Figure 5.4 Affinity is the strength with which a monovalent ligand interacts with a binding site.

When the antigen consists of many repeating identical epitopes or when antigens are multivalent, the association between the entire antigen molecules and antibodies depends not only on the affinity between each epitope and its corresponding antibody but also on the sum of the affinities of all the epitopes involved. The term **avidity** is used to denote the overall binding between antibodies and a multivalent antigen. So, when complex Ag having multiple repeating epitopes, are mixed with Ab having multiple binding sites, the interaction of such type between multivalent Ab and Ag is called, the avidity. Thus, in general, IgM antibodies with 10 antigen binding sites are of higher avidity than IgG antibodies, although the binding of each Fab in the IgM antibody with ligand may be of the same affinity as that of the Fab from IgG.

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Physical form of the antigen

The physical form of the antigen influences the reaction with an antibody. When the antigen is particulate, the reaction of an antibody with the antigen can be detected by agglutination (clumping) of the antigen. If the antigen is a soluble, one generally looks for the precipitation of the antigen after the production of large insoluble antigen-antibody complexes.

Adjuvant

An adjuvant (from Latin word *adjuvare* means *to help*) is a substance that when mixed with an immunogen and injected with it, enhances the immune response against the immunogen. Adjuvants are often used to boost the immune response in case of weak immunogens or when antigens are available in small amount. A carrier for a hapten and an adjuvant is different. A hapten will become immunogenic when conjugated covalently to a carrier; it will not become immunogenic if mixed with an adjuvant. Thus an adjuvant enhances the immune response to immunogens but does not confer immunogenicity on haptens. How adjuvants augment the immune response has remained enigmatic.

While many adjuvants have been developed in animal models and tested experimentally in humans, only one has been used for routine vaccination. Currently, aluminum potassium sulfate (alum) is a common adjuvant used for human vaccines. Alum prolongs the persistence of antigen. When an antigen is mixed with alum, the salt precipitates the antigen. Another commonly used adjuvant is *Freund's complete* and *incomplete adjuvant* . Freund's incomplete adjuvant contains antigen in aqueous solution, oil, and an emulsifying agent such as mannide monooleate. Freund's *complete adjuvant* contains heat-killed *Mycobacteria* as an additional ingredient.

Comparison of antigen recognition by T-cells and B-cells

The recognition of antigens by T-cells and B-cells is fundamentally different. B-cells recognize soluble antigens whereas most T-cells recognize only peptides combined with MHC molecules.

Table 5.3 Antigen recognition by B and T-cells

Characteristics	*B-cells*	*T-cells*
Antigen interaction	B-cell receptor (membrane Ig) binds Ag	T-cell receptor binds Ag and MHC
Nature of antigens	Protein, polysaccharide, glycolipid	Peptide
Binding soluble antigens	Yes	No
Epitopes recognized	Accessible areas of protein structure containing sequential amino acids and non-sequential amino acids	Antigens processed internally and presented as linear peptides bound to MHC molecules
MHC molecules	Not required	Required to display processed antigen

Adapted and modified from Immunology, Goldsby et. al, Freeman Publication.

T-dependent and T-independent antigens

T-dependent or *thymus dependent* (TD) *antigens* are those that require direct contact with T_H cells, not simply exposure to T_H -derived cytokines to stimulate B-cells for the production of antibodies. Proteins are T-dependent antigens. *T-independent* or *thymus independent* (TI) *antigens* are antigens that can activate B cells in the absence of any direct contact with T_H cells. Examples of TI antigens are lipopolysaccharide (LPS) of bacterial outer membrane, bacterial flagellin and cell-wall and capsular polysaccharides.

Unlike the T-dependent antigens, the T-independent antigens:

- Do not produce isotype switching (IgM is almost exclusively produced).
- Do not demonstrate affinity maturation (in which antibodies of progressively higher affinity are produced).
- Do not show immunological memory hence secondary responses.

Pages 457 to 462 are not shown in this preview.

5.7 Immunoglobulins : Structure and function

Antibodies, the antigen-binding *glycoproteins* are synthesized exclusively by B-cells and in billions of forms, each with a different amino acid sequence and a different antigen binding site. Collectively called *immunoglobulins* (Ig), they are among the most abundant protein components in the blood, constituting about 20% of the total protein components in the blood plasma. Antibodies (Ab) are present on the B-cell membrane and also are secreted by plasma cells.

5.7.1 Basic structure of antibody molecule

The simplest antibodies are Y-shaped molecules with two identical antigen-binding sites, one at the tip of each arm of the Y. Because of their two antigen-binding sites, they are described as *bivalent*.

Antibody has a common structure of 4 polypeptide chains. It is a heterodimer and consists of two identical **light (L) chains** (each containing about 220 amino acids residues, about 25000 MW) and two identical **heavy (H) chains** (each usually containing about 440 amino acids residues, about 50,000 MW). Each light chain is bound to heavy chain by disulfide bridges and other non-covalent linkages. Thus, antibody is a dimer of H—L chain.

All species studied have the two major classes of light chains: κ and λ. Any one individual of a species produces both types of light chain. However, in any one immunoglobulin molecule, the light chains are always either both κ or both λ, never one of each. While there are two types of light chains, the immunoglobulins of virtually all species have been shown to consist of five different types of heavy chains- α, γ, δ, ε and μ. These five different types of heavy chains are called *isotypes*. The heavy-chains of a given antibody molecule determine the class of that antibody: IgM (μ), IgG (γ), IgA (α), IgD (δ) or IgE (ε). Each class can have either κ or λ light chains. Any individual of a species makes all heavy chains, but in any one antibody molecule, both heavy chains are identical. Thus an antibody molecule of the IgG class could have the structure $\kappa 2\gamma 2$ with two identical κ light chains and two identical γ heavy chains. Alternatively, it could have the structure $\lambda 2\gamma 2$ with two identical λ light chains and two identical γ heavy chains.

Immunoglobulin heavy chain isotypes

Isotype	*Heavy chain*
IgM	μ
IgD	δ
IgG	γ
IgA	α
IgE	ε

Minor differences in the amino-acid sequences of the α and the γ heavy chains led to further classification of the heavy chains into subclasses. In humans, there are two subclasses of α heavy chains (α_1 and α_2) and four subclasses of γ heavy chains (γ_1, γ_2, γ_3 and γ_4).

Both light and heavy chains have a variable sequence at their N-terminal ends but a constant sequence at their C-terminal ends. Light chains have a **constant region** (C_L) about 110 amino acids long and a **variable region** (V_L) of the same size. The variable region (V_H) of the heavy chains (at their N-terminus) is also about 110 amino acids long, but the heavy-chain constant region (C_H) is about three to four times longer (330 or 440 amino acids), depending on the class. It is the N-terminal ends of the light and heavy chains that come together to form the antigen-binding site.

The diversity in the variable regions of both light and heavy chains is for the most part restricted to three small *hypervariable regions* (each ~10 amino acid residues long) in each chain called **complementarity determining regions** (CDR); the remaining parts of the variable region, known as **framework regions**, are relatively constant. Proceeding from either the V_L or V_H amino terminus, these regions are called CDR1, CDR2 and CDR3. The CDR3 is the most variable of the CDRs.

Deduction of Ab structure

Porter and Edelman elucidated the basic structure of the Ig molecule but their approaches were different. Porter found that proteolytic treatment with the enzyme *papain* split the Ig molecule (molecular mass 150,000Da) into three fragments of about equal size. Two of these fragments were found of equal size, each of molecular mass 45,000Da called **Fab fragments** (fragment antigen binding) because they had Ag binding activity. Fab fragments are considered to be univalent, possessing one binding site each and being in every way identical to each other. The third fragment of molecular weight 50000 called the **F_C fragment** (fragment crystallizable). It cannot bind antigen. At about the same time, Edelman discovered that when γ-globulin was extensively reduced by treatment with mercaptoethanol, the molecule fell apart into four chains: two identical light chains with a molecular mass of about 53,000Da each and two others of about 22,000 Da each. The large molecules were designated **heavy** (H) **chains** and the smaller ones, **light** (L) **chains**. On the basis of these results, the structure of immunoglobulin molecules, was proposed.

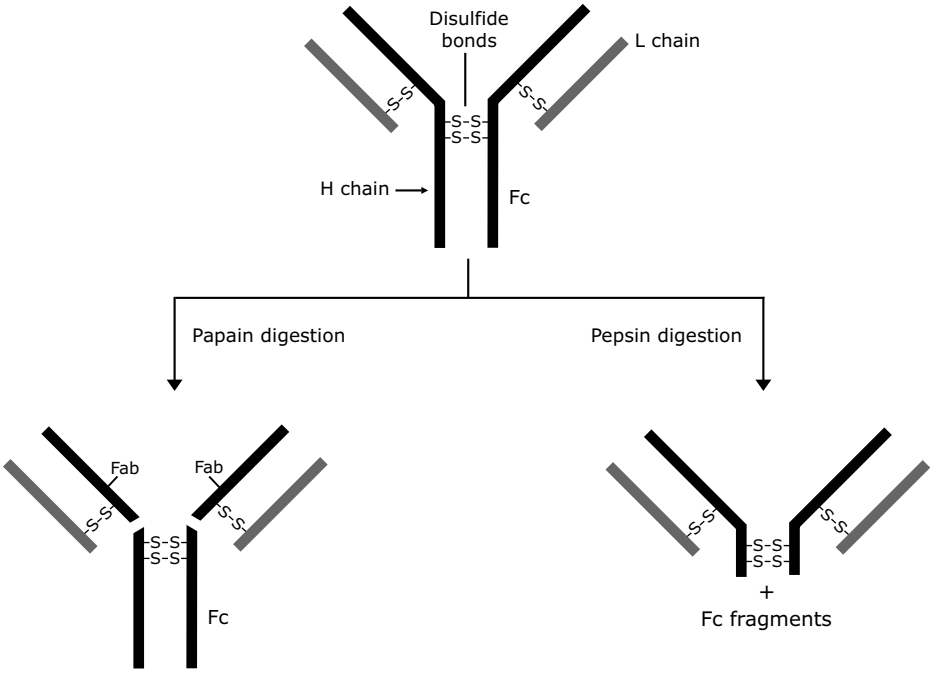


Figure 5.12 Prototype structure of IgG, showing chain structure and interchain disulfide bonds. The fragments produced by various treatments are also indicated. Light (L) chains are depicted in gray and heavy (H) chains in black.

Another researcher, Nisonoff used *pepsin* to digest the Ig. It generated a single 100,000 MW fragment composed of two Fab-like fragments and designated $F(ab')_2$. Unlike Fab-fragments, $F(ab')_2$ fragment was able to precipitate antigens. The F_C fragment was not recovered from pepsin digestion, because it had been digested into multiple fragments.

Immunoglobulins domains

The structure of the immunoglobulin molecule is determined by the primary, secondary, tertiary and quaternary organization of the protein. The *primary structure* i.e. the amino-acid sequence, accounts for the variable and constant regions of the heavy and light chains. The *secondary structure* is formed by folding of the extended polypeptide chain back and forth upon itself into an *antiparallel β-pleated sheet*. The chains are then folded into a tertiary structure of compact globular domains, which are connected to neighboring domains by continuations of the polypeptide chain that lie outside the β-pleated sheets. Finally, the globular domains of adjacent heavy and light

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5.15 The complement system

The complement (term coined by Paul Ehrlich) system, made up of approximately 30 serum and cell surface proteins/glycoproteins, is an important effector arm of both the innate and antibody-mediated acquired immune responses. The complement systems perform various functions. These functions include: *Lysis of cells, opsonization, viral neutralization, inflammatory responses* and *solubilization* and *clearance of immune complexes* which remove the immune complexes from the circulation and deposit them in the spleen and liver.

Complement components

Complement components include a group of proteins/glycoproteins. These components are designated by numerals (C1–C9), by letter symbols (e.g. factor D), or by trivial names (e.g. homologous restriction factor). Designation C1, C2,…C9 denote the order in which the components were discovered, rather than their position in the activation sequence.

Complement components are synthesized mainly by liver hepatocytes. Monocytes, tissue macrophages, and epithelial cells of the gastrointestinal tract also produce significant amounts of complement components. In the circulation, the concentration of all complement proteins is about 3 mg/mL. Some complement components are found at high concentrations (e.g. C3 at about 1 mg/mL), while others (such as factor D and C2) are found in only trace amounts.

Complement activation pathways

Most complement components are present in the serum in functionally inactive forms as proenzymes or zymogens. Complement activation pathways involve the activation of one component that triggers the activation of the next component in the sequence.

Upon activation, individual components are split into fragments, designated by lower-case letters. The smaller of the cleaved fragments is generally designated with a lower-case *a,* and the large fragment with a lower-case *b*; for historical reasons, however, the large cleavage fragment of C2 is usually referred to as C2a and the smaller fragment C2b. The larger fragments bind to the target near the site of activation, and the smaller fragments diffuse from the site and can initiate localized inflammatory responses. The complement fragments interact with one another to form functional complexes. Those complexes that have enzymatic activity are designated by a bar over the number or symbol (e.g. $\overline{\text{C3bBb}}$). There are three pathways of complement activation. These are the **classical**, **lectin** and **alternative pathways**.

Classical pathways

The classical pathway was so named because it was the first complement pathway to be worked out. The classical pathway commonly begins with the formation of antigen-antibody complexes. Thus this pathway is a major effector pathway of the humoral adaptive immune response. Other activators include some viruses, necrotic cells and subcellular membranes (e.g. from mitochondria), aggregated immunoglobulins, and beta-amyloid, found in Alzheimer disease plaques.

The classical pathway is initiated when C1 binds to the antibody in an antigen-antibody complex. C1 is a complex of three different proteins: C1q (made up of six identical subunits) combined with two molecules each of C1r and C1s. IgM and certain subclasses of IgG (human IgG1, IgG2, and IgG3) are efficient complement-activating antibodies. Among the human immunoglobulins, the ability to bind and activate C1 is, in decreasing order, IgM > IgG3 > IgG1 >> IgG2. IgG4, IgD, IgA, and IgE do not have C1q receptors; these antibodies do not bind or activate C1 and thus do not activate the classical complement pathway. C1q of C1 binds to the Fc region of antibodies. This activates C1r, a *serine protease* that initiates the cascade of the classic pathway. First, C4 is proteolytically activated into C4b, which in turn cleaves C2 into C2a and C2b. C4b and C2a together form *C3 convertase*.

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Chapter 06

Diversity of Life

6.1 Taxonomy

Taxonomy (arrangement by the rules) is the branch of biology that deals with *identification* (placement of a new organism into a previously described group), *nomenclature* (the naming of organisms) and *classification* (ordering of organisms into groups- can be phenetic or phylogenetic) of organisms. **Systematics** is the process of organizing taxonomic information about organisms into a logical classification that provides the framework for all comparative studies. It is the scientific study of biological diversity and its evolutionary history. Systematics and taxonomy are collectively referred to as the *systematic biology*.

Levels of taxonomy

There are three levels of taxonomy:

Alpha taxonomy	:	It is concerned with finding, describing and naming of organisms. This is the first and most basic step in taxonomy.
Beta taxonomy	:	It includes identification of natural groups and biological classes.
Gamma taxonomy	:	It includes study of evolutionary processes and patterns.

Organisms were first classified more than 2,000 years ago by Greek philosopher *Aristotle*. He classified organisms as either plant or animal. Modern biological classification began with the eighteenth century Swedish naturalist *C. Linnaeus.* He established a simple system for classifying and naming organisms. He developed a hierarchy (a ranking system) for classifying organisms that is the basis for modern taxonomy.

6.1.1 Nomenclature

Nomenclature is the formal naming of a particular organism according to some standardized system. The fundamental principle of nomenclature is that each organism must have only one scientific name. In contrast to scientific names, many organisms also bear *common names* (also called vernacular names), which are generally used by people within a limited geographic region. Presently, the criteria for scientific naming of plants, algae and fungi are based on the rules and recommendations of the International Code of Nomenclature or ICN. It was formerly called the International Code of Botanical Nomenclature (ICBN); the name was changed at the International Botanical Congress in July 2011. Similarly, International Code of Zoological Nomenclature (ICZN) provides rules for naming animals. The scientific names of species are **binomials** (literally meaning two names). Swedish botanist Linnaeus (referred to as the *father of taxonomy*) first proposed the system of *binomial nomenclature* or *binary nomenclature*. According to binomial nomenclature, the first name of the binomial is the *genus name* (generic name). The second name of the binomial is the *specific epithet* or *species epithet* (*specific name* - a term used only in zoology). The first word denoting the genus starts with a capital letter while the specific epithet starts with a small letter. Binomial names are always either italicized or underlined to indicate their Latin origin. For example, *Mangifera indica* is the scientific

name of mango. In this name *Mangifera* represents the genus while *indica,* is a specific epithet. A binomial name in which the name of the genus and that of the species are identical is called *tautonym*.

Nomenclature types

The scientific names must be associated with some physical entity, known as a *nomenclatural type* or simply *type*. A nomenclatural type is one particular specimen of an organism to which the scientific name of that organism is formally attached. Types are usually physical specimens that are kept in a museum or herbarium research collection, but it may also be an illustration. The type serves the purpose of acting as a reference for the name, upon which the name is based. There are different types of nomenclatural type:

Holotype : A holotype is the one specimen or illustration upon which a name is based.

Isotype : An isotype is a duplicate specimen of the holotype, collected at the same time by the same person from the same population.

Lectotype : A lectotype is a specimen that is selected from the original material to serve as the type when no holotype was designated at the time of publication, if the holotype is missing, or if the original type consisted of more than one specimen or taxon.

Neotype : A specimen later selected to serve as the single type specimen when an original holotype has been lost or destroyed, or where the original author never cited a specimen.

6.1.2 Classification

Classification is the arrangement of taxa into some type of order. The purpose of classification is to provide a system for cataloguing and expressing relationships between taxa. Classification is not a single step process but involves hierarchy of steps in which each step represents a category termed **rank**. Rarely, the term *taxonomic category* is used instead of *rank*. Every organism can be classified at seven taxonomic ranks- *kingdom*, *phylum, class, order, family, genus* and *species*. Each rank contains organisms with similar characteristics. The kingdom is the largest unit of classification. It splits into smaller units called *phyla* (singular, *phylum*). Phyla splits into *classes,* classes into *orders,* orders into *families*, families into *genera* (singular, *genus*) and genera into *species.*

Example:

Seven taxonomic ranks	*Taxa*
Kingdom	Plantae
Phylum (Division)	Magnoliophyta
Class	Liliopsida
Order	Arecales
Family	Arecaceae
Genus	*Cocos*
Species	*Cocos nucifera*

The various units of classification – kingdom, phylum, class and so on — are called *taxonomic categories* and together they make up the *taxonomic hierarchy.* A **taxon** is a defined group of organisms typically treated at a given rank. In the above example, Magnoliophyta is a taxon placed at the rank of phylum; Liliopsida is a taxon placed at the rank of class. As one progresses down the hierarchy, the number of organisms in each taxon decreases, and the similarities between them increases.

6.1.3 Biological species concept

Biologists almost universally agree that the species is a fundamental natural unit. However, biologists have not been able to agree on exactly how species should be defined. Taxonomists practically define species by means of morphological or phenetic characters. If one group of organisms consistently differs from other organisms, it will be defined as a separate species.

However, the most widely accepted species concept is the *biological species concept*. It defines species in terms of interbreeding. According to Ernst Mayr, *species are groups of interbreeding natural populations that are reproductively isolated from other such groups.* Reproductively isolated means that members of the species do not interbreed with members of other species. The biological species concept explains why the members of a species resemble one another, and differ from other species. When two organisms breed within a species, their genes pass into their offspring. As this process is repeated, the genes of different organisms are constantly shuffled around the species gene pool. The shared gene pool gives the species its identity. By contrast, genes are not (by definition) transferred to other species, and different species therefore take on a different appearance. If two species are very similar to each other morphologically so much so that it is difficult to distinguish but are nevertheless reproductively isolated are termed as **sibling species** (sometimes called *cryptic species*).

Subspecies

Subspecies is a taxonomic subdivision of a species often distinguished by special phenotypic characters and by its origin or localization in a given geographical region. Members of a subspecies would share a unique, geographic locale, a set of phylogenetically concordant phenotypic characters, and a unique natural history relative to other subdivisions of the species. Although subspecies are not reproductively isolated, they will normally be allopatric and exhibit recognizable phylogenetic partitioning.

The terms *race* and *subspecies* are potentially misleading and are most often used synonymously. Although the term race is normally used in case of human populations. However, when a distinction is made, 'race' generally implies a lower level of differentiation.

A species which consists of a single uniform population, devoid of differentiated and semi-isolated local populations (subspecies) is *monotypic* (i.e. all one basic kind). A species is *polytypic* if it is subdivided into two or more subspecies.

6.1.4 Phenetics

Phenetic means 'as observed in the phenotype'. Phenetics, also known as *taximetrics*, is an attempt to classify organisms based on overall similarity, usually in morphology or other observable traits, regardless of their phylogeny or evolutionary relationship. Phenetic characters refer to all the observable or measurable characters of an organism. Phenetic classification schemes do not distinguish between homologous characters and analogous characters. The result of phenetic relationship is often summarized with a tree-like network called a *phenogram*. Phenogram is a type of dendrogram which is based on phenetic data. Lines called phenon lines, drawn at right angles to the dichotomously branching dendrogram, represent lines of percentage similarity of phenetic features between organisms. (*Dendrogram* is a diagram that represents relationships among groups of taxa, with the highest taxon at the base of a vertical line from which lower taxa branch at appropriate levels).

A common misconception equates phenetics with numerical taxonomy, but in fact most phenetic taxonomy is not numerically based. Phenetics is not characterized by any particular methodology. *Numerical taxonomy* is a classification system which developed as a part of multivariate analysis. It aims to create a taxonomy using numeric algorithms like cluster analysis rather than using subjective evaluation of their properties. It is based on quantifying observable characteristics of organisms. The closeness between different groups is determined by the total number of individual's characteristics which they have in common.

Determination of phenetic relationship (overall similarity) between Operational Taxonomic Units

The character states are compared from one OTU (Operational Taxonomic Unit) to the next. A number of methods (or coefficients) exist for making such comparisons between OTUs and one of the simplest method is called the *simple matching coefficient*. This is calculated by counting the number of character states for each character which are common between two OTUs and then dividing this number by the total number of characters used. The simple matching coefficients are then arranged to form a *similarity matrix*. This is a matrix in which the rows and columns represent organisms and each value is an association coefficient measuring the similarity of two different organisms so that each organism is compared to every other one in the table.

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To see how this analysis is done, consider the set of characters for each of five vertebrates– a leopard, turtle, salamander, tuna, and lamprey. As a basis of comparison, an outgroup is selected. An **outgroup** is a species or group of species from an evolutionary lineage that is known to have diverged before the lineage that includes the species being studied (the **ingroup**). An example of such a character is a structure called a notochord, a flexible rod running the length of the animal. Lancelets have notochords throughout their lives, but in vertebrates, the notochord is present only in embryos and is replaced later in development by the backbone. The species making up the ingroup display a mixture of shared ancestral and shared derived characters. The outgroup comparison enables us to focus on just those characters that were derived at the various branch points of vertebrate evolution. Note that *all* the vertebrates in the ingroup have backbones; this is a shared primitive character that was present in the ancestral vertebrate, though not in the outgroup.

Cladogram

A branching diagram (dendrogram) that is constructed by cladistic principle and method is called a *cladogram.* Any branching of the cladogram represents lineage divergence, the formation of two separate lineages from one common ancestor. The *point* of divergence is termed a *node* and the region between two nodes is called an *internode* or internal branch. Taxa placed at the ends of branches are called *terminal taxa* and the branches leading to them are called *terminal branches*. The two taxa that split from the node are called *sister taxa*. The sister taxa must be more closely related to each other than to any other group because they share a close common ancestor.

The term *phylogenetic tree* is often used synonymously with *cladogram.* However, *phylogenetic tree and cladogram are not same*. A phylogenetic tree (*phylogram*) is a specific type of cladogram where the branch lengths are proportional to the predicted evolutionary time between organisms or sequences. Cladograms are branched diagrams, similar in appearance to family trees, that illustrate patterns of relatedness where the branch lengths are not necessarily proportional to the evolutionary time between related organisms or sequences.

Apart from phenetic and phylogenetic taxonomy, some scientists advocate evolutionary taxonomy. *Evolutionary taxonomy* incorporates both phenetic and phylogenetic elements. In terms of the kinds of taxonomic groups, evolutionary classification recognizes paraphyletic and monophyletic but not polyphyletic groups. In terms of the kinds of characters used to infer phylogeny, it forms groups by homologies rather than homoplasies, but does not distinguish ancestral from derived homologies.

6.2 The five–kingdom system

In the early days of biology, all organisms were classified as either plants or animals. In year 1969, Robert H. Whittaker, an American taxonomist, described a five-kingdom classification. The five kingdoms are *Monera* (prokaryotes)*, Protista, Fungi, Plantae* and *Animalia*. His classification is based on two main criteria - the level of organization of the organisms and their method of nutrition. He recognized two levels of organization - prokaryotes (which are without nucleus) and eukaryotes (which are with nucleus). Eukaryotes are further subdivided into unicellular and multicellular eukaryotes. The methods of nutrition were: heterotrophic (which are further subdivided into ingestive and absorptive) and autophototrophic.

Monera : Prokaryotic cellular organization.

Protista : Unicellular eukaryotes.

Fungi : Multicellular eukaryotes which feed heterotrophically by absorption.

Plantae : Multicellular eukaryotes which are photoautotrophic.

Animalia : Multicellular eukaryotes which feed heterotrophically by ingestion.

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creating a level of classification larger than a kingdom – a new category called a *domain*. The two domains of prokaryotes are Bacteria and Archaea. The domain Eukarya encompasses all of the kingdoms of eukaryotes. The *three-domain system* of biological classification was introduced by Carl Woese on the basis of differences in the signature sequences of 16S/18S rRNA genes of eubacteria, archaebacteria and eukaryotes.

History of taxonomic concepts

Linnaeus, 1735	2 Kingdoms	Animalia and Vegetabilia
Haeckel, 1866	3 Kingdoms	Protista, Plantae and Animalia
Chatton, 1937	2 Empires	Prokaryota and Eukaryota
Copeland, 1956	4 Kingdoms	Monera (prokaryotes), Protista, Plantae and Animalia
Whittaker, 1969	5 Kingdoms	Monera, Fungi, Protista, Plantae and Animalia
Woese *et al*, 1977	6 Kingdoms	Eubacteria, Archaea, Protista, Fungi, Plantae and Animalia
Woese and Fox, 1999	3 Domain systems	Eubacteria, Archaea and Eukaryotes

6.3 Protists

The German zoologist Ernst Haeckel first used the term *protista*. Protists are mostly unicellular (some are colonial and multicellular) eukaryotic organisms. Most eukaryotes are single-celled organisms. Protists are difficult to characterize because of the great diversity of the kingdom. These organisms vary in body form, nutrition and reproduction. Protists are capable of sexual, as well as asexual reproduction. They can be free-living, or may live symbiotically with another organism. The symbiosis can be mutualistic, where both partners benefit, or parasitic, where the protist uses its host as a source of food or shelter while providing no advantage to the other organism. Protists are more nutritionally diverse than other eukaryote groups. Some protists are photoautotrophs. Some are heterotrophs, absorbing organic molecules or ingesting larger food particles. Still others, called mixotrophs, combine photosynthesis and heterotrophic nutrition.

Despite the great diversity evident in this kingdom, scientists have been able to classify the protists into several groups. The protists can be classified into one of three main categories, animal-like, plant-like and fungus-like. Grouping into one of the three categories is based on an organism's mode of reproduction, method of nutrition and motility. The animal-like protists are known as the *protozoa*, the plant-like photosynthetic protists are the *algae*, and the fungus-like protists are the *slime molds* and *water molds*.

6.3.1 Protozoan protists

The protozoa are all unicellular heterotrophs, mostly free living. They obtain their nutrition by ingesting other organisms or dead organic materials. The word 'protozoa' comes from the *Latin* word for 'first animals'.

Features of protozoa:

- Unicellular (single celled) organisms.
- Aquatic, free-living or parasitic organisms.
- Nutrition is heterotrophic.
- Locomotion with the help of pseudopodia, cilia or flagella. The protozoans are grouped into various phyla based on their modes of locomotion. They may use cilia, flagella or pseudopodia. Some protozoans are non motile. The protozoans that use pseudopodia to move are known as *amoeboid protozoans*, those that use flagella are called *flagellated protozoans*, those that use cilia are known as the *ciliated protozoans* and those that do not move are called the *sporozoans*.
- Both asexual and sexual reproduction occur among the protozoa. One of the simplest and most common forms of asexual reproduction is *binary fission*, a mitotic process. Other forms of asexual reproduction are *budding* and *multiple fission* (or schizogony). In multiple fission, a large number of daughter cells form from the division of a single protozoan.

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A generalized life cycle of bryophytes

During sexual life cycle, sporophyte produces spores in sporangium. Haploid spores formed by meiosis begins the gametophytic generation. The spore germinates to form a gametophyte, which is more diverse in bryophytes than in any other group of plants. The gametophyte bears antheridia and archegonia. Antheridia produce sperms, and archegonia produce eggs. Sexual reproduction in bryophytes requires free water, because the sperm must swim to the egg. Sperms released from an antheridium do not swim randomly; rather, they are attracted by a gradient of a still-unidentified chemotactic substance(s) produced by an archegonium. A sperm fertilizes an egg and forms a diploid zygote, thus beginning the sporophytic generation of the life cycle. Bryophytes have *heteromorphic* alternation of generation; that is, the sporophyte and gametophyte are distinctly different.

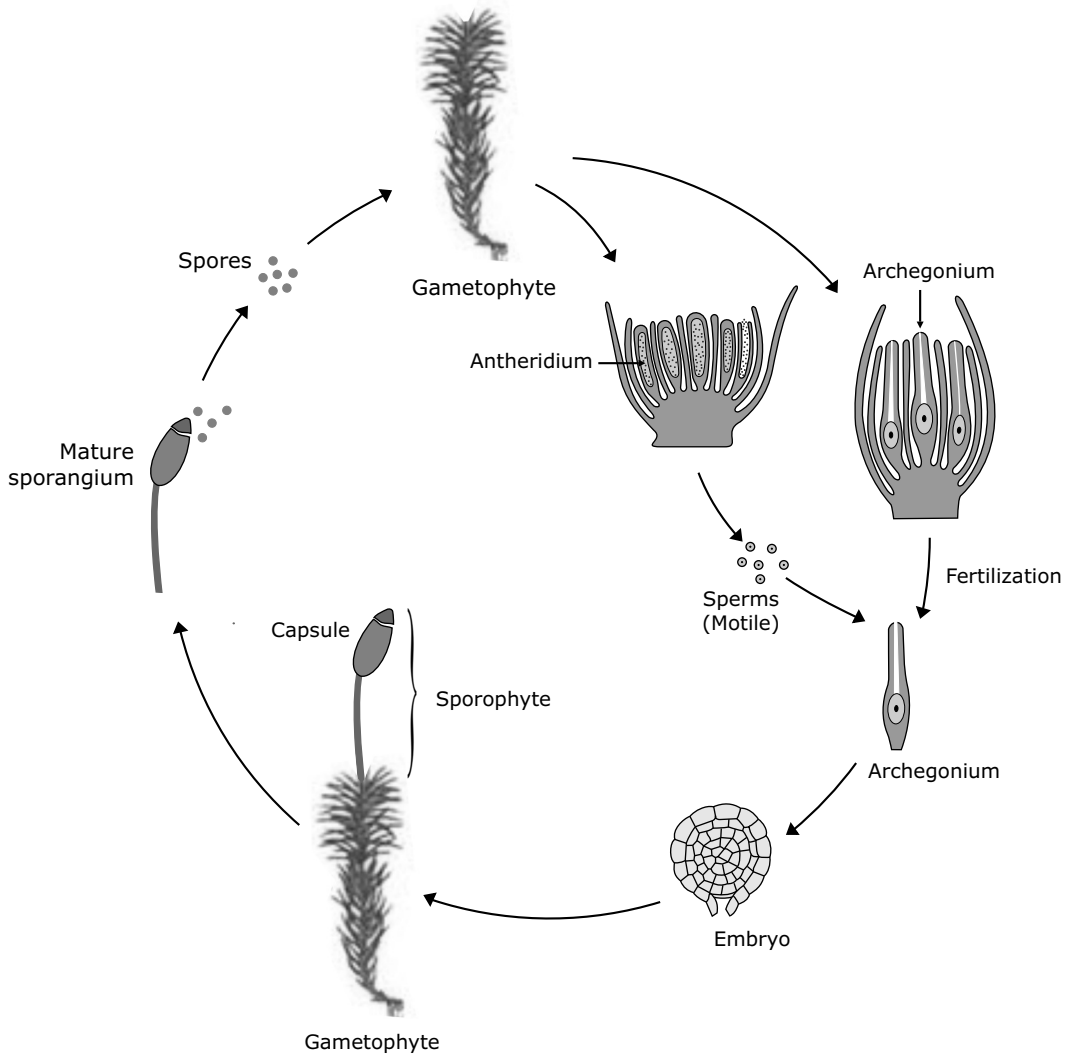
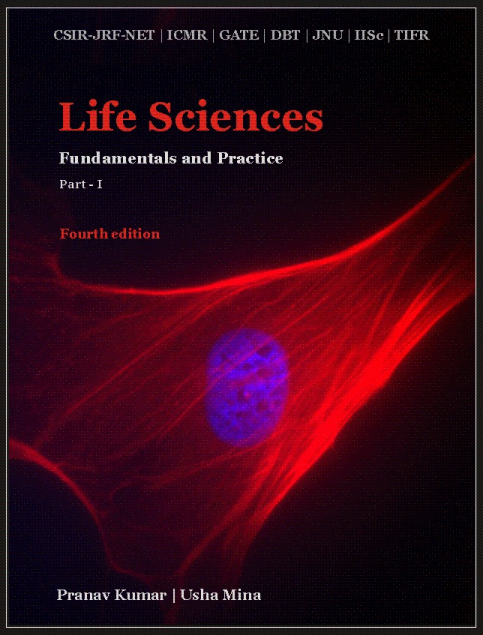


Figure 6.9 Sexual life cycle in *Funeria*. The spore germinates and produces a protonema (not shown in the figure) and the leafy stage of the gametophyte develops from the protonemal stage. The male and female sex organs, the antheridia and the archegonia respectively, are produced on the gametophytic plant. Haploid sperm are released from the antheridia and fertilize with a haploid egg present in an archegonium. The zygote develops into the diploid sporophyte. Typically sporophyte is differentiated into a foot, seta and spore forming capsule.

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